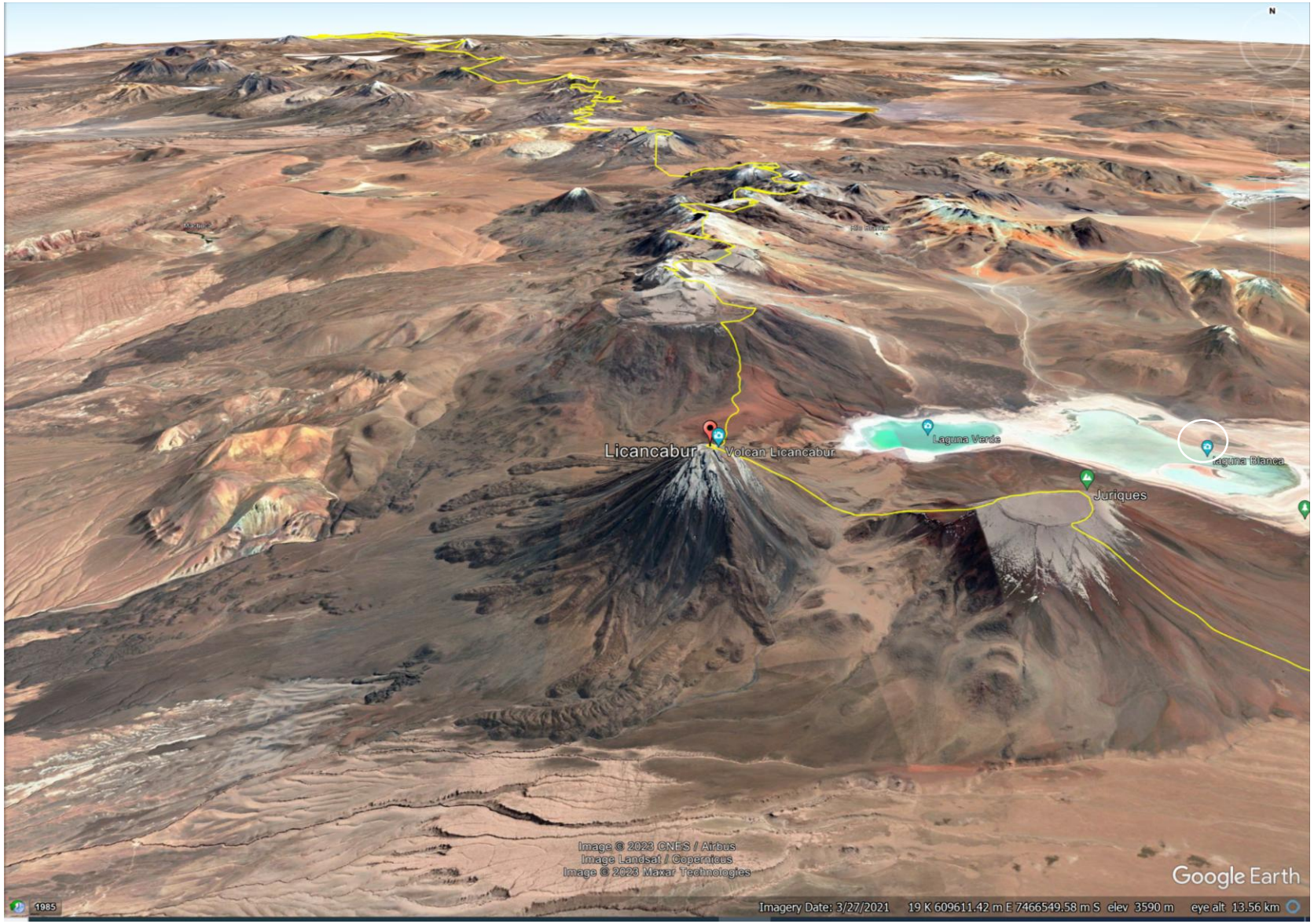


LICANCABUR LAHARZ Exercise





Atacama desert with almost no precipitation all year

Licancabur

6km

12km

15km

Image © 2023 CNES / Airbus
Image © 2023 Maxar Technologies

Google Earth

Imagery Date: 11/9/2021 22°51'18.30" S 67°56'11.54" W elev 3693 m eye alt 19.11 km



Atacama desert with almost no precipitation all year

Licancabur

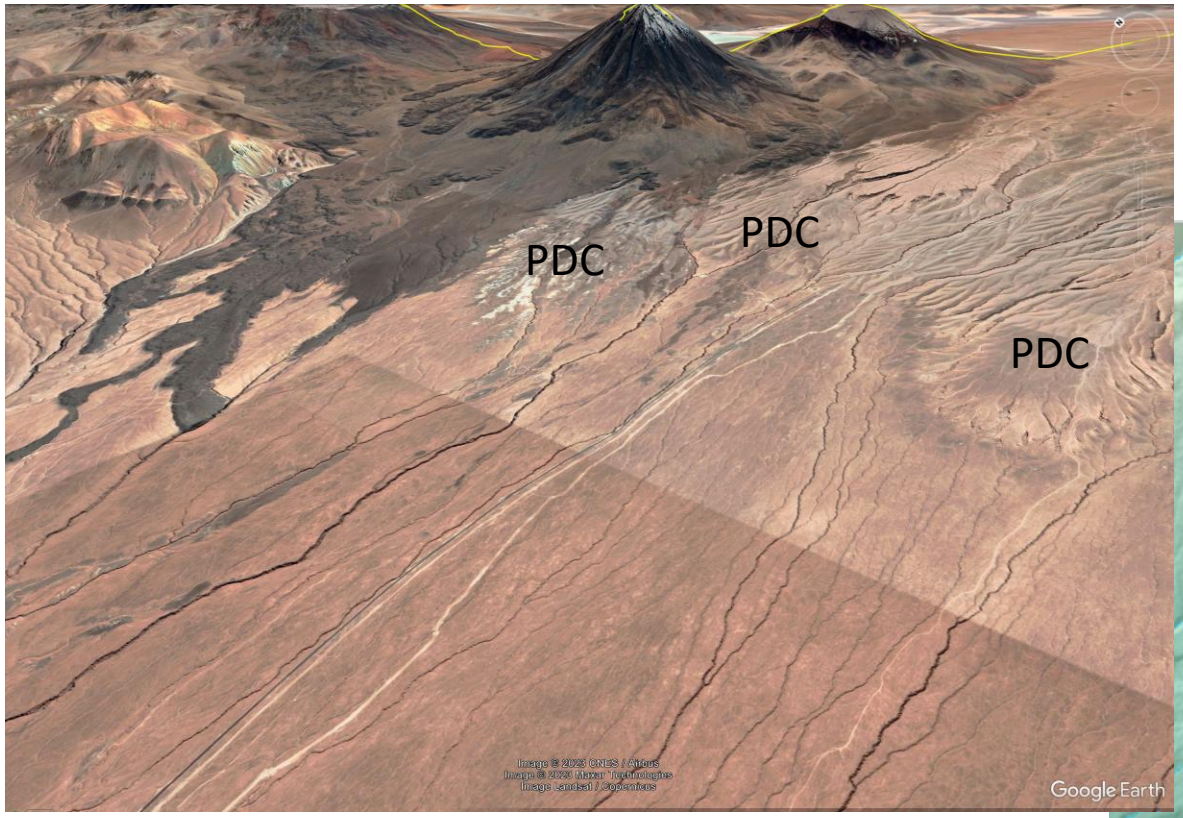
Licancabur

CHILE

15km

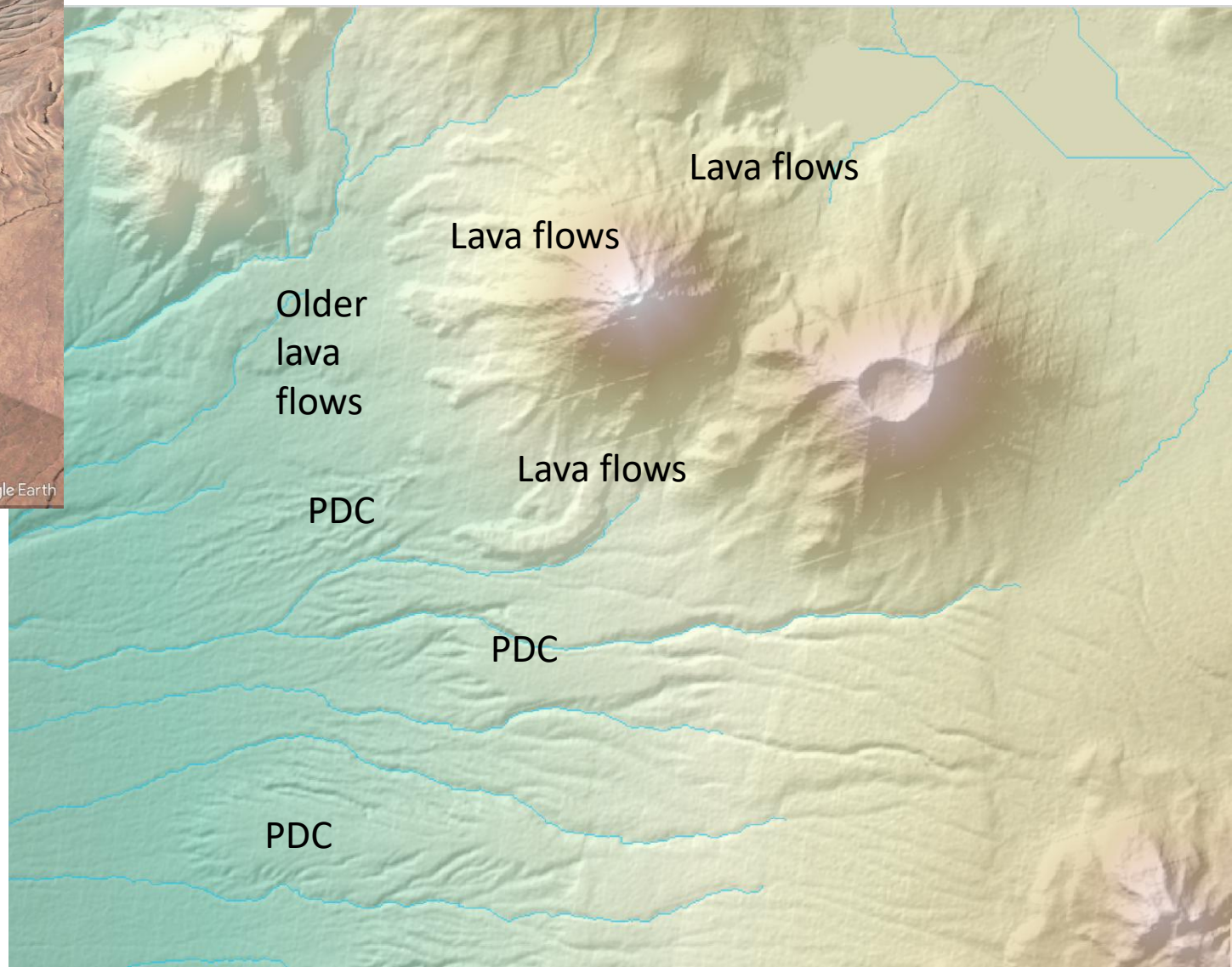
Google Earth

Imagery Date: 11/9/2021 22°51'18.30" S 67°56'11.54" W elev 3693 m eye alt 19.11 km

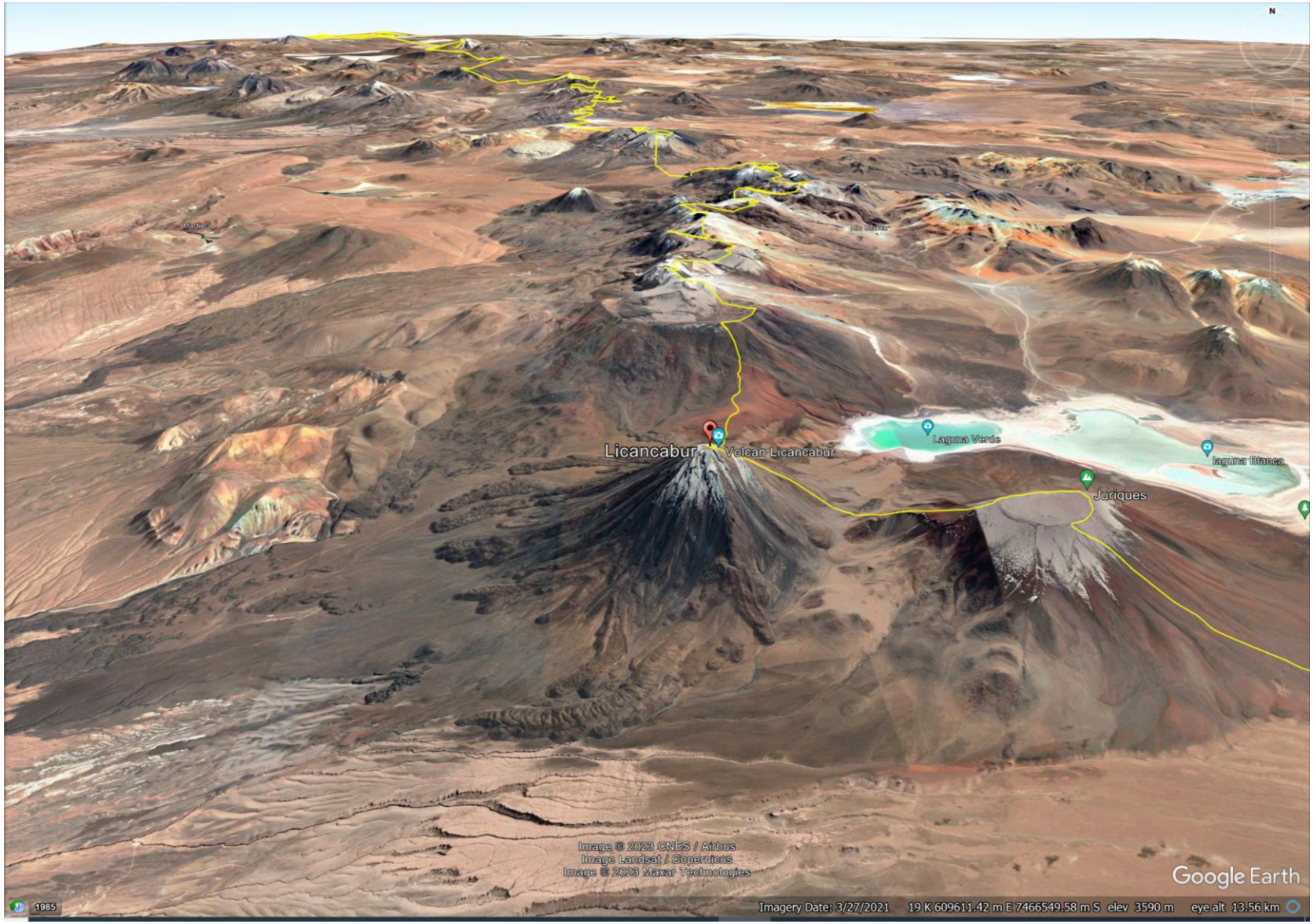


Google Earth and satellite imagery

Geomorphology seen in DEM and shaded-relief map



PDCs = Pyroclastic density currents

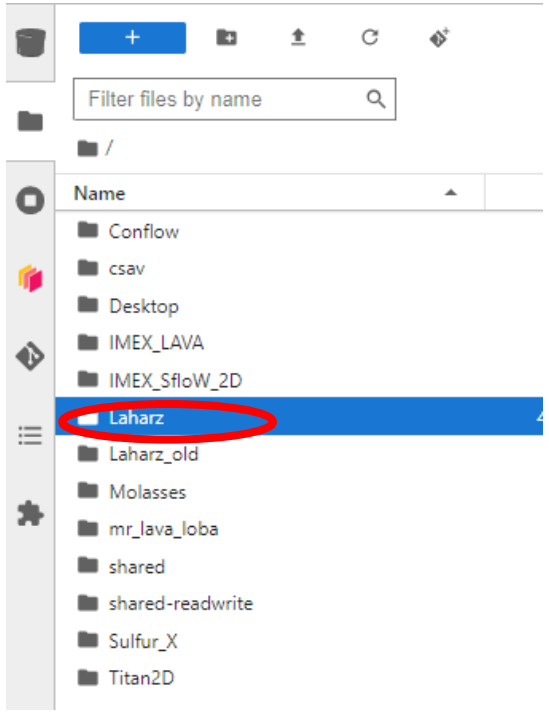


These exercises just use Licancabur as an example!

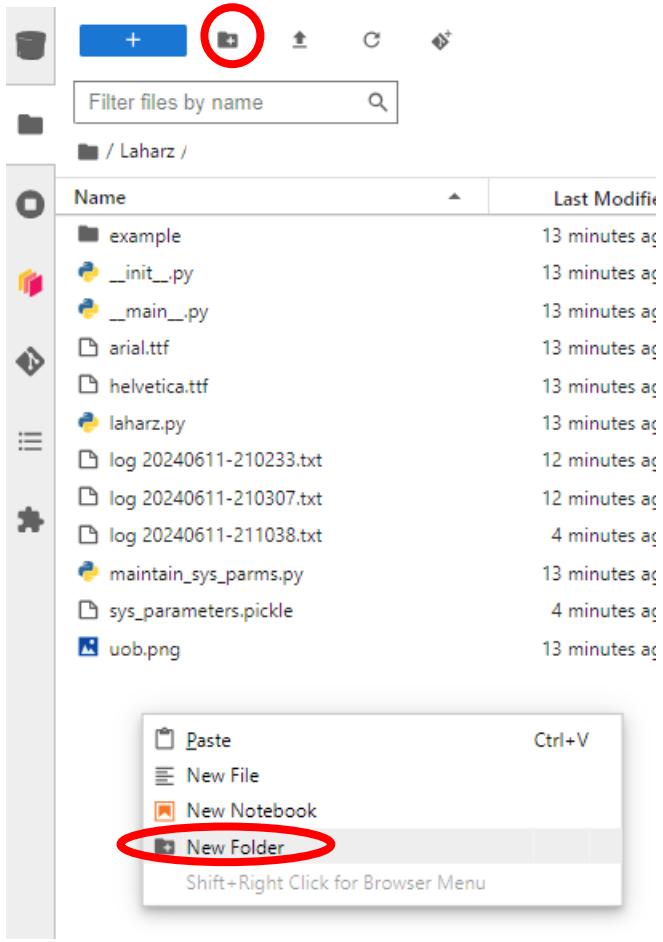
The model parameters in all the following exercises may or may not be realistic.

Our exercise hazard maps are certainly worse than any made by Chileans!!!!!

Let’s make a folder to work in!

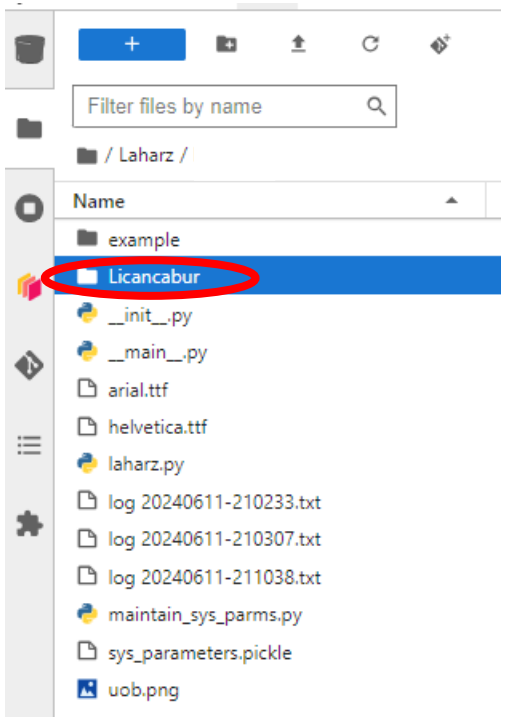


Double click on “Laharz”

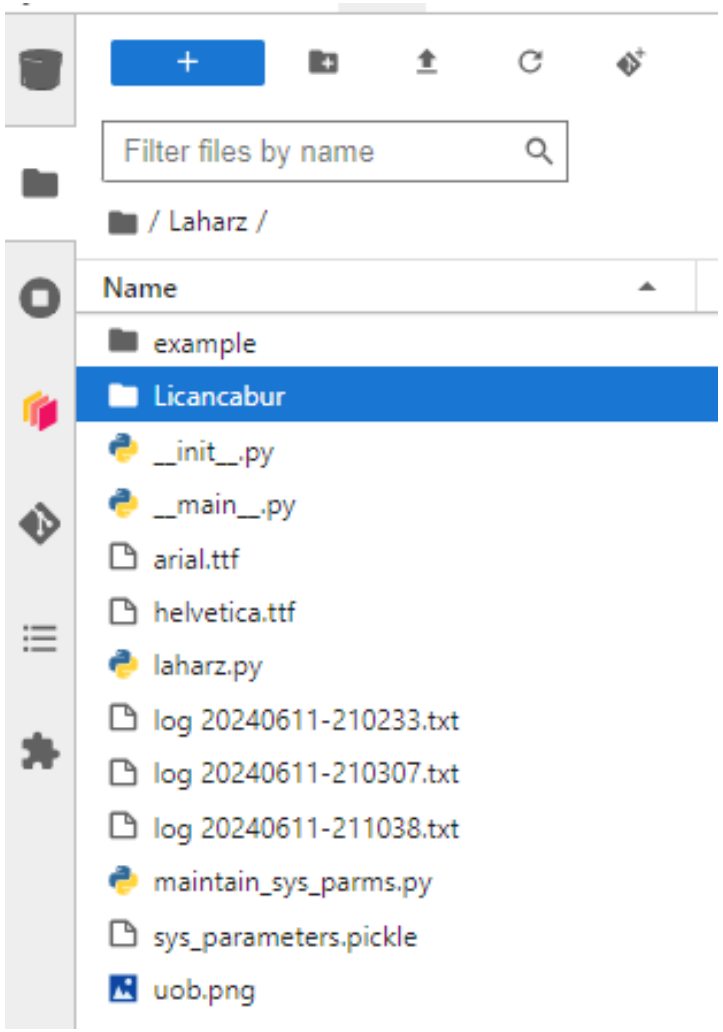


Right click in a blank space and choose “New Folder”

Or click the “New folder“ icon at the top



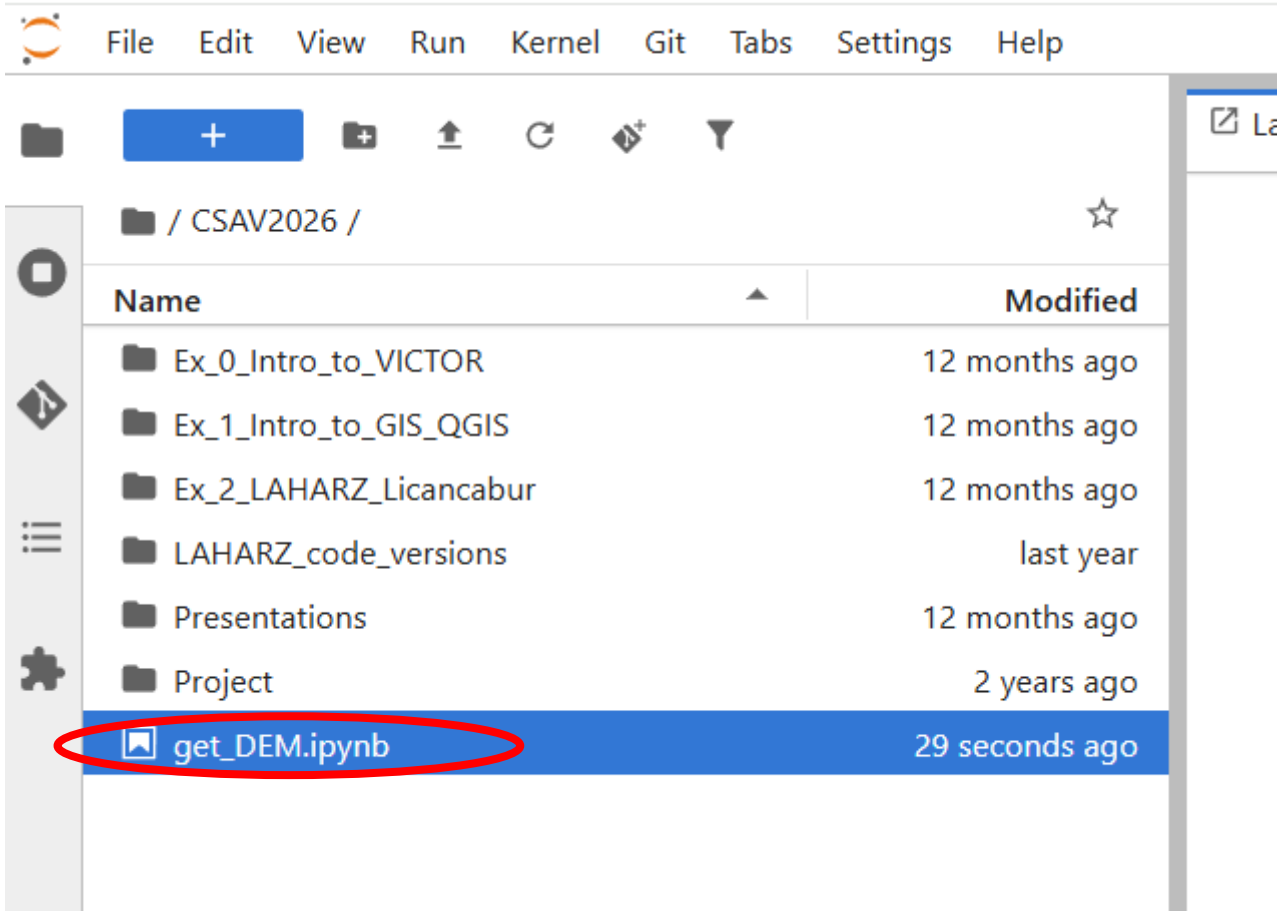
Name the new folder “Licancabur”



When you work on your projects, you'll want to make a new folder here too /Laharz/new_volcano_name (so if my project is on Mt. St. Helens I might make a folder called /Laharz/Helens)

Download a DEM

- Run the “get_DEM.ipynb” notebook to download a DEM



Download a DEM

- Run the “get_DEM.ipynb” notebook to download a DEM
- Run the import libraries cells, then set the input parameters as seen below

Setting input parameters

In the following cell we use the lat/lon of a target volcano’s summit, obtained from the Global Volcano Project database from the volcano’es name, to set the bounding box.

[8]:
```
#Choose a volcano and DEM options

volcano = victor.search_volcano('Licancabur') # Get volcano information from the GVP database

center_lat = volcano[1][0]  # Get volcano latitude
center_lon = volcano[2][0]  # Get volcano longitude

# Set the extent of the region of interest, in meters in each direction from the summit
eastwest_extent = 40000
northsouth_extent = 40000

# Set bounding box in lat/lon
utms = utm.from_latlon(center_lat, center_lon)
[north_lat, east_lon] = utm.to_latlon(utms[0]+eastwest_extent, utms[1]+northsouth_extent, utms[2], utms[3])
[south_lat, west_lon] = utm.to_latlon(utms[0]-eastwest_extent, utms[1]-northsouth_extent, utms[2], utms[3])

# Select a DEM format: Input "ascii" for ASCII format (.asc) and "tiff" for geotiff format
outputFormat = "tiff"

# Enter the name of the output file (without extension)
name = "Licancabur"

dataset = "SRTMGL1"

print(center_lat, center_lon)
```

```
-22.83 -67.88
```

Download a DEM

- Run the Obtain DEM cell to download the DEM

```
-22.83 -67.88
```

Obtain DEM

This cells obtains the DEM from OpenTopograhy using the input options. It will print out the resulting file name.

```
[4]: dem = victor.download_dem(north_lat, south_lat, east_lon, west_lon, outputFormat, dataset)
     print (dem)
```

```
DEM downloaded as DEM_22S_67W_SRTMGL1_11062026.geotiff
DEM_22S_67W_SRTMGL1_11062026.geotiff
```

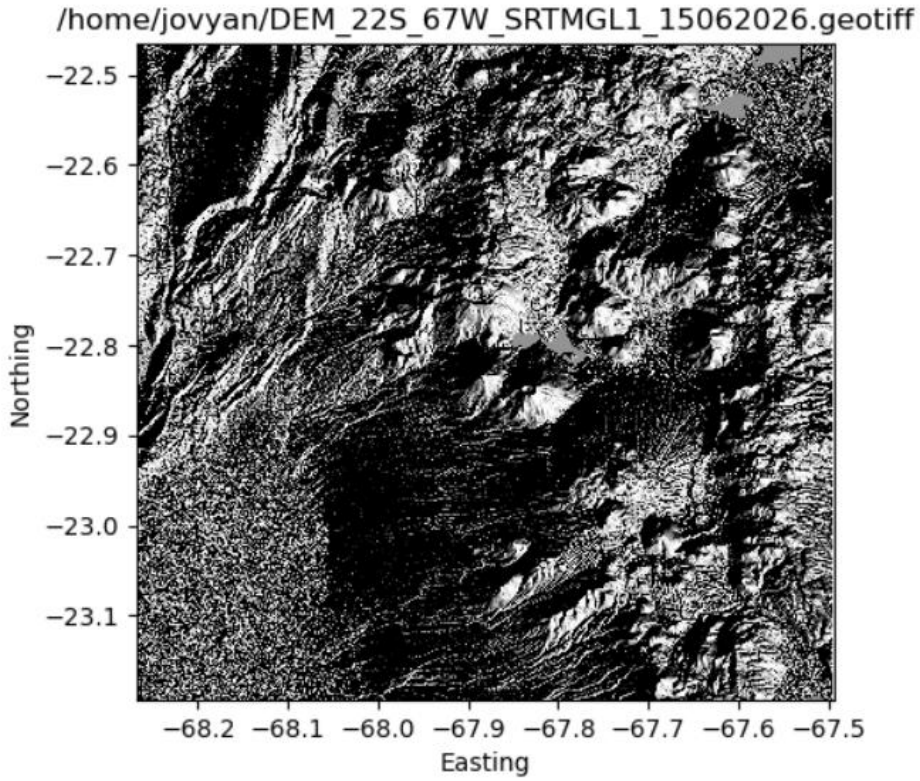
Download a DEM

- Run the DEM visualization cell to make sure it downloaded the right thing

DEM Visualization

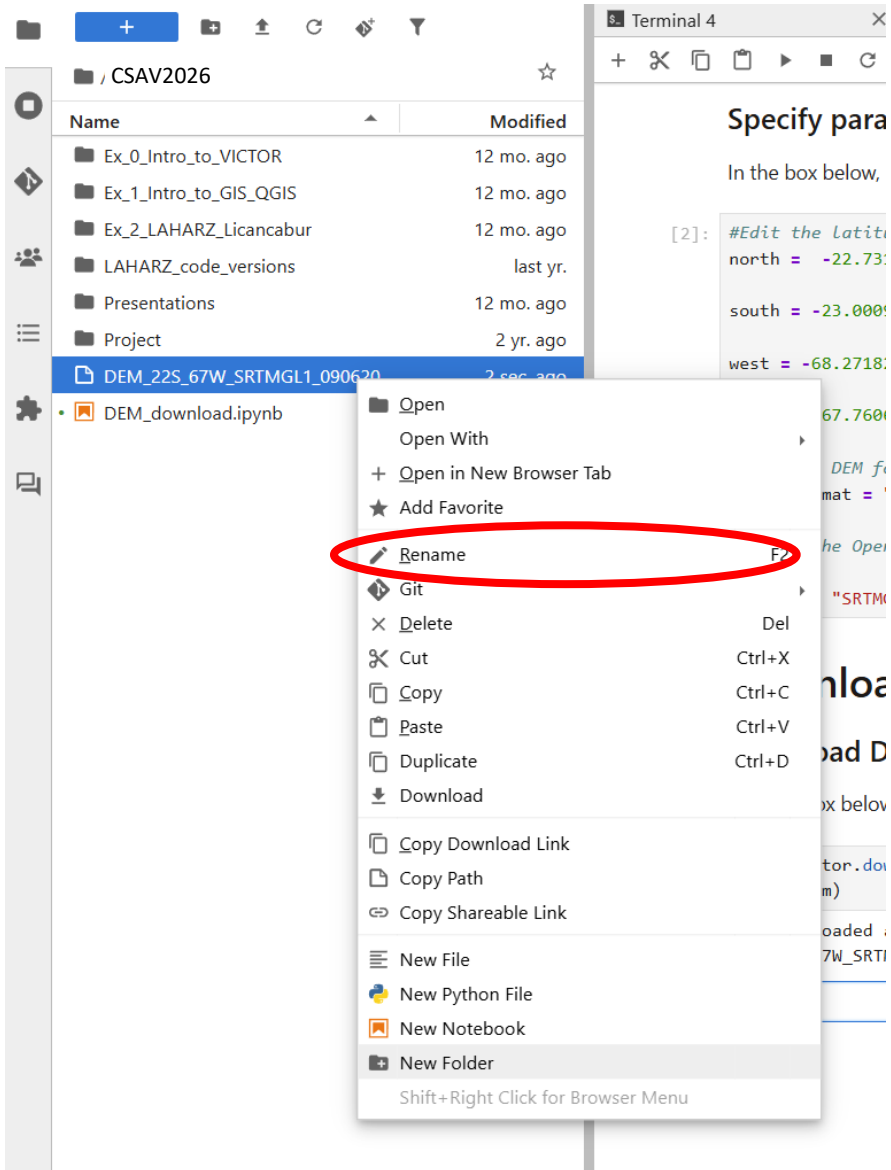
To make sure we got what we wanted, the next cell will plot the DEM using the victor.py DEM visualization function

[10]:
```
victor.plot_dem('/home/jovyan/'+dem)
plt.savefig(name)
```



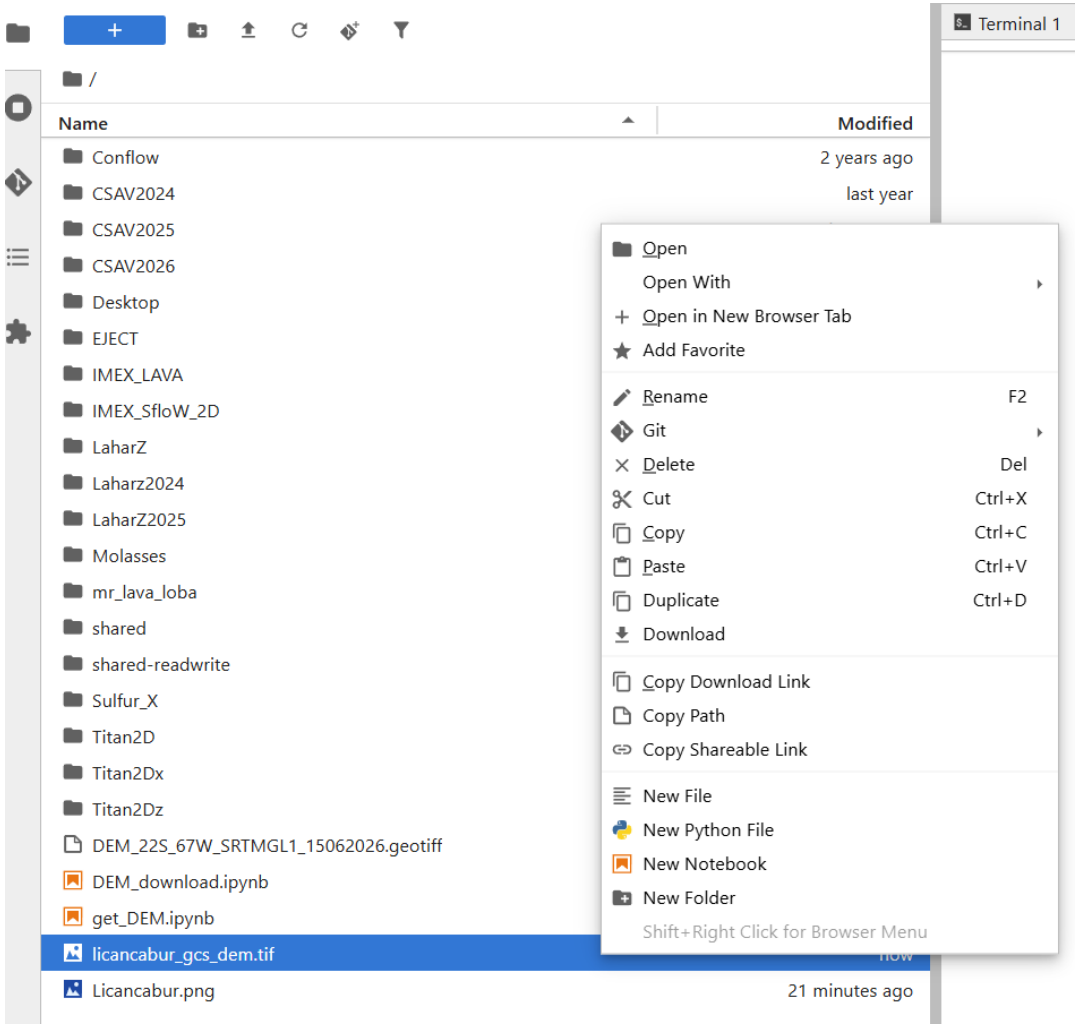
[ ]:

Rename the DEM

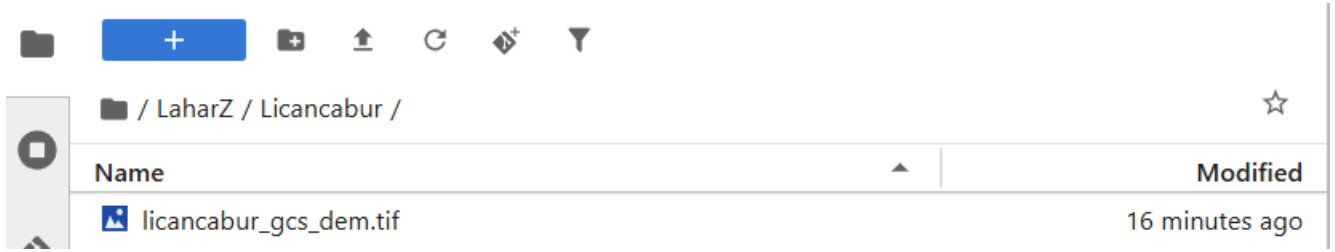


- The downloaded DEM should now be in your CSAV2026 folder
- Rename it by right clicking and choosing “Rename”
- Name it “licancabur_gcs_dem.tif”
- GCS=geographic coordinate system
- Make sure you change ‘geotiff’ to ‘tif’

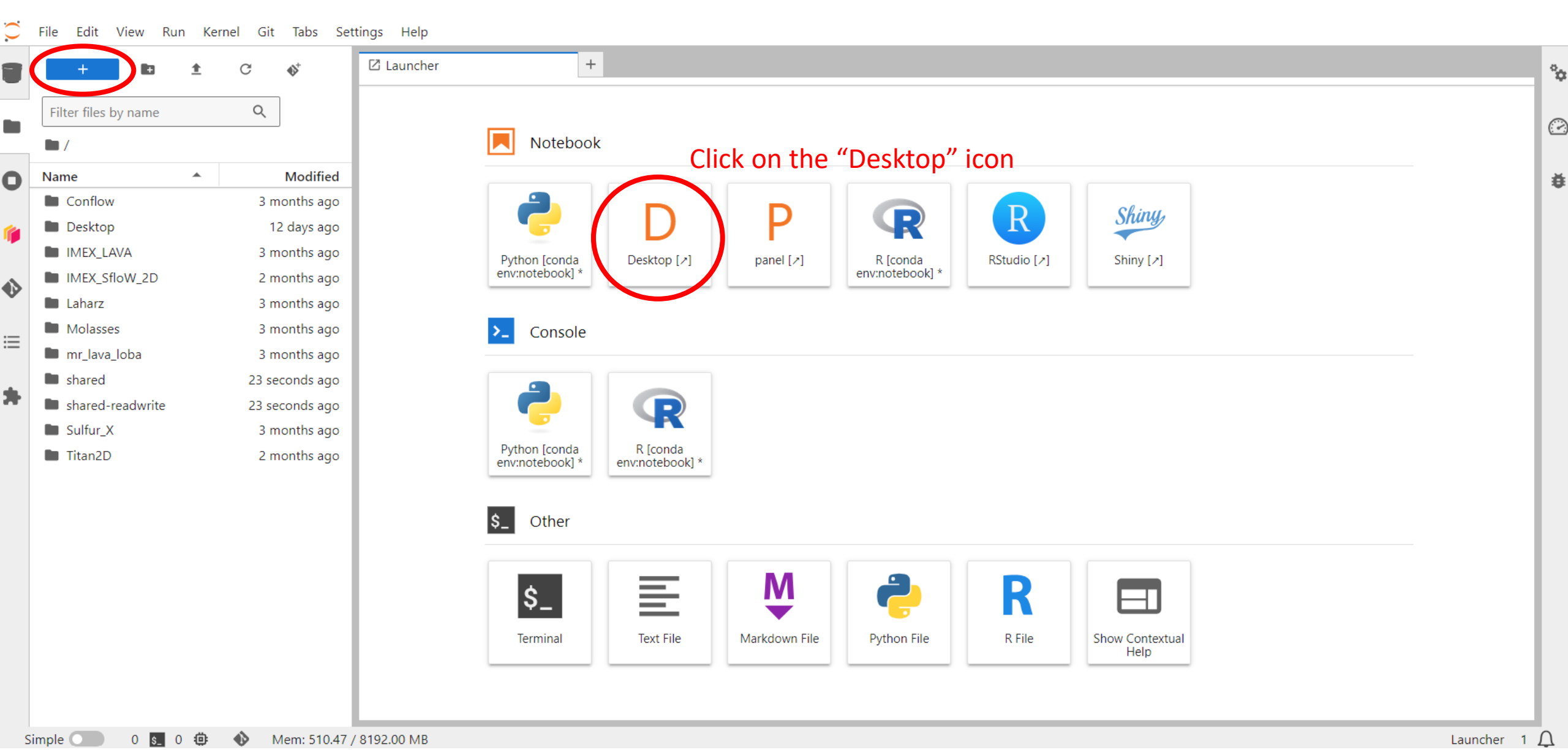
Move the downloaded Licancabur DEM to the new folder



Right click on your DEM and choose “Copy”

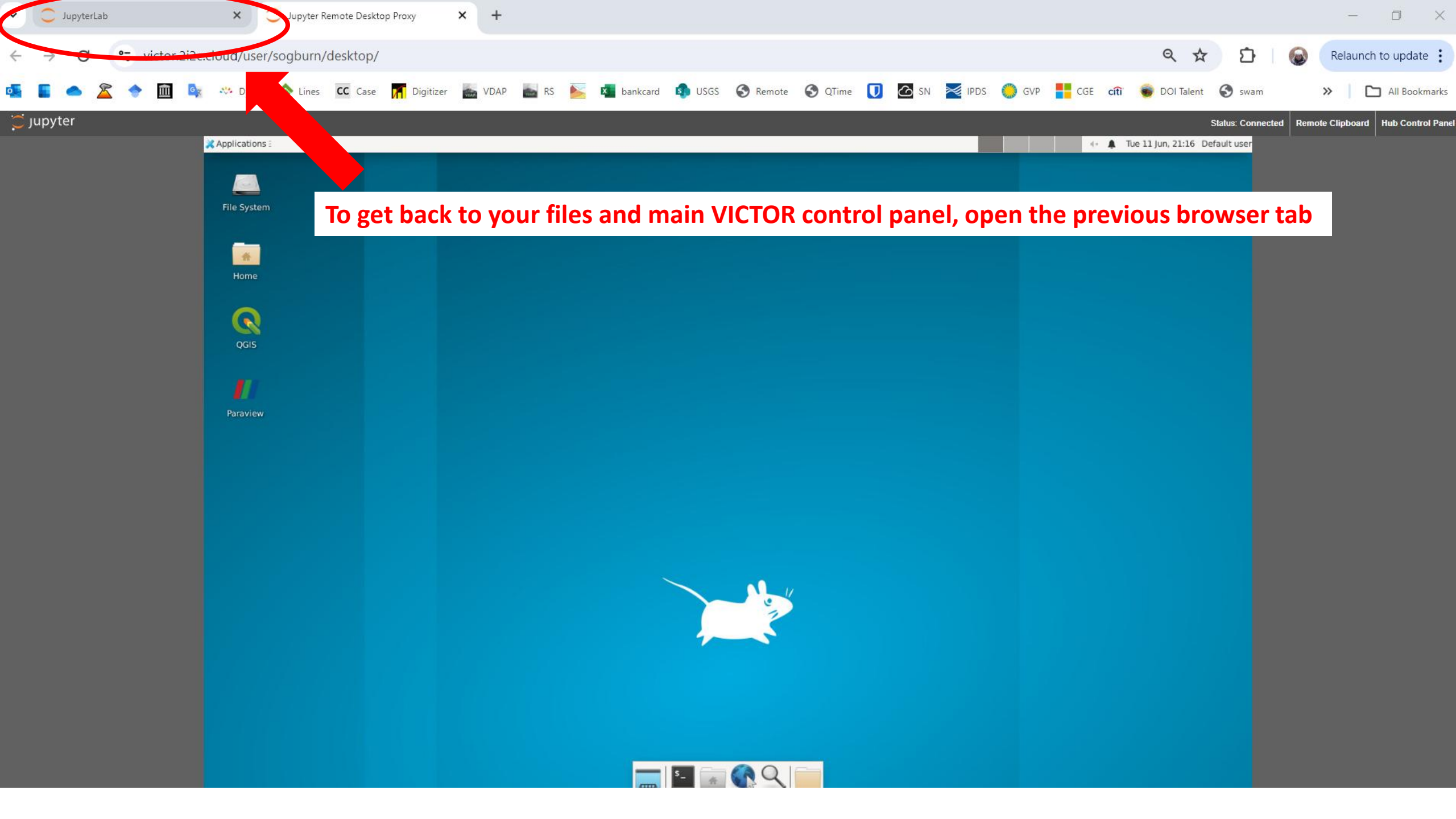


Navigate to LaharZ/Licancabur, right click, and choose “Paste”

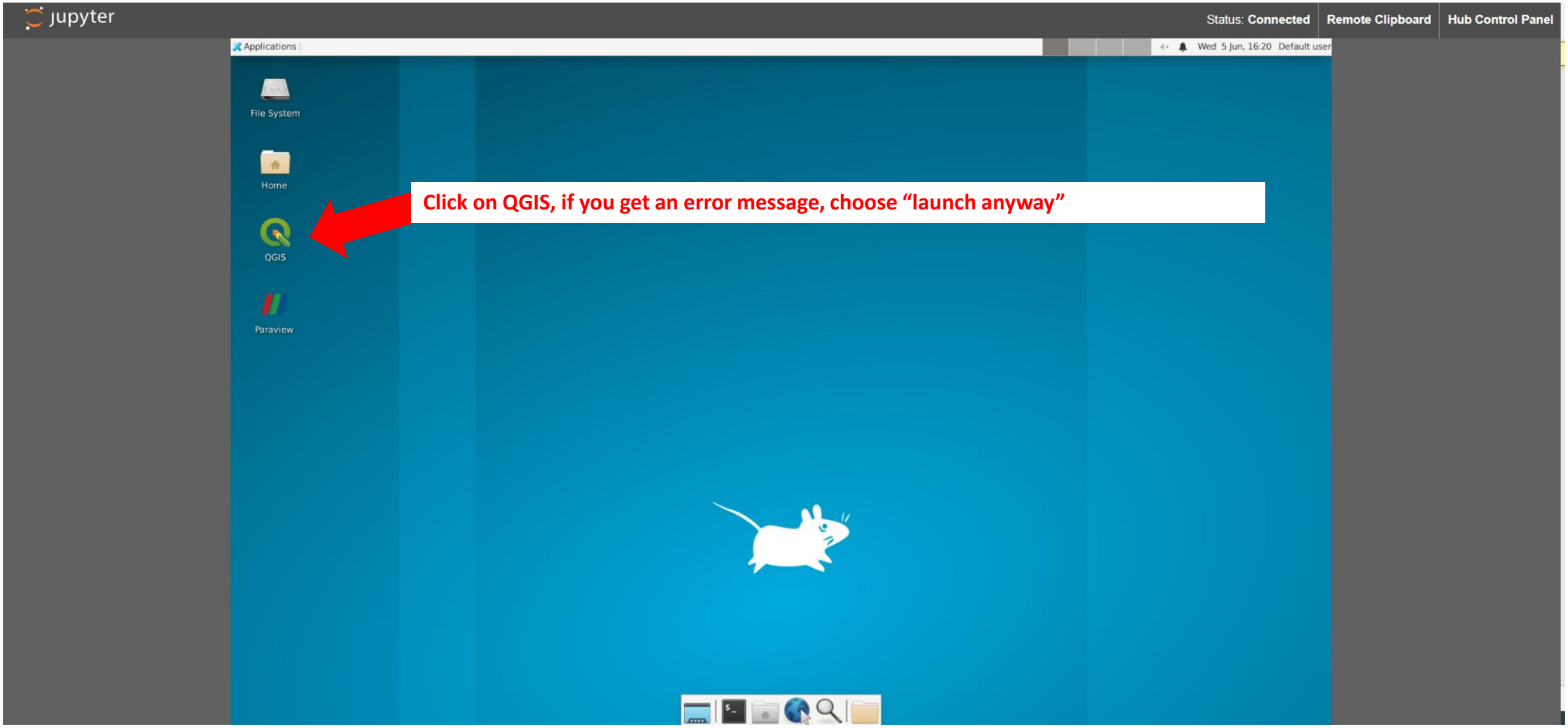


Click on the “Desktop” icon

Tip: If you lose your “Launcher” screen, click the “plus” icon at the top left for “New Launcher”


JupyterLab
Jupyter Remote Desktop Proxy
victor.2i2c.cloud/user/sogburn/desktop/
Relaunch to update
Lines
Case
Digitizer
VDAP
RS
bankcard
USGS
Remote
QTime
SN
IPDS
GVP
CGE
DOI Talent
swam
All Bookmarks
jupyter
Status: Connected
Remote Clipboard
Hub Control Panel
Applications
Tue 11 Jun, 21:16
Default user
File System
Home
QGIS
Paraview
To get back to your files and main VICTOR control panel, open the previous browser tab

Let’s make a filled DEM.



Applications | LaharZ 2.1.1 | Untitled Project — QGIS | Terminal | Tue 11 Jun, 23:04 Default use

Untitled Project — QGIS

Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help

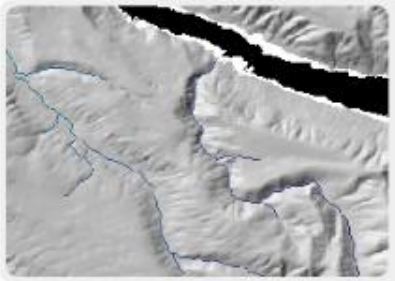
Click on “New Project”

Browser

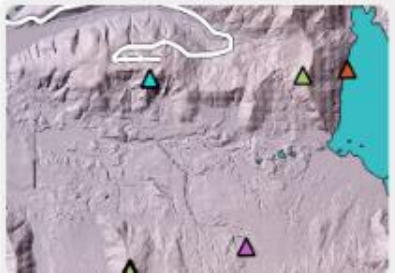
Favorites
Spatial Bookmarks
Home
/
GeoPackage
SpatiaLite
PostgreSQL
MS SQL Server
WMS/WMTS
Vector Tiles
XYZ Tiles
WCS
WFS / OGC API - Features
ArcGIS REST Servers



ol
yan/Laharz/laharz/Licancabur/large_vol.qgz
EPSG:4326 (EPSG:4326 - WGS 84)

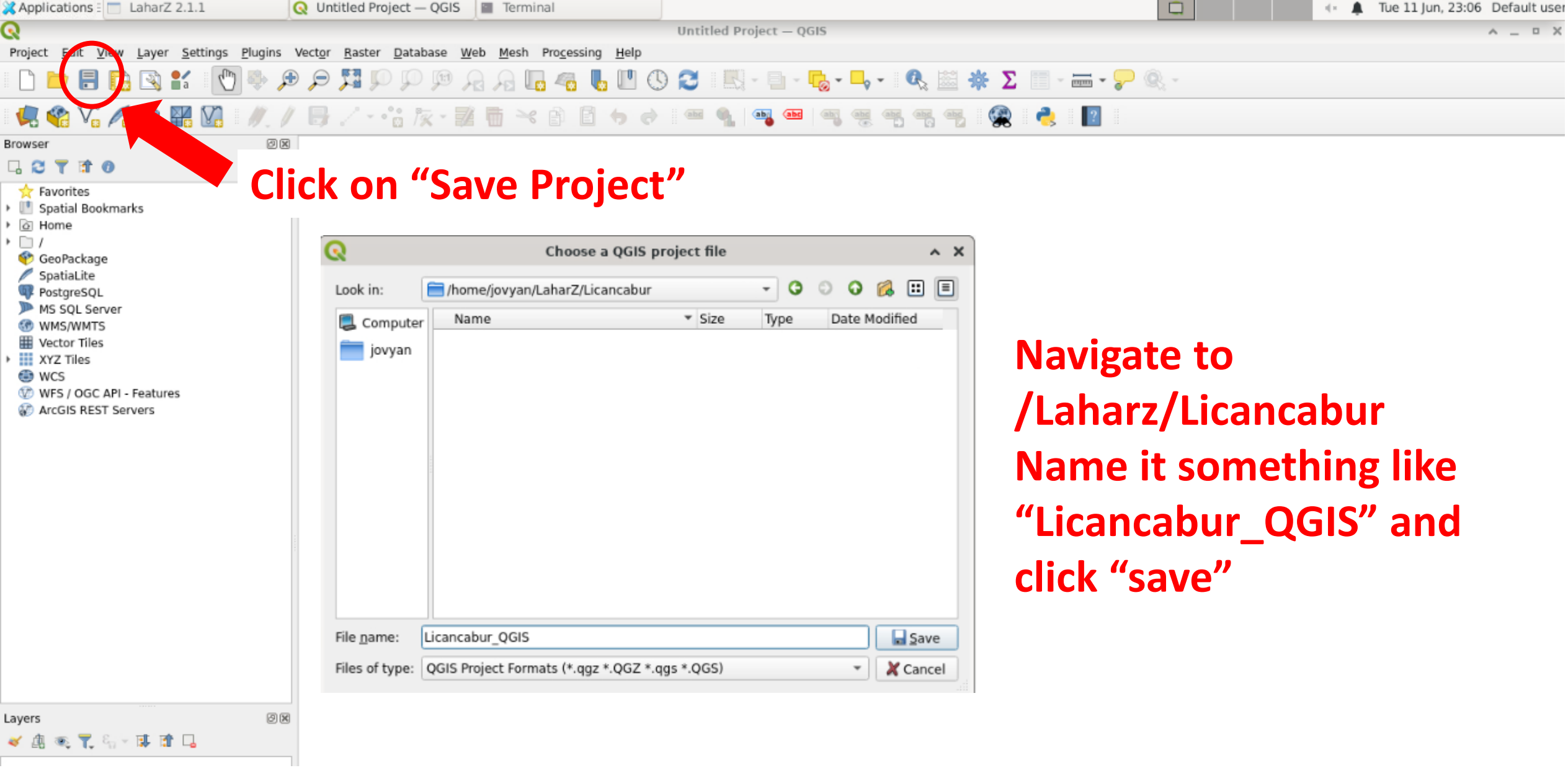


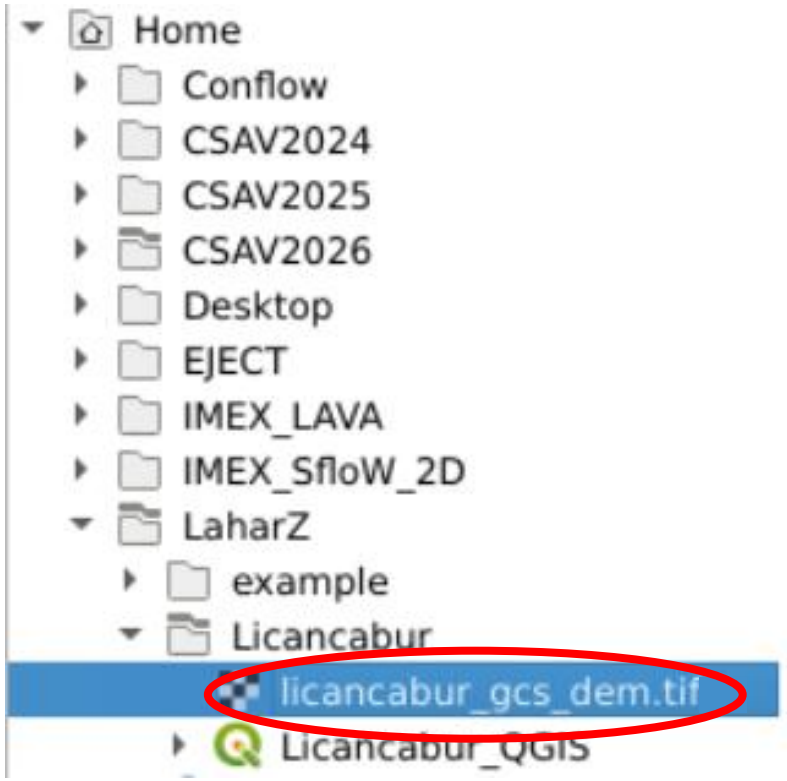
wgs_outputs
/home/jovyan/Laharz/laharz/rainier/wgs_outputs.qgz
EPSG:4326 (EPSG:4326 - WGS 84)



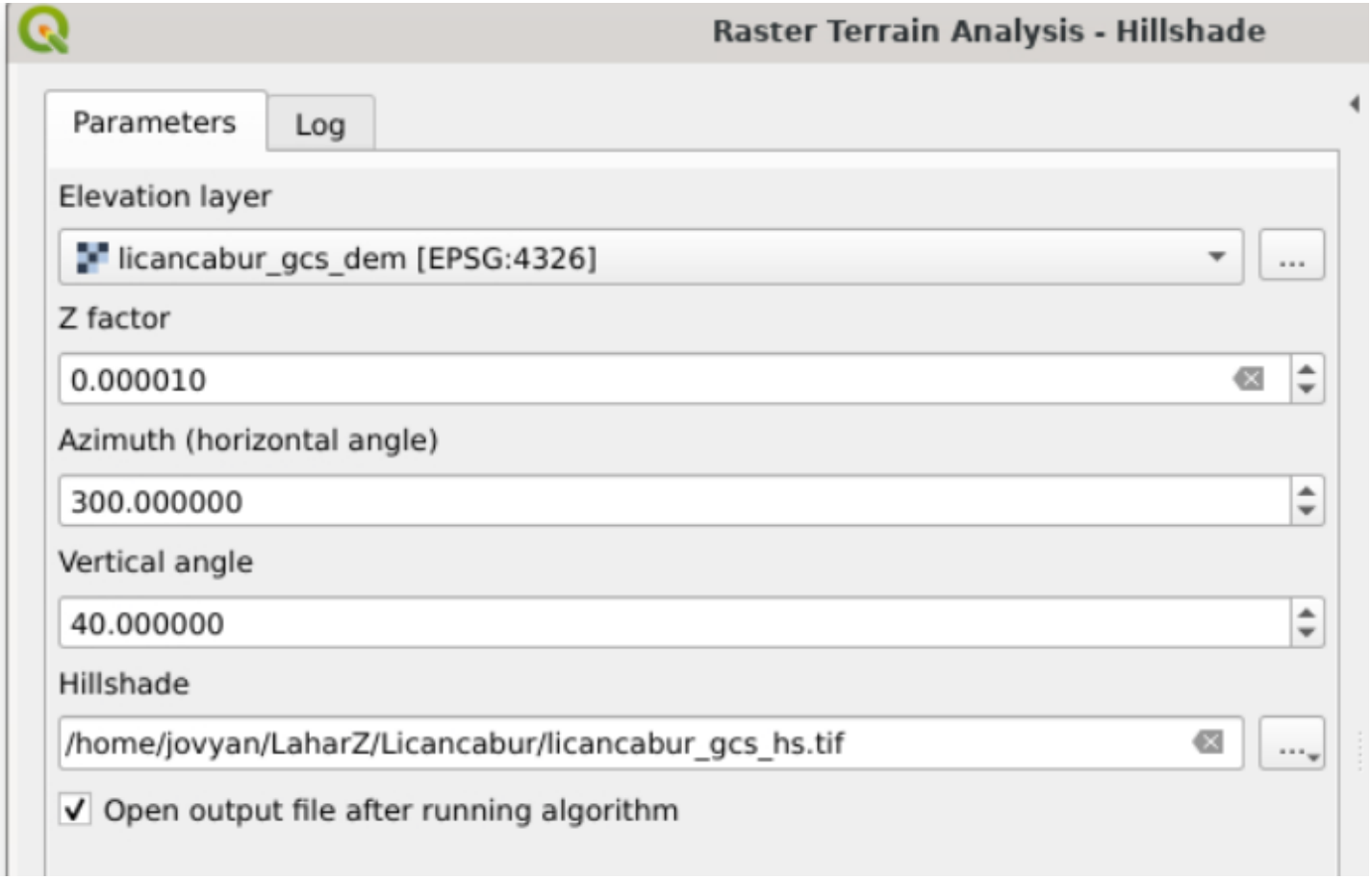
MSH_GIS
/home/jovyan/Laharz/GIS_exercise/MSH_GIS.qgz
EPSG:26710 (EPSG:26710 - NAD27 / UTM zone 10N)

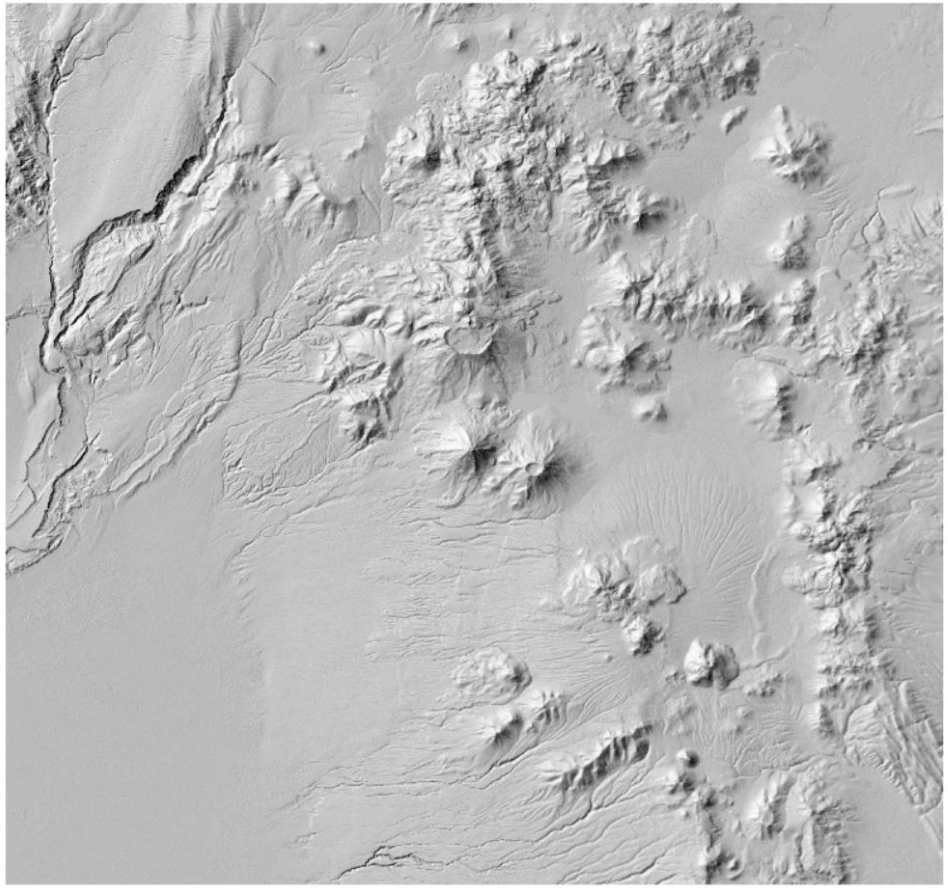
Project Templates


Applications
LaharZ 2.1.1
Untitled Project — QGIS
Terminal
Tue 11 Jun, 23:06
Default user
Untitled Project — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Favorites
Spatial Bookmarks
Home
/
GeoPackage
SpatiaLite
PostgreSQL
MS SQL Server
WMS/WMTS
Vector Tiles
XYZ Tiles
WCS
WFS / OGC API - Features
ArcGIS REST Servers
Layers
Click on “Save Project”
Choose a QGIS project file
Look in:
/home/jovyan/LaharZ/Licancabur
Computer
jovyan
Name
Size
Type
Date Modified
File name:
Licancabur_QGIS
Save
Files of type:
QGIS Project Formats (*.qgz *.QGZ *.qgs *.QGS)
Cancel
Navigate to
/Laharz/Licancabur
Name it something like
“Licancabur_QGIS” and
click “save”

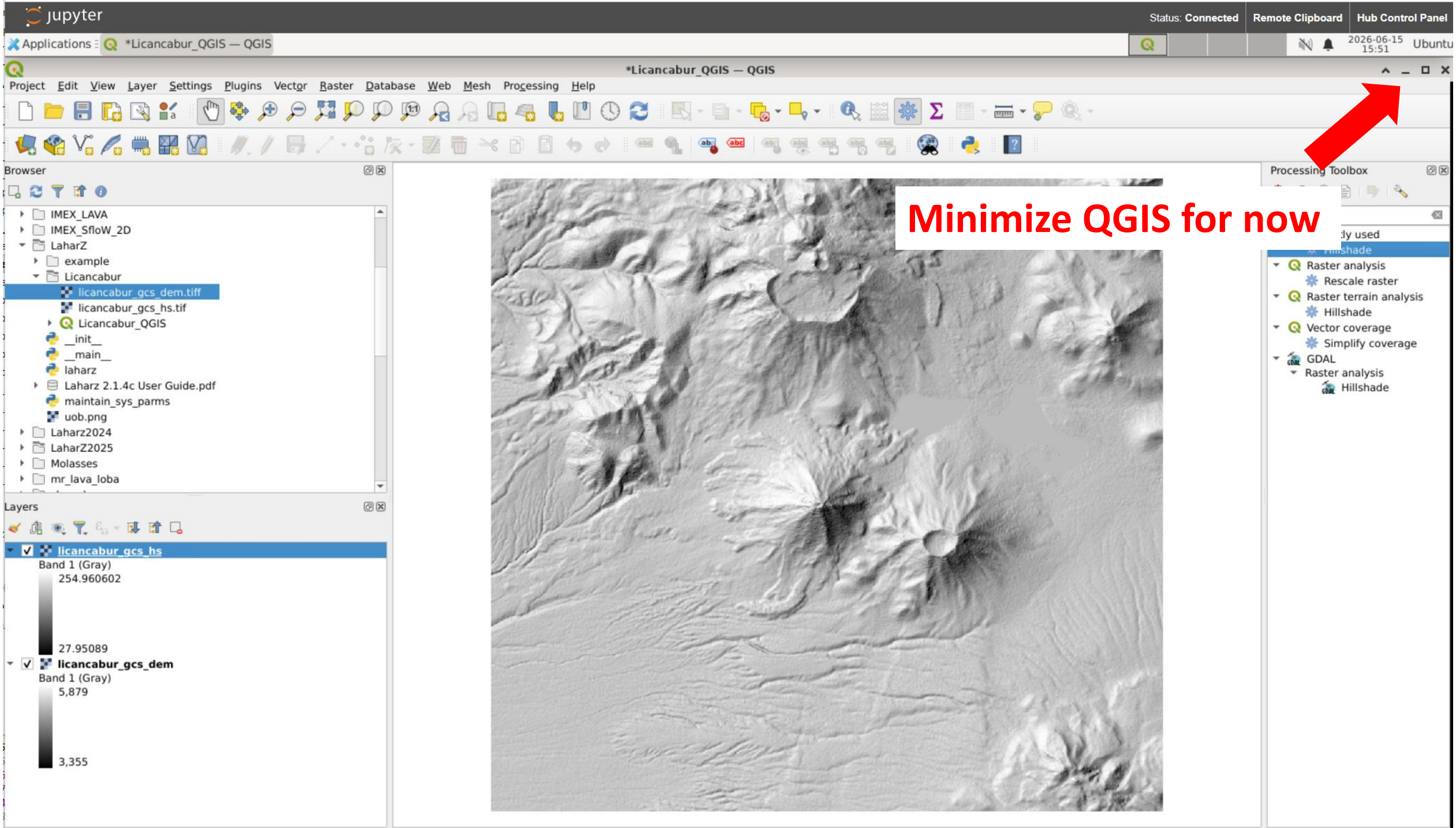


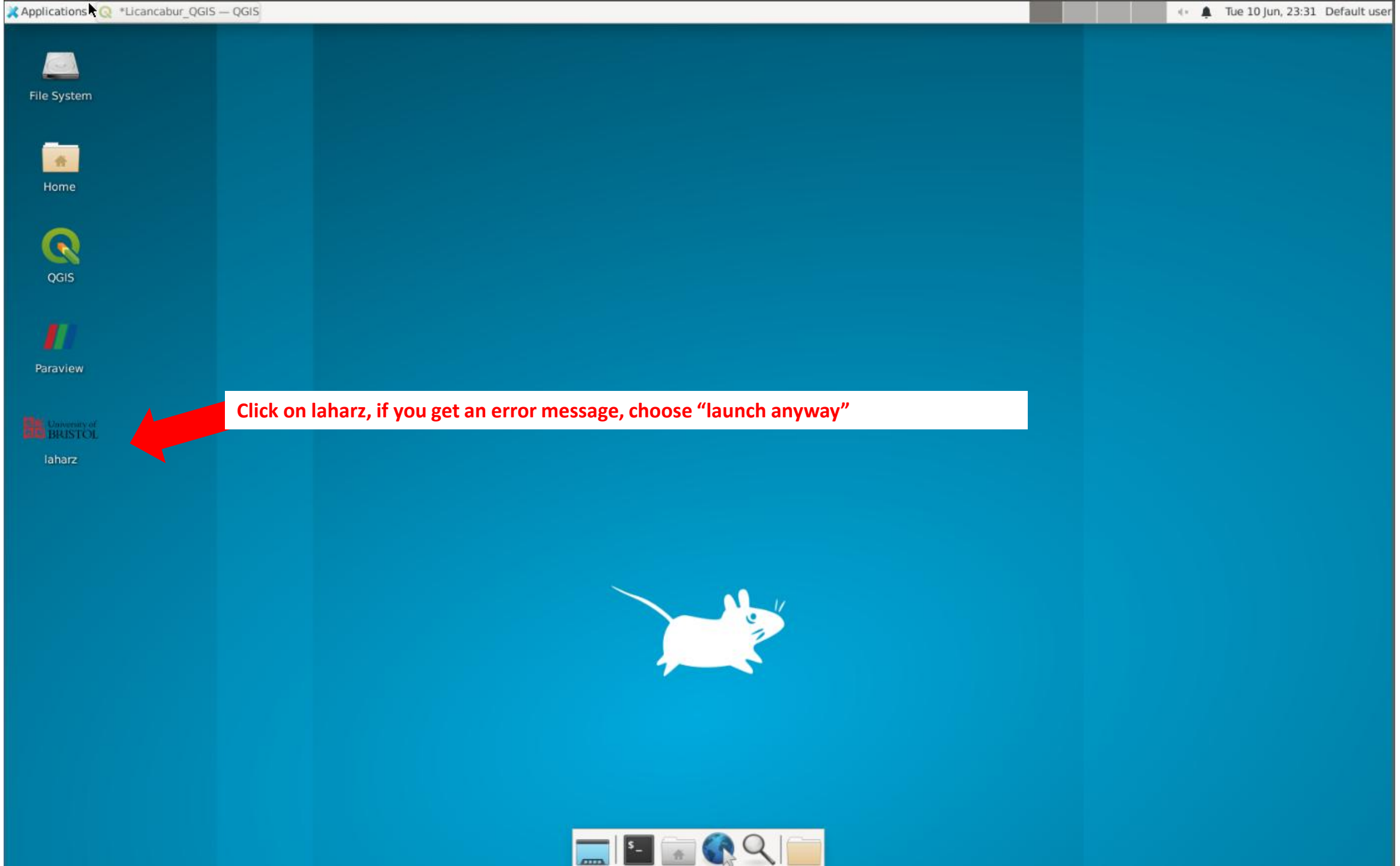
Navigate to the Laharz/Licancabur folder
Drag and drop the “licancabur_gcs_dem.tif” DEM file onto your map to check that it looks good





- Now make a hillshade of the DEM
- Open the Raster Terrain Analysis →Hillshade tool
- Remember to change the Z factor to 0.00001 because the raster is in degrees lat/long
- Save the file in /LaharZ/Licancabur as “licancabur_gcs_hs.tif”


jupyter
Status: Connected
Remote Clipboard
Hub Control Panel
Applications
*Licancabur_QGIS — QGIS
2026-06-15 15:51
Ubuntu
*Licancabur_QGIS — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
IMEX_LAVA
IMEX_SfloW_2D
LaharZ
example
Licancabur
licancabur_gcs_dem.tiff
licancabur_gcs_hs.tif
Licancabur_QGIS
__init__
__main__
laharz
Laharz 2.1.4c User Guide.pdf
maintain_sys_parms
uob.png
Laharz2024
LaharZ2025
Molasses
mr_lava_loba
Layers
licancabur_gcs_hs
Band 1 (Gray)
254.960602
27.95089
licancabur_gcs_dem
Band 1 (Gray)
5,879
3,355
Minimize QGIS for now
Processing Toolbox
Raster analysis
Rescale raster
Raster terrain analysis
Hillshade
Vector coverage
Simplify coverage
GDAL
Raster analysis
Hillshade


Applications
*Licancabur_QGIS — QGIS
Tue 10 Jun, 23:31
Default user
File System
Home
QGIS
Paraview
laharz
Click on laharz, if you get an error message, choose “launch anyway”

LaharZ 2.1.4

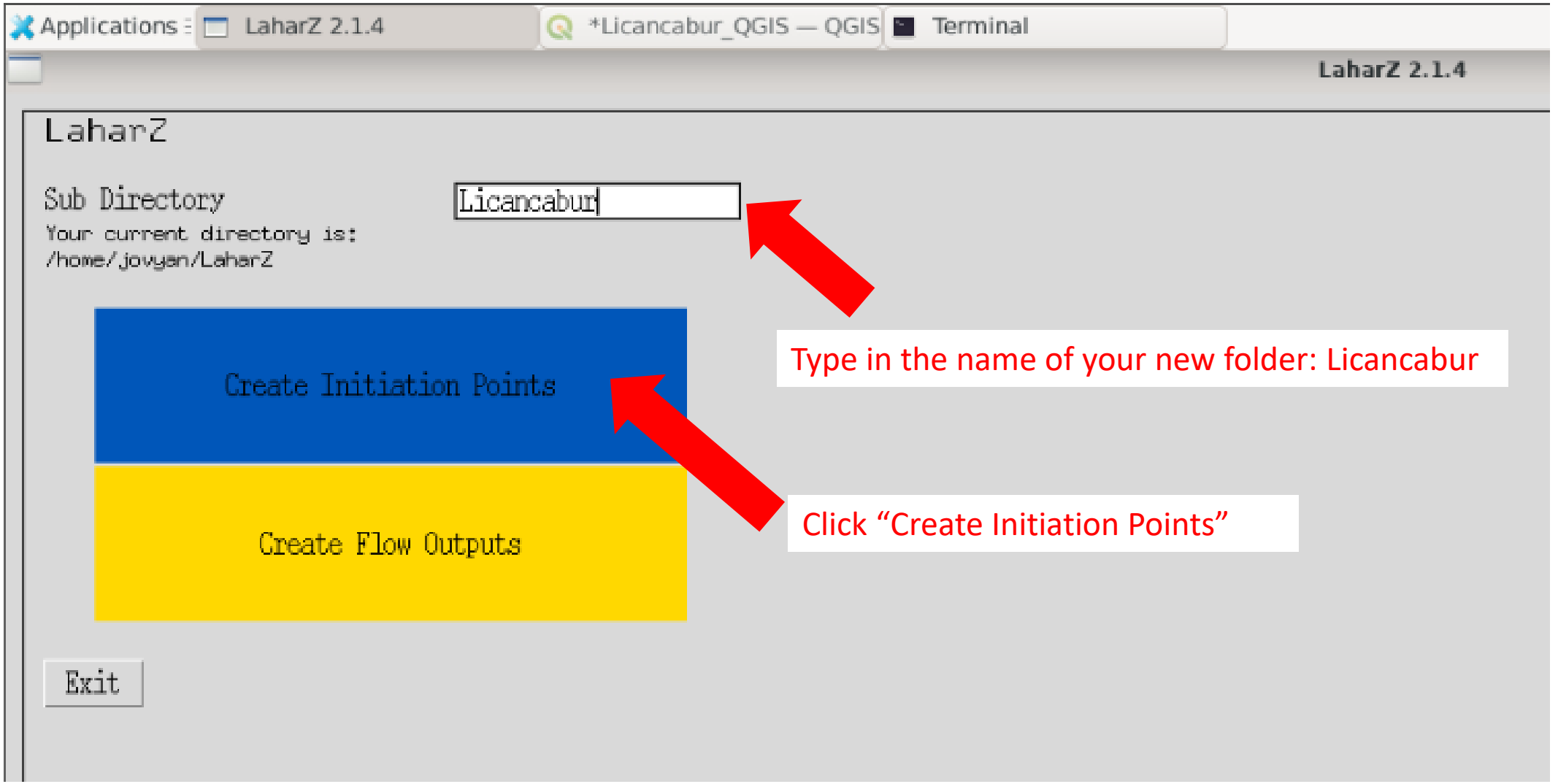
LaharZ



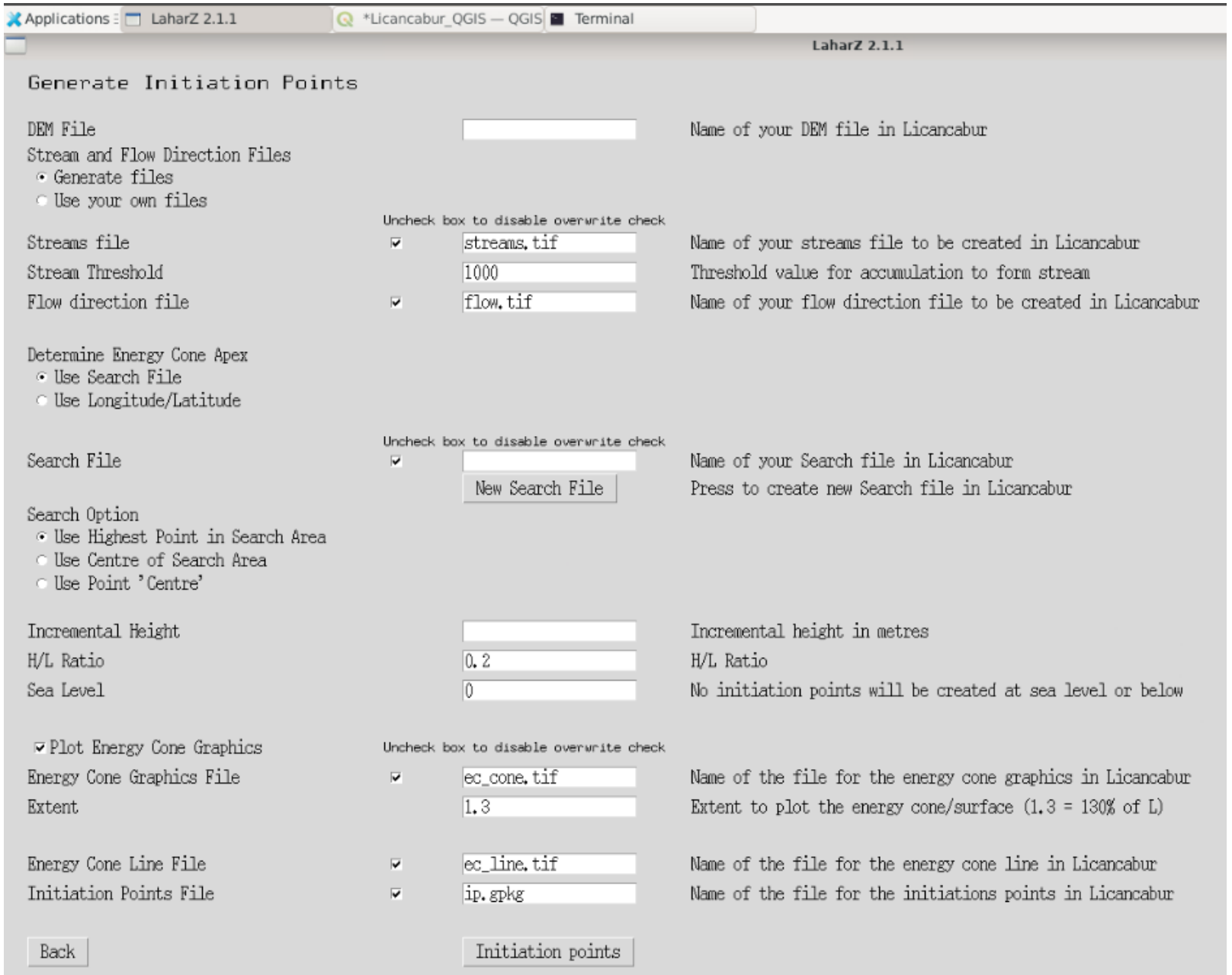
University of BRISTOL

Version: 2.1.4 stand alone

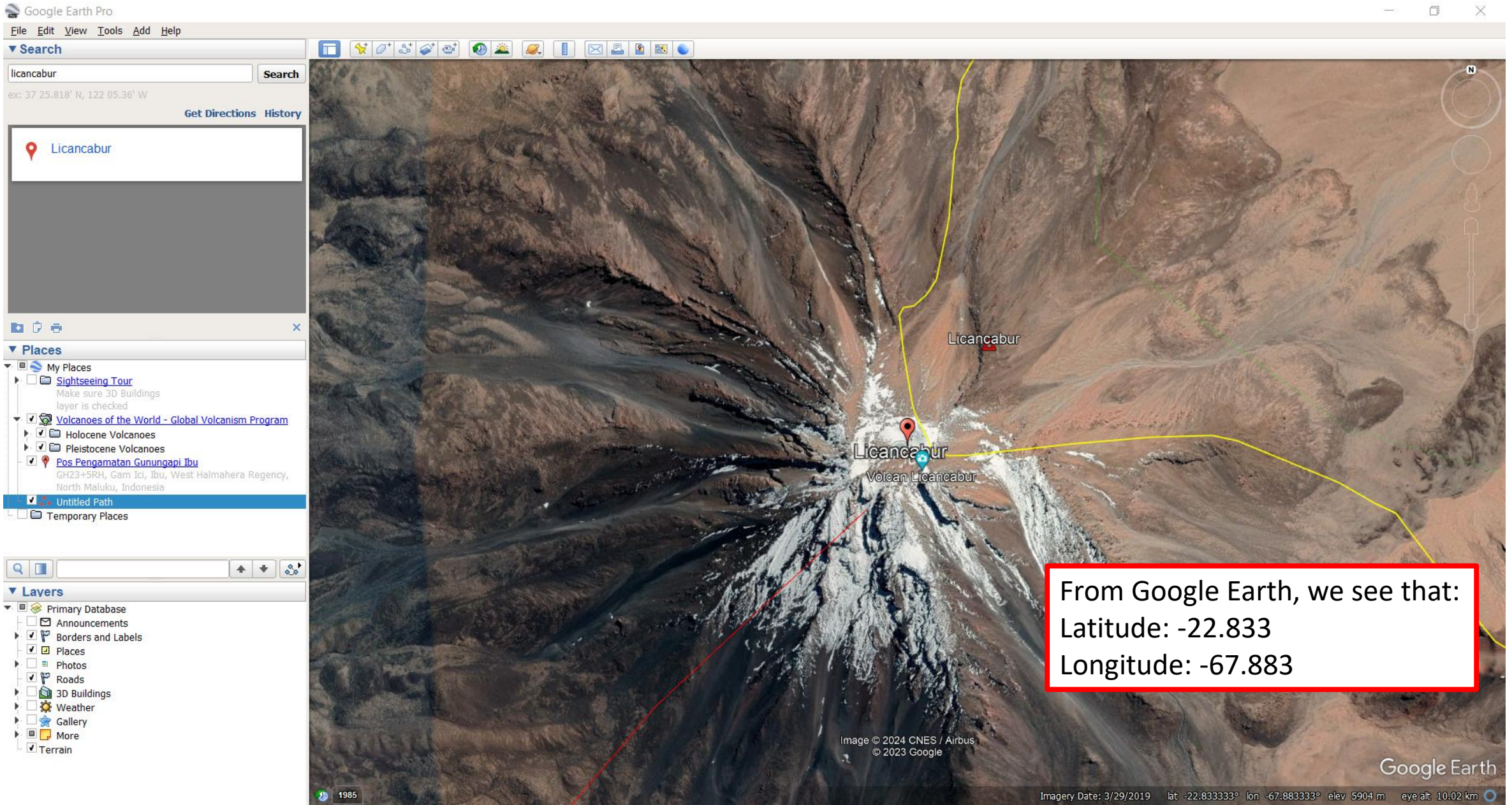
Let’s make a stream network, an H/L cone, and start points

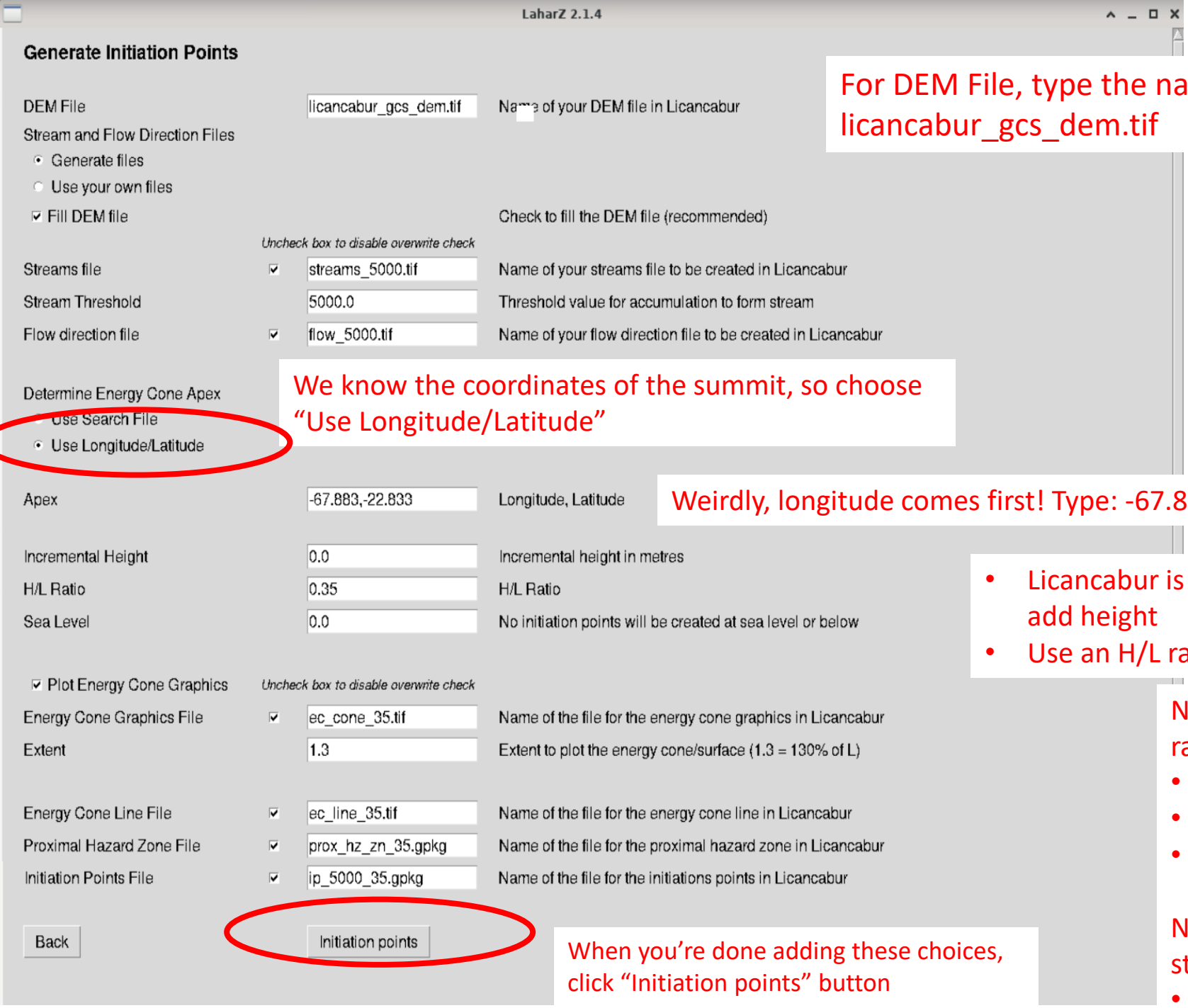


We have some fields to fill in! There are some default values to change.



For this exercise, we need coordinates of the summit





For DEM File, type the name of your DEM: licancabur_gcs_dem.tif

For this step, we're going to try a stream threshold of 5000.

- Streams file: streams_5000.tif
- Stream threshold: 5000
- Flow direction file: flow_5000.tif

We know the coordinates of the summit, so choose "Use Longitude/Latitude"

Weirdly, longitude comes first! Type: -67.883,-22.833

- Licancabur is nice and conical, so don't add height
- Use an H/L ratio of 0.35

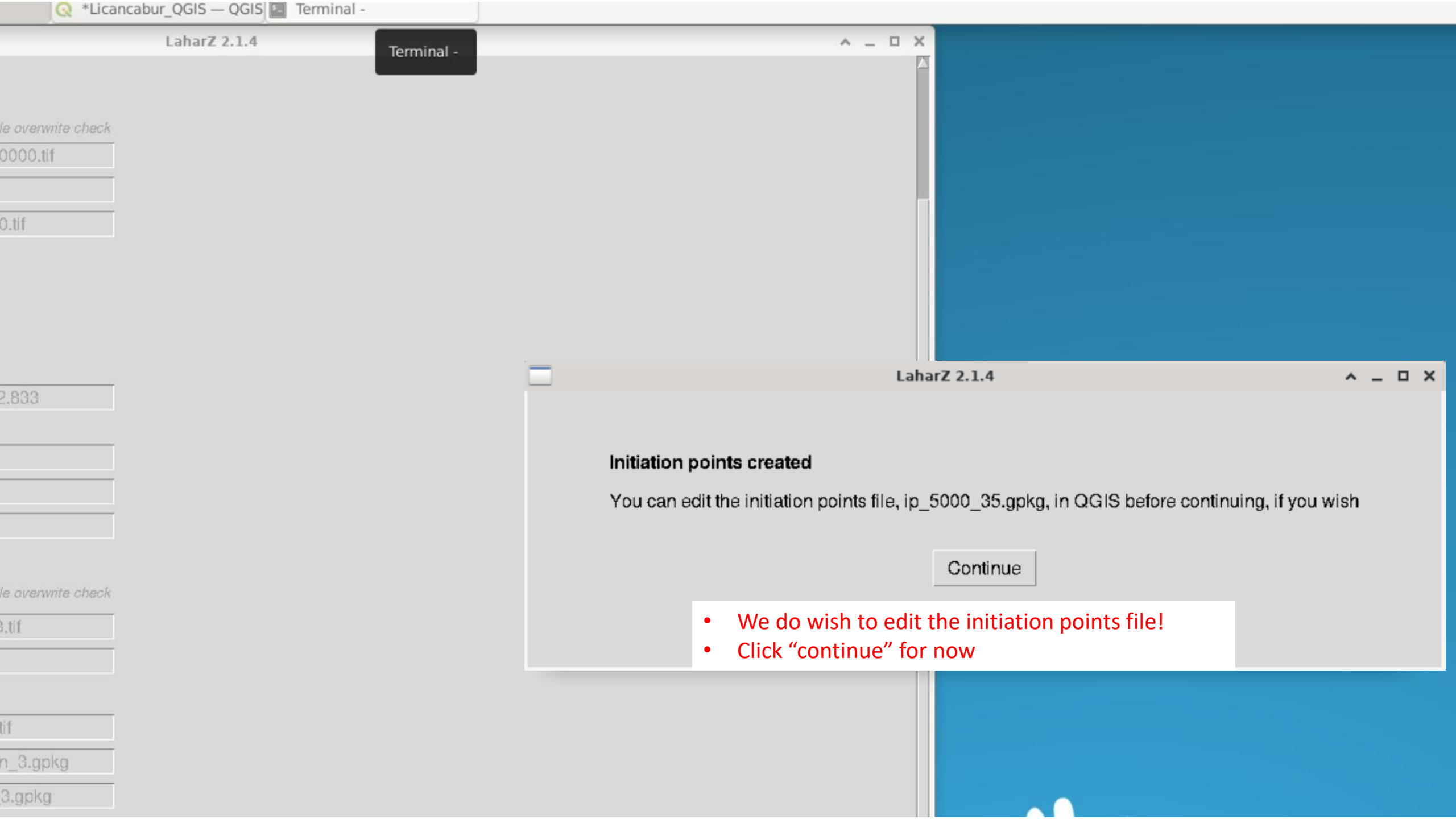
Name the energy cone files with the H/L ratio used:

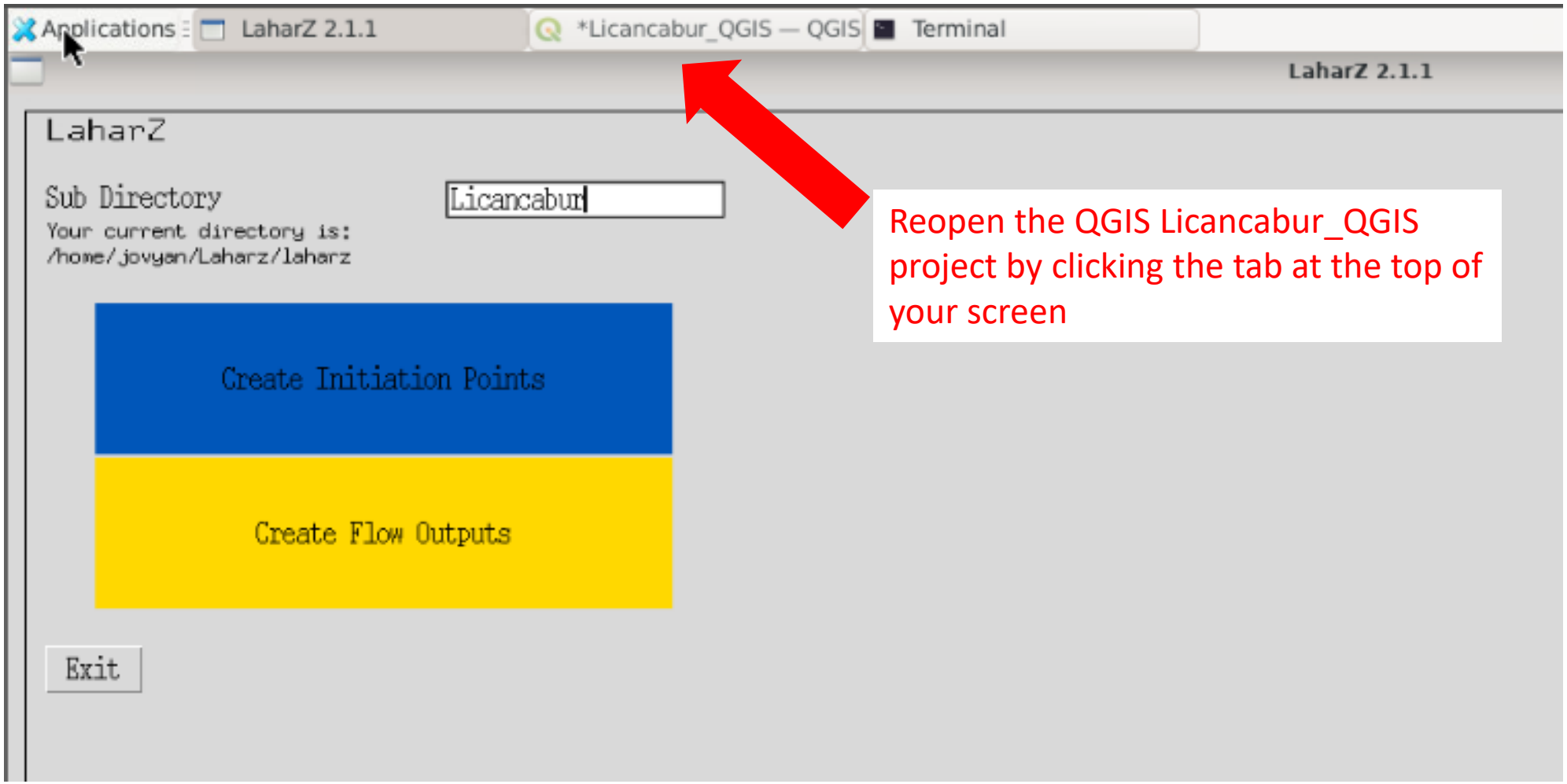
- ec_cone_35.tif
- ec_line_35.tif
- prox_hz_zn_35.gpkg

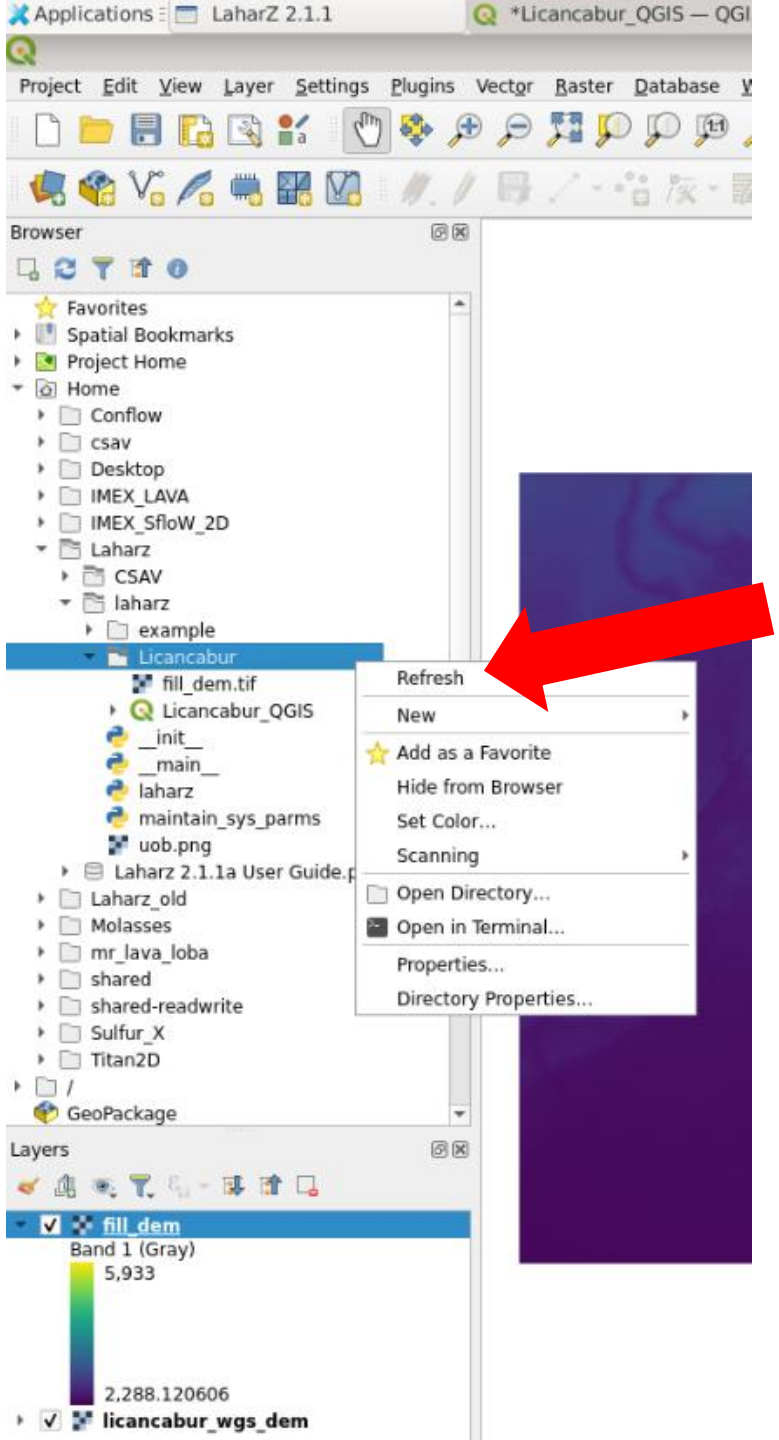
Name the initiation points with the stream threshold and H/L ratio:

- ip_5000_35.gpkg

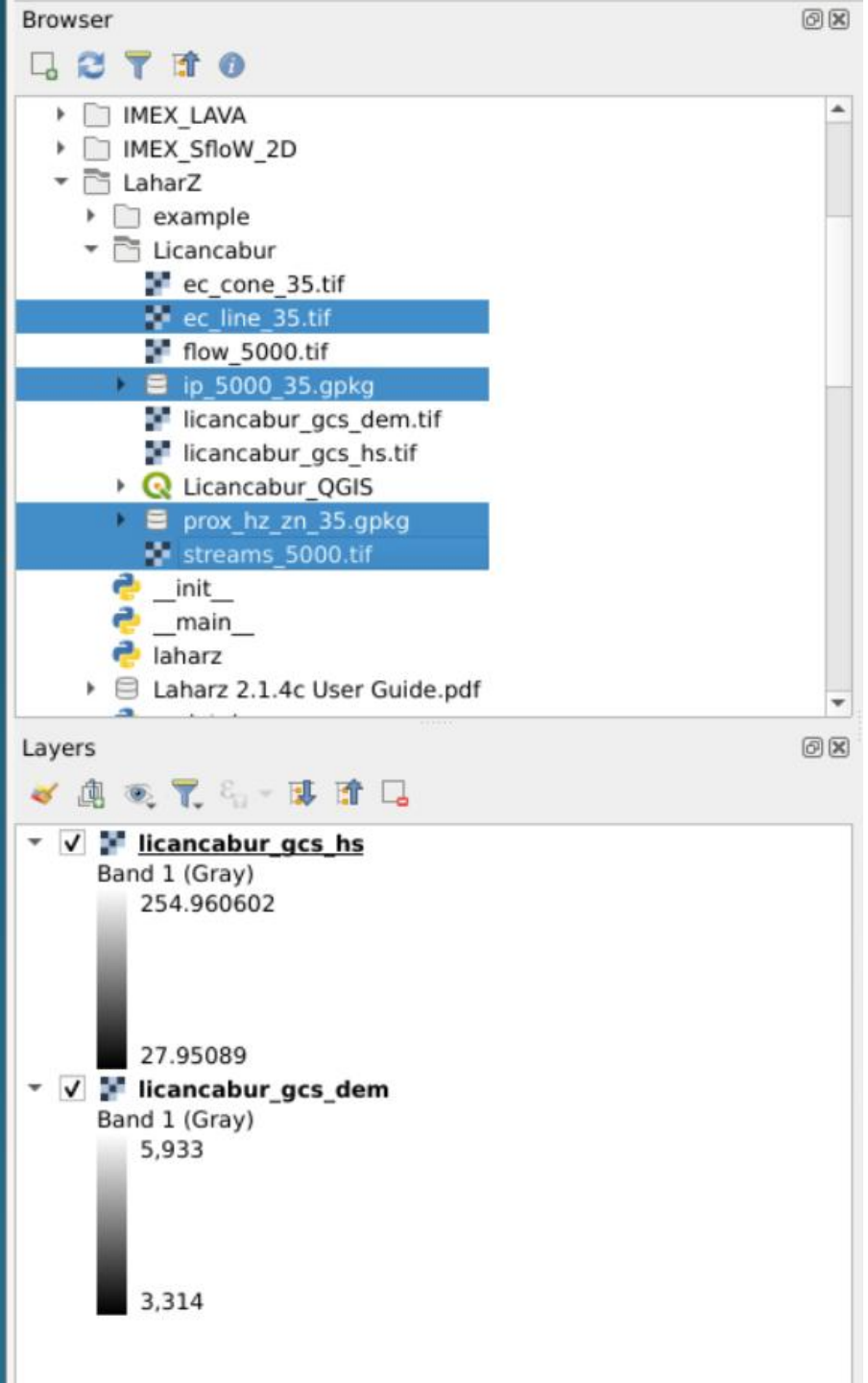
When you're done adding these choices, click "Initiation points" button


*Licancabur_QGIS — QGIS
Terminal -
LaharZ 2.1.4
Terminal -
le overwrite check
0000.tif
0.tif
2.833
le overwrite check
.tif
tif
n_3.gpkg
3.gpkg
LaharZ 2.1.4
Initiation points created
You can edit the initiation points file, ip_5000_35.gpkg, in QGIS before continuing, if you wish
Continue
• We do wish to edit the initiation points file!
• Click “continue” for now


Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
LaharZ 2.1.1
LaharZ
Sub Directory
Licancabur
Your current directory is:
/home/jovyan/Laharz/laharz
Create Initiation Points
Create Flow Outputs
Exit
Reopen the QGIS Licancabur_QGIS project by clicking the tab at the top of your screen

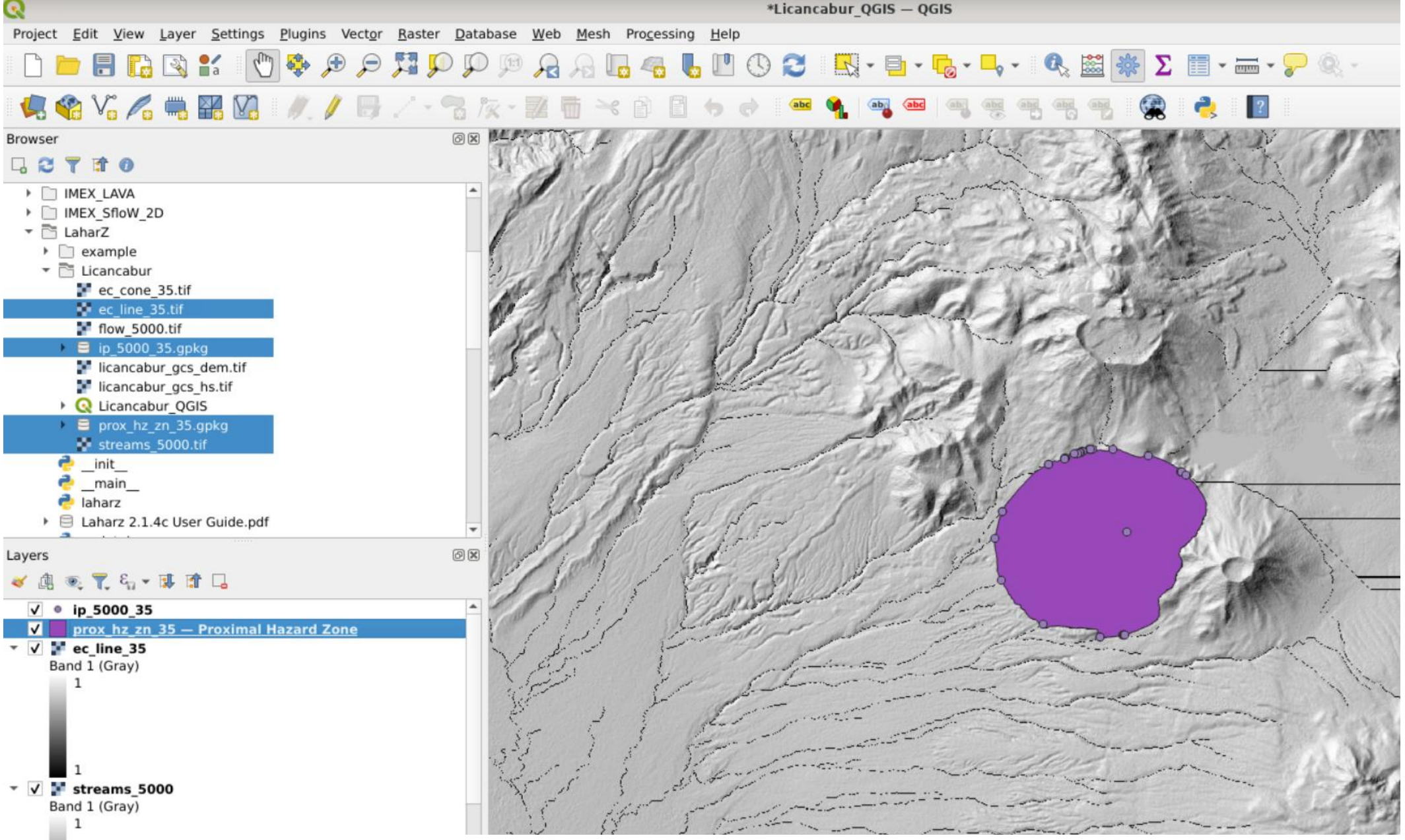


Right click on the “Licancabur” folder and click “Refresh” to see your new files

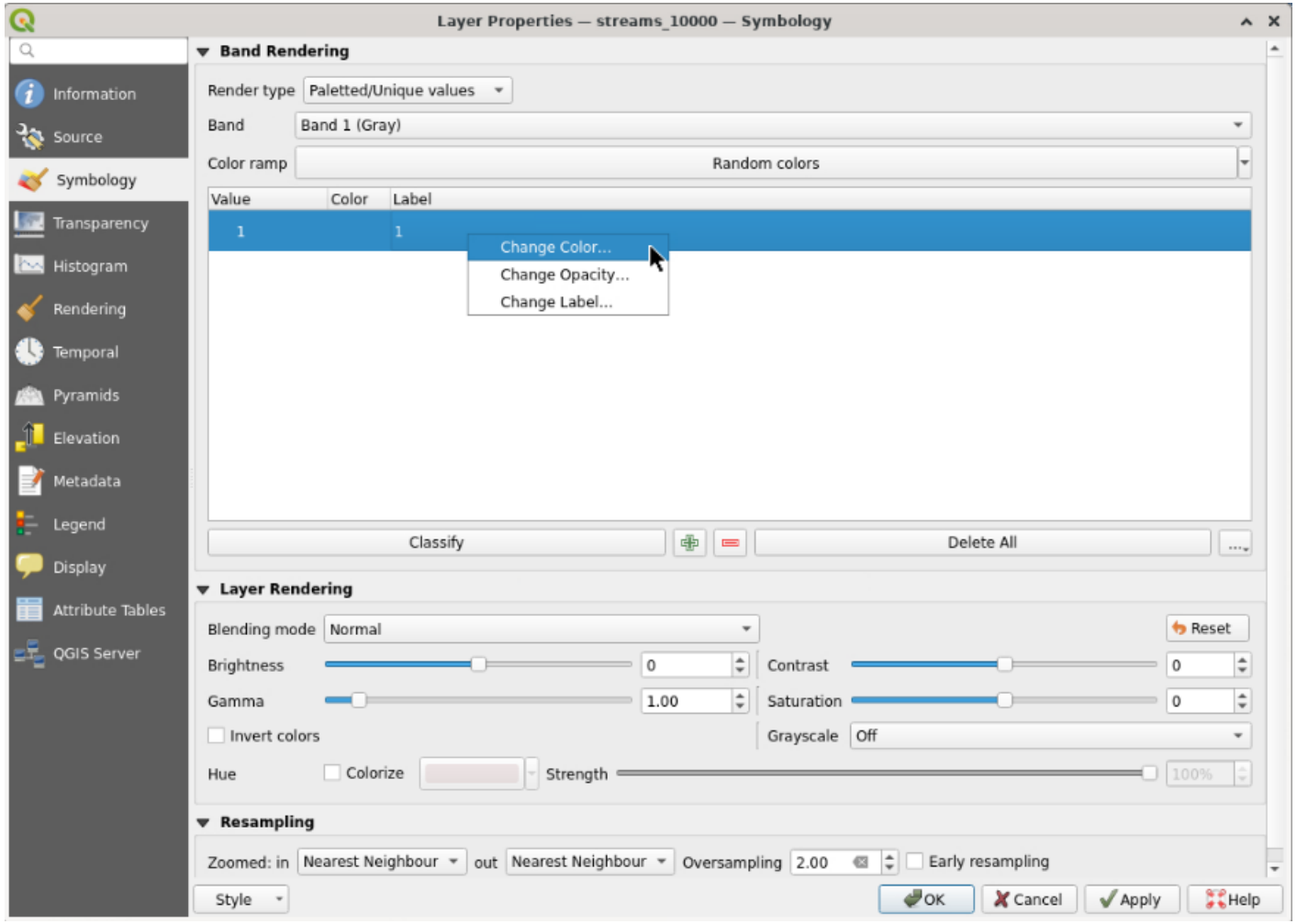


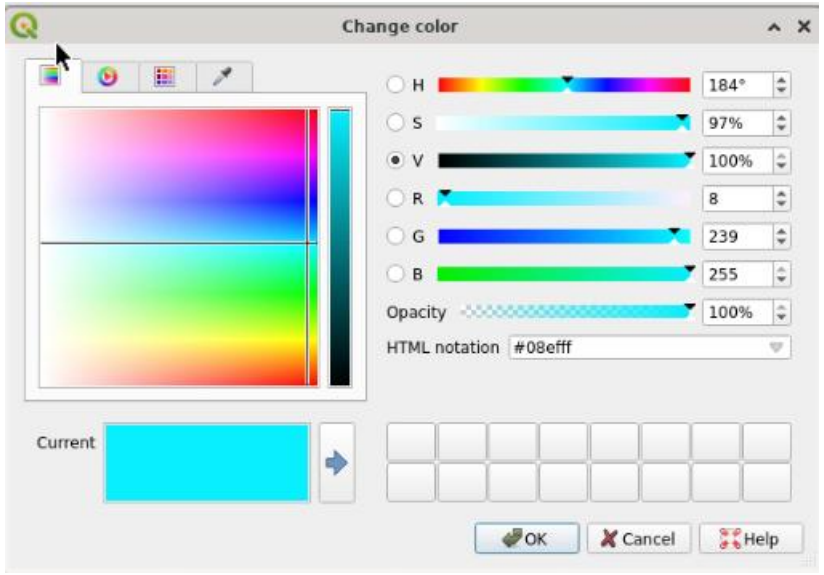
Let's look at the stream network, H/L cone (ec_line), and the initiation points we created!

Add these files to your map:
streams_5000.tif
ec_line_35.tif
ip_10000_35.gpkg
prox_hz_zn_35.gpkg

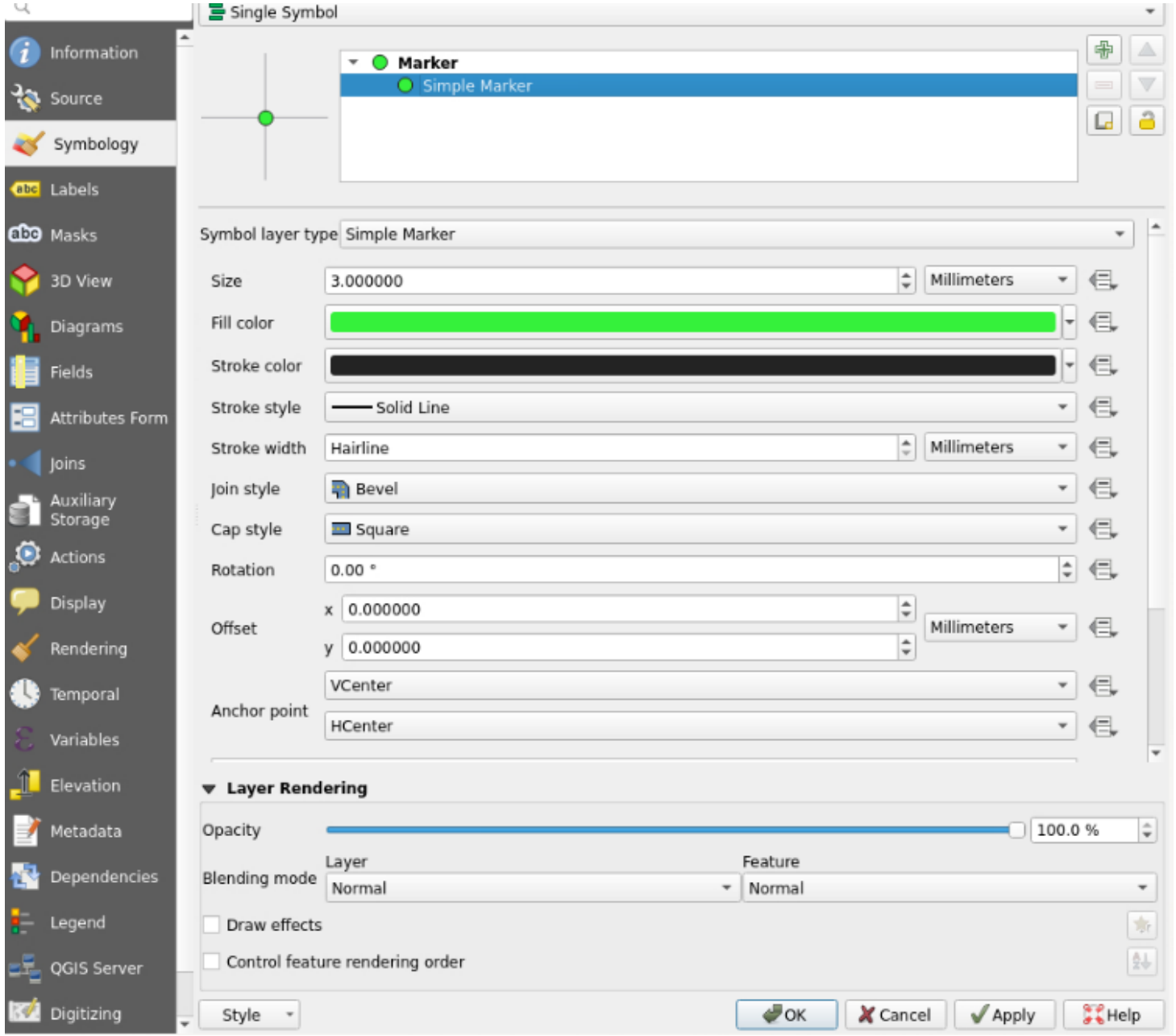


If the default colors are hard to see, change them (right click on layers, choose “Properties”)



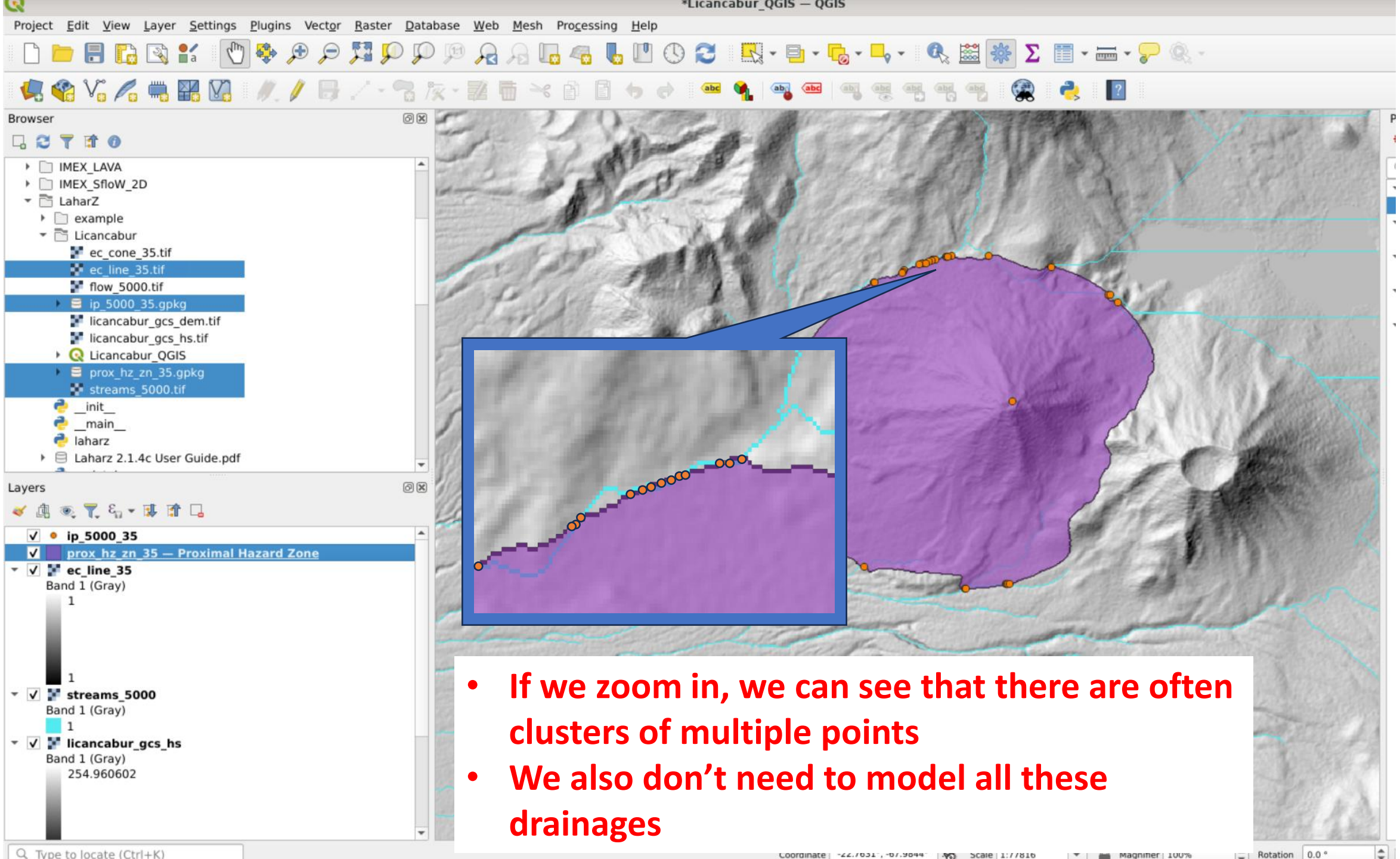


For the stream network change the “render type” to “Paletted/unique values” and click “classify”. Then right click on the listed value and choose “change color”

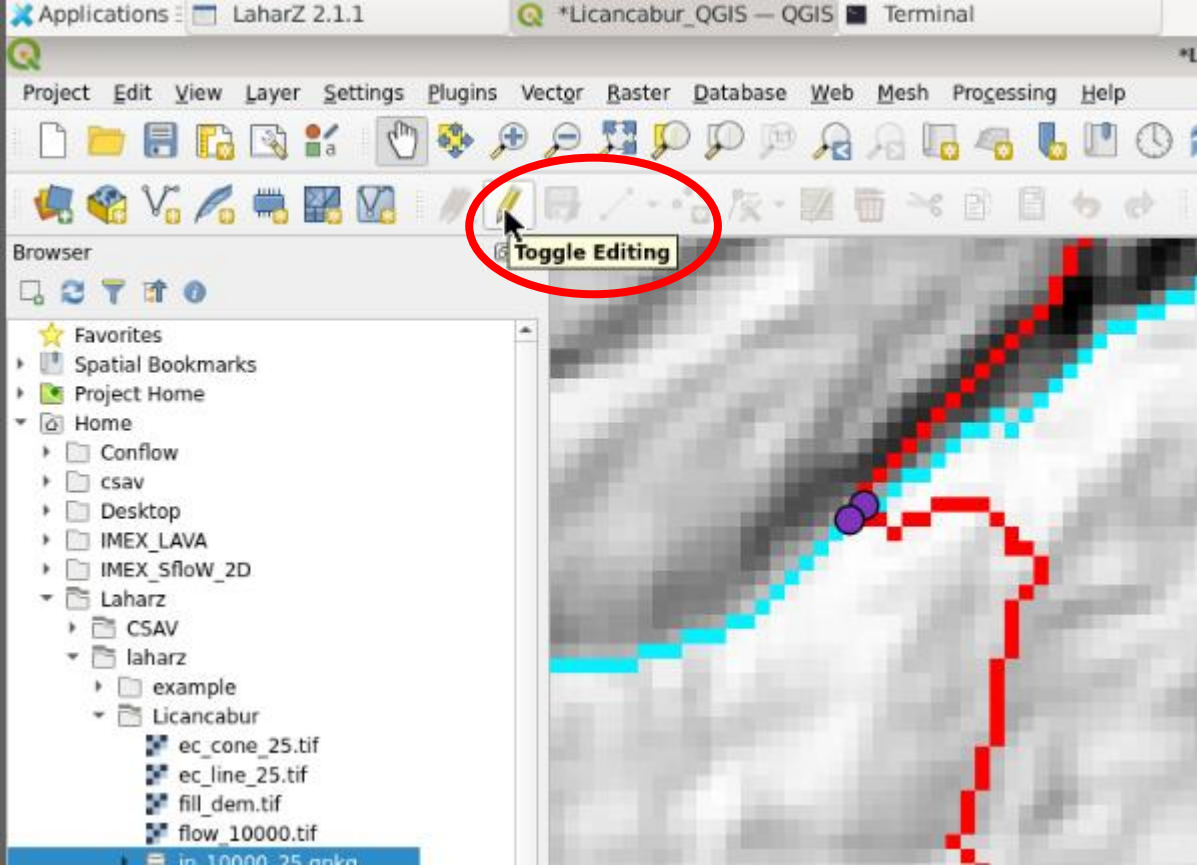


For the initiation points, change the color to something you like.

You can also change the size of the points if you want.

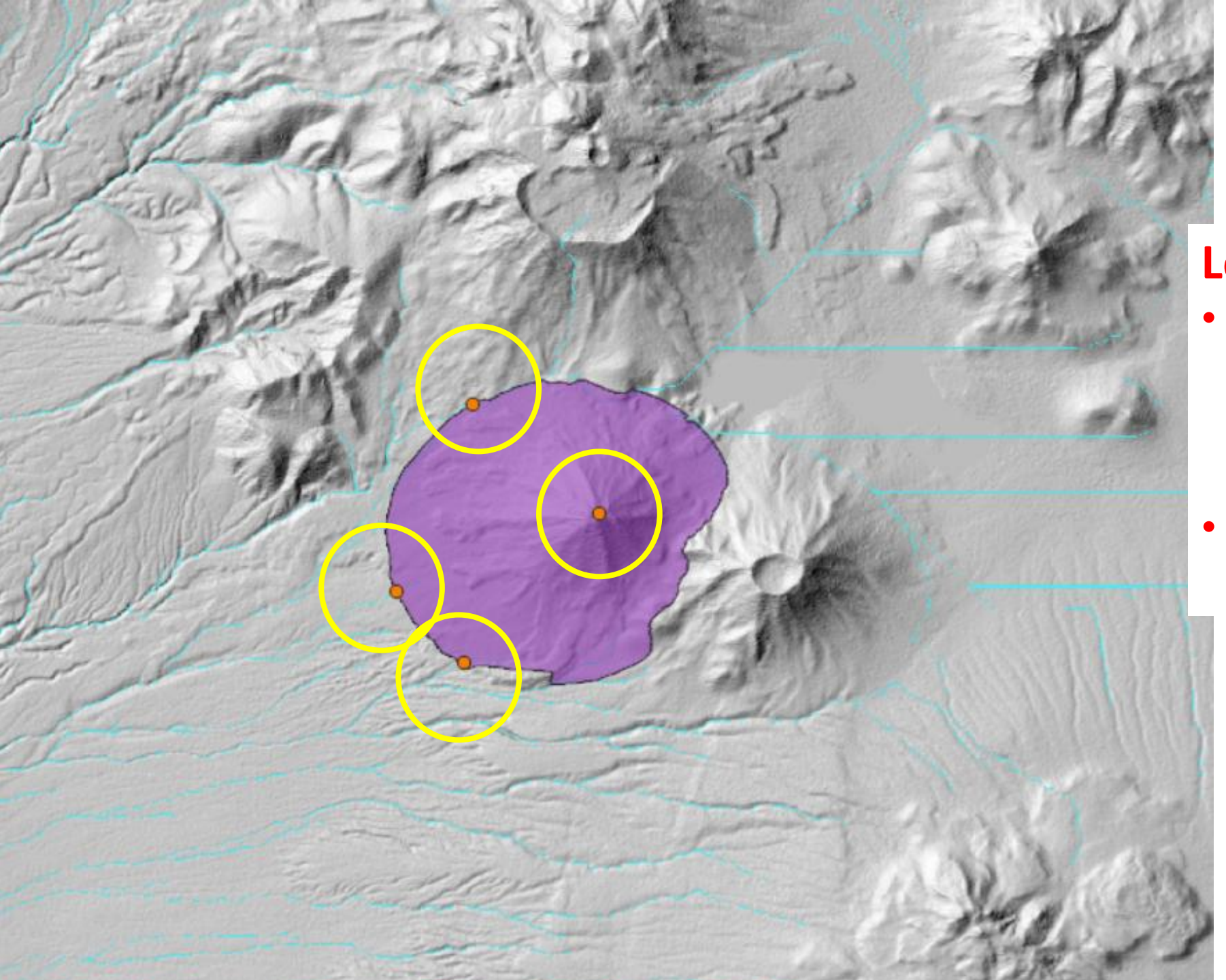


- If we zoom in, we can see that there are often clusters of multiple points
- We also don't need to model all these drainages



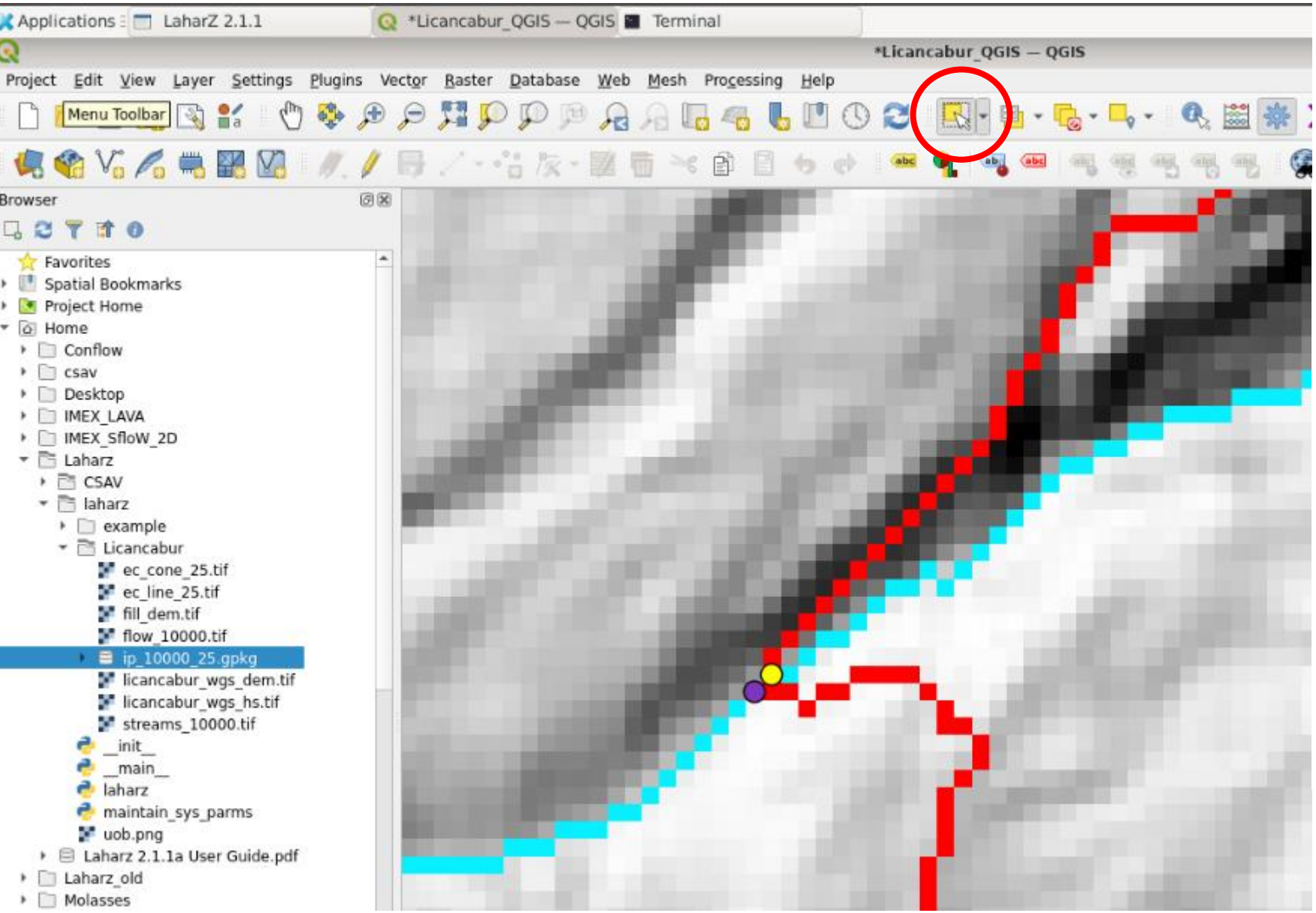
Let's delete some points

- Turn editing on by clicking the pencil "Toggle editing"



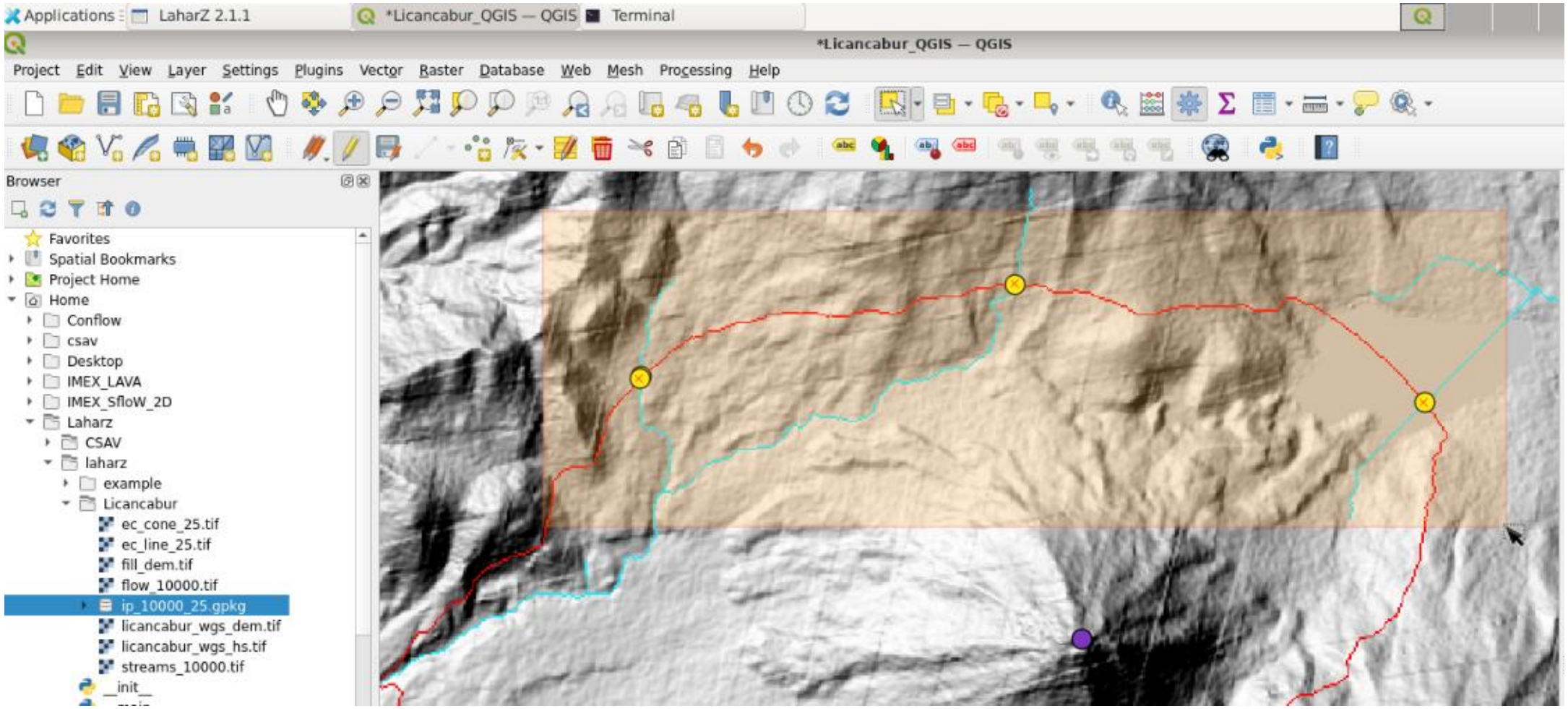
Let’s delete some points

- Let’s keep just one point in each of the areas circled and delete everything else
- Also keep the summit point



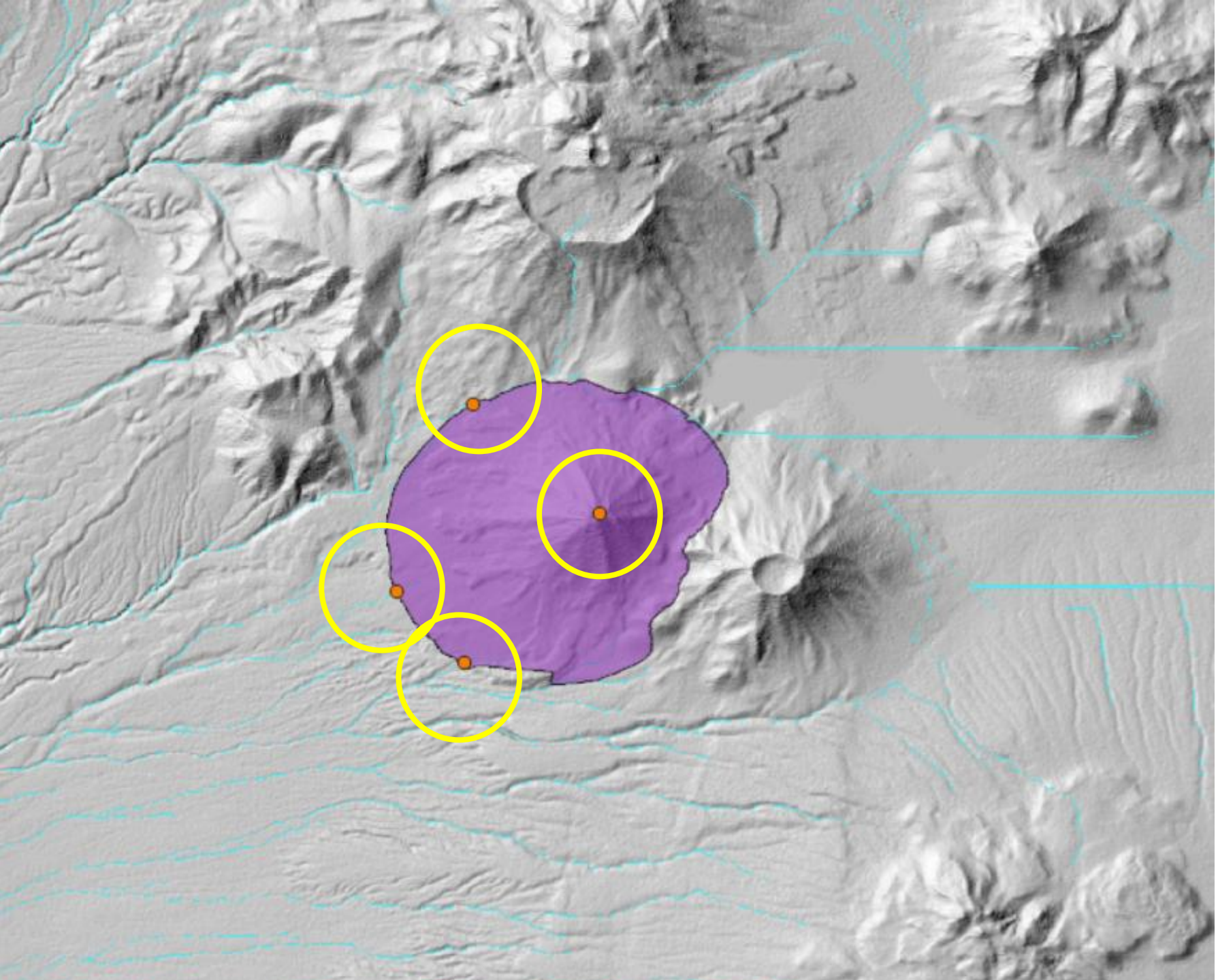
Let's delete some points

- Turn on the "Select features by area" tool and zoom in to each point cluster
- Click on the point you want to delete
- It will turn yellow
- The press the "delete" key on your keyboard

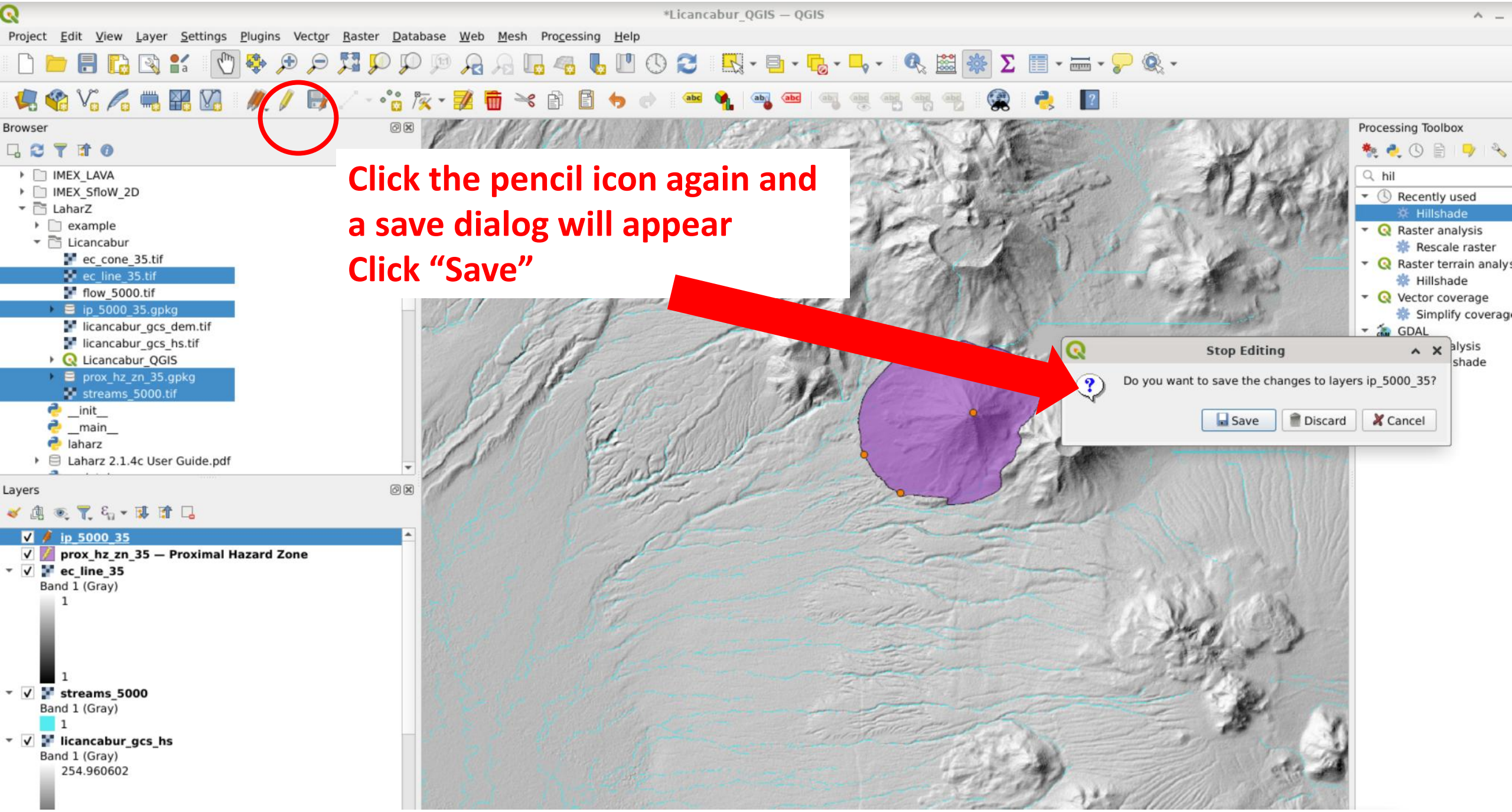


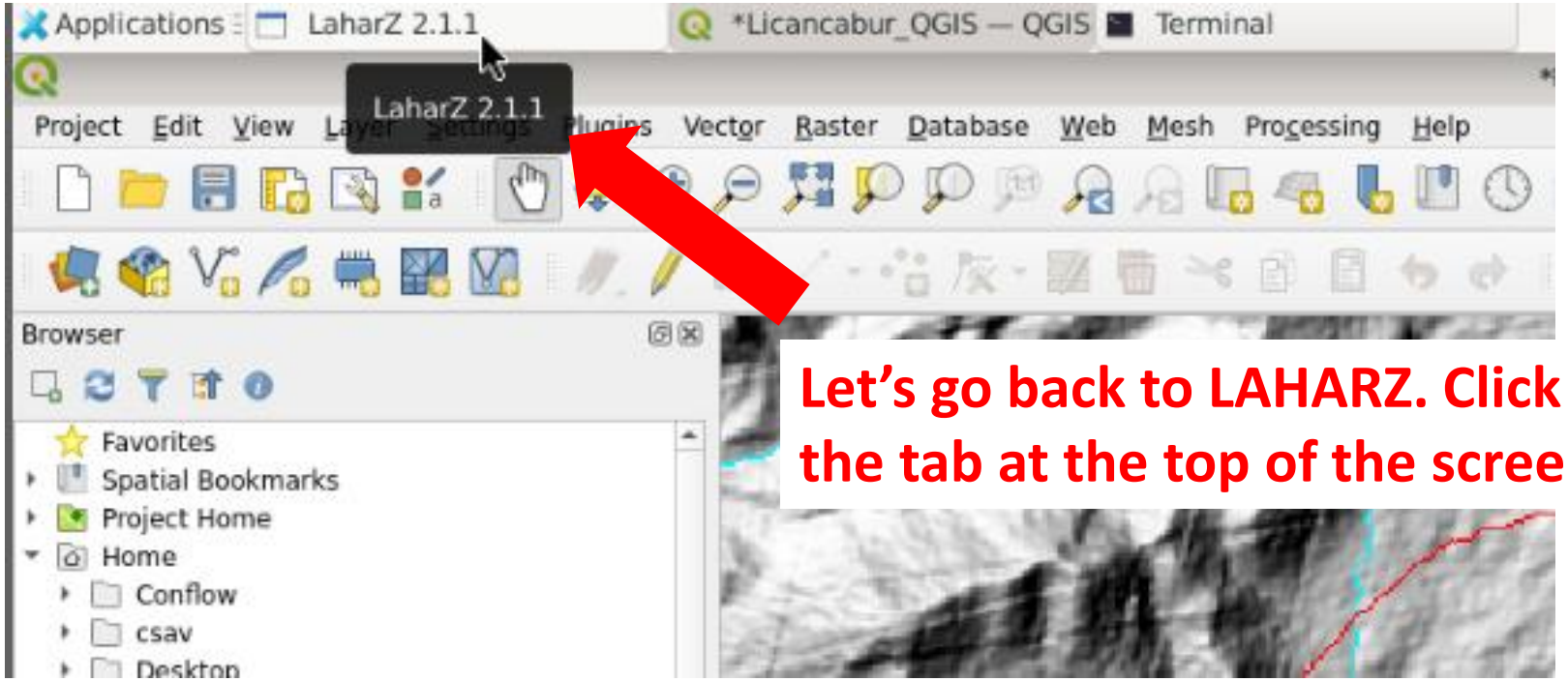
Let's delete some points

- You can also use the "Select features by area" tool to drag a selection box around multiple points
- They will turn yellow
- The press the "delete" key on your keyboard

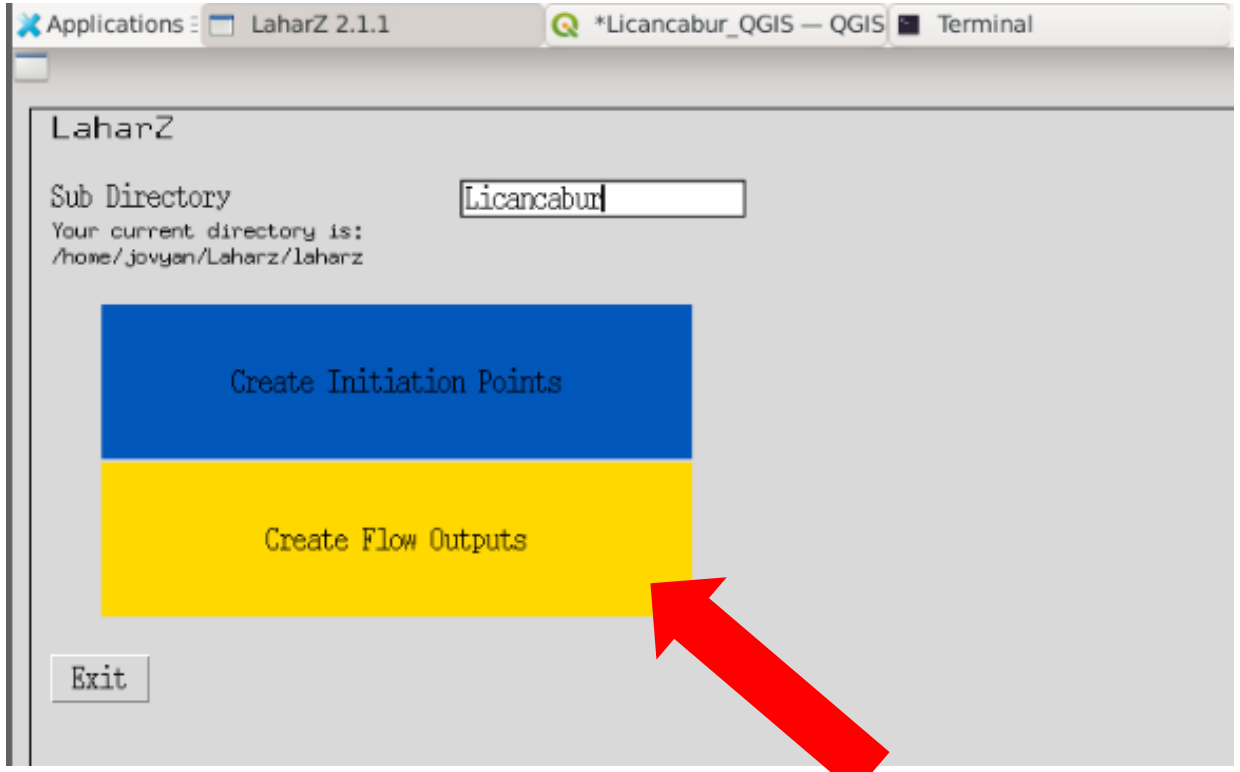


- We should now have 3 start points and 1 summit point left
- Click the pencil icon again and a save dialog will appear

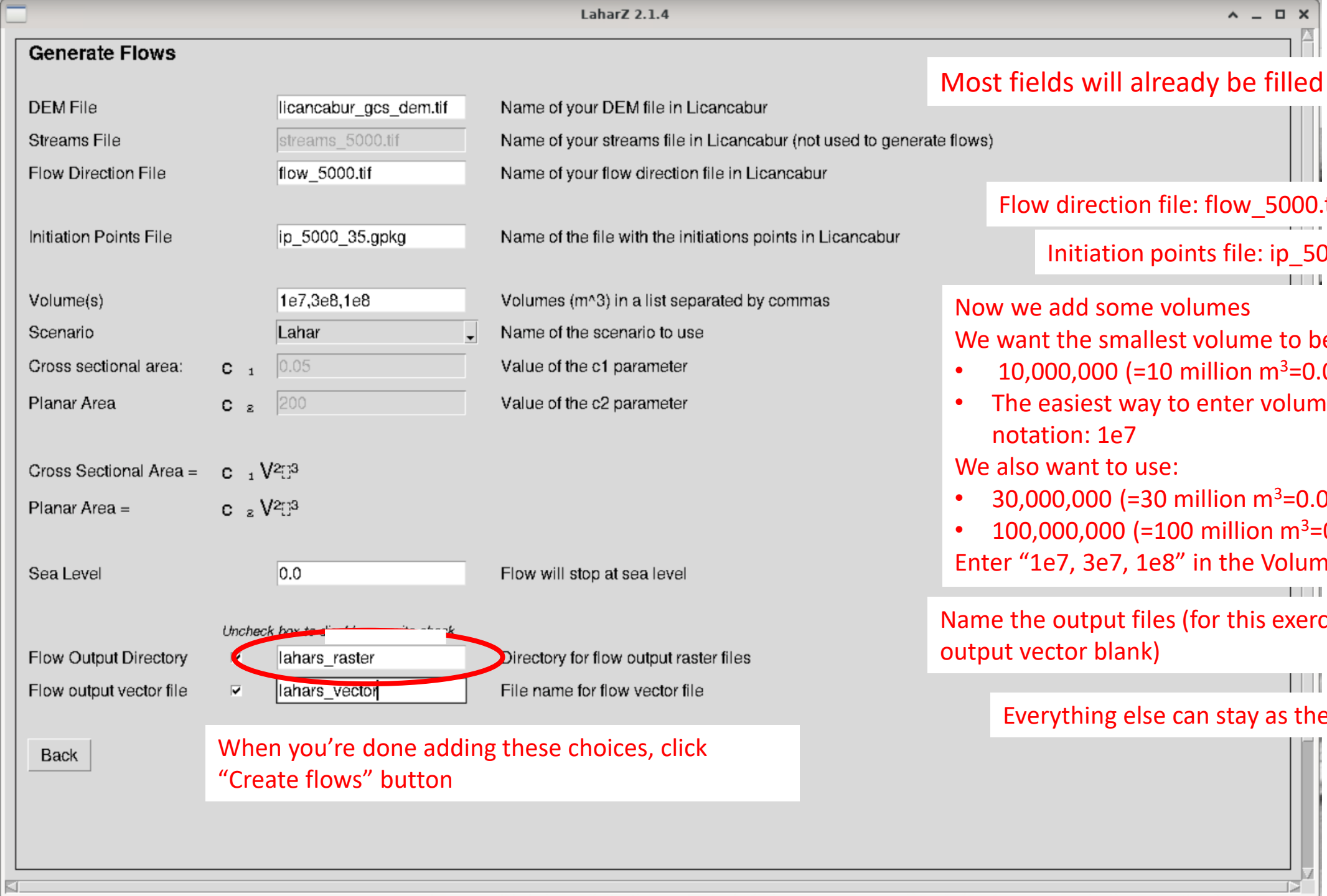

*Licancabur_QGIS — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
IMEX_LAVA
IMEX_SfloW_2D
LaharZ
example
Licancabur
ec_cone_35.tif
ec_line_35.tif
flow_5000.tif
ip_5000_35.gpkg
licancabur_gcs_dem.tif
licancabur_gcs_hs.tif
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prox_hz_zn_35.gpkg
streams_5000.tif
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Laharz 2.1.4c User Guide.pdf
Layers
ip_5000_35
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ec_line_35
Band 1 (Gray)
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streams_5000
Band 1 (Gray)
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Band 1 (Gray)
254.960602
Click the pencil icon again and
a save dialog will appear
Click “Save”
Stop Editing
Do you want to save the changes to layers ip_5000_35?
Save
Discard
Cancel
Processing Toolbox
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Recently used
Hillshade
Raster analysis
Rescale raster
Raster terrain analys
Hillshade
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Simplify coverag
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Applications
LaharZ 2.1.1
*Licancabur_QGIS — QGIS
Terminal
LaharZ 2.1.1
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Favorites
Spatial Bookmarks
Project Home
Home
Conflow
csav
Desktop
Let’s go back to LAHARZ. Click the tab at the top of the screen

Let’s run LAHARZ!



Click “Create Flow Outputs”



Most fields will already be filled in

Flow direction file: flow_5000.tif

Initiation points file: ip_5000_35.gpkg

Now we add some volumes

We want the smallest volume to be:

- 10,000,000 (=10 million m^3=0.01km^3=1x10^7=1e7)
- The easiest way to enter volumes is using scientific notation: 1e7

We also want to use:

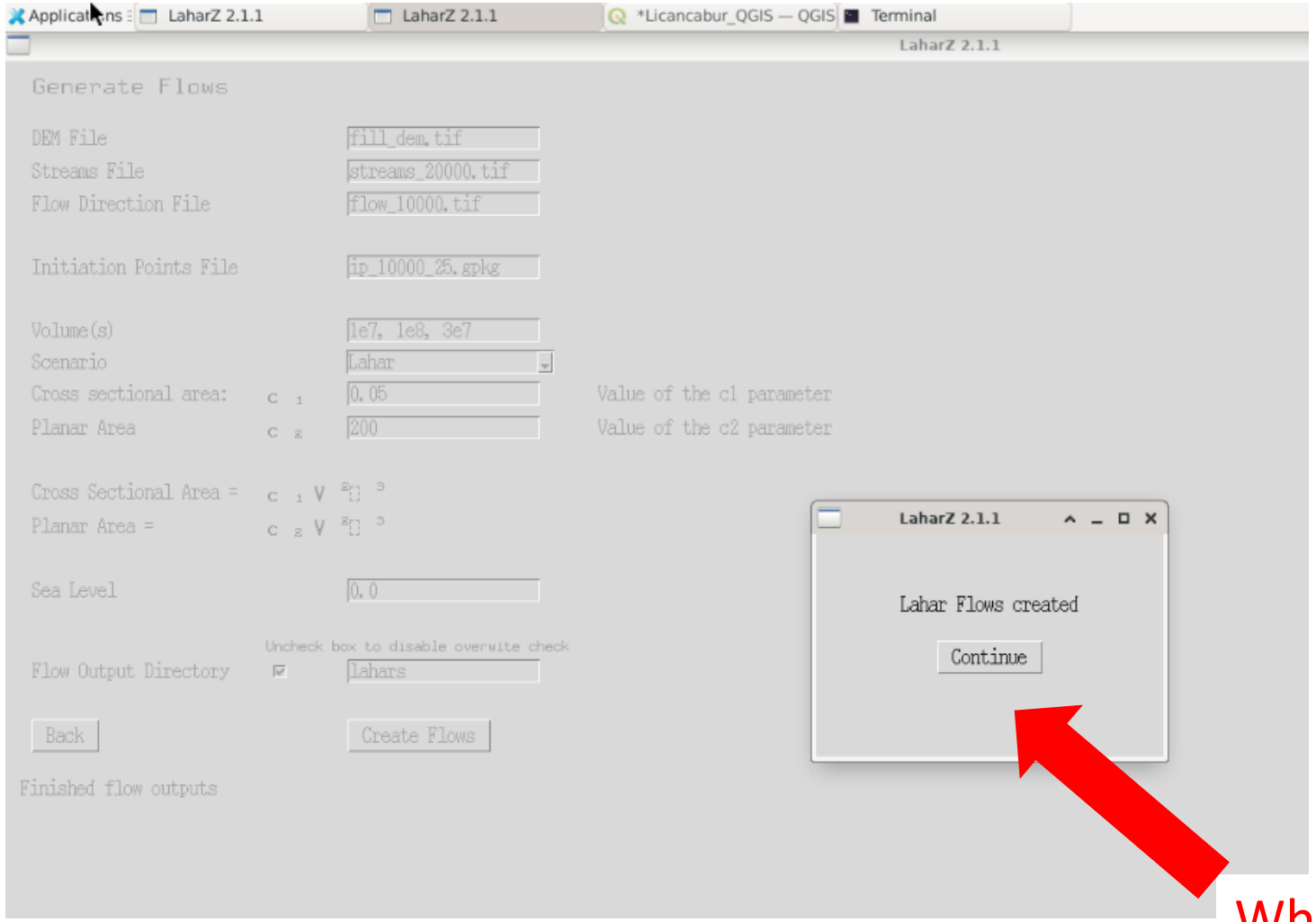
- 30,000,000 (=30 million m^3=0.03 km^3=3x10^7=3e7)
- 100,000,000 (=100 million m^3=0.1km^3=1x10^8=1e8)

Enter “1e7, 3e7, 1e8” in the Volumes box

Name the output files (for this exercise, we’ll leave output vector blank)

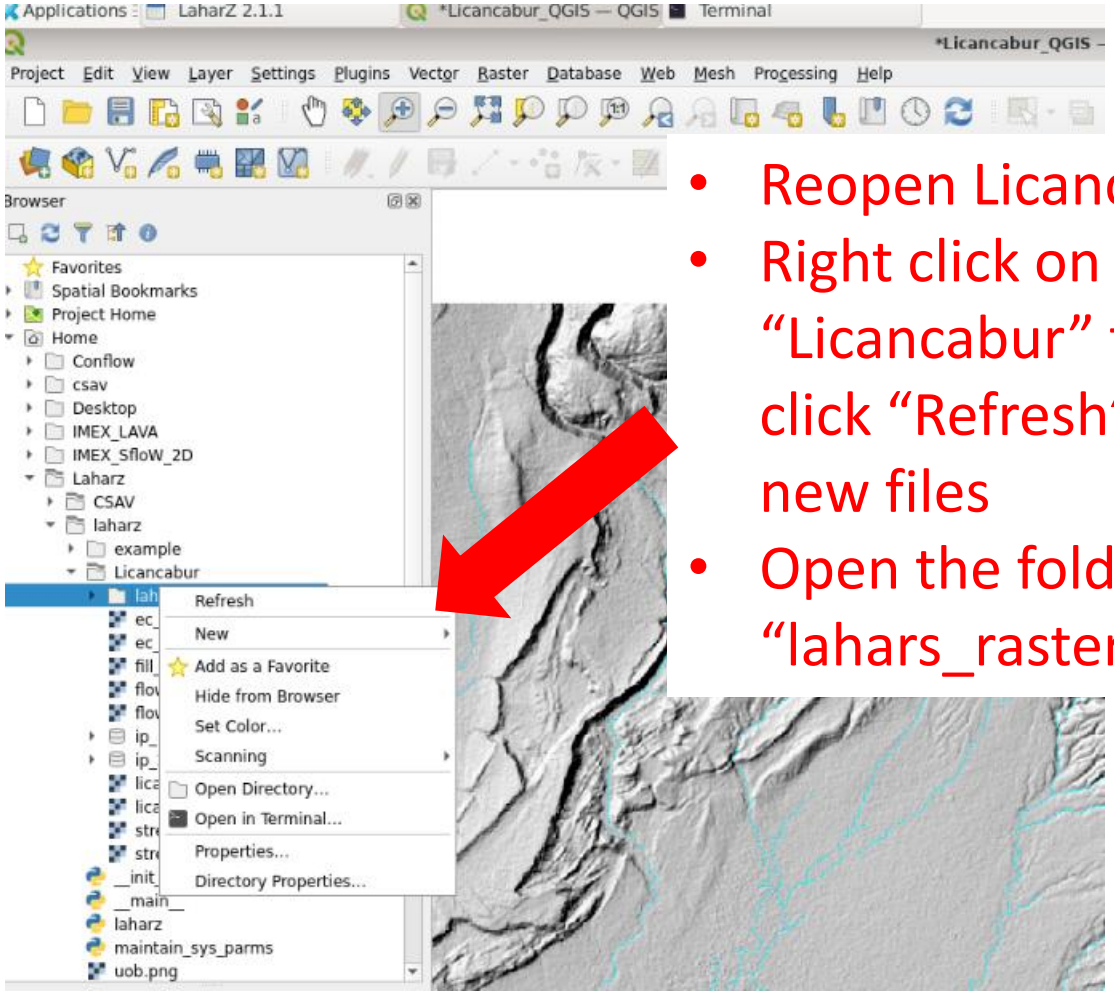
Everything else can stay as the default

When you’re done adding these choices, click “Create flows” button

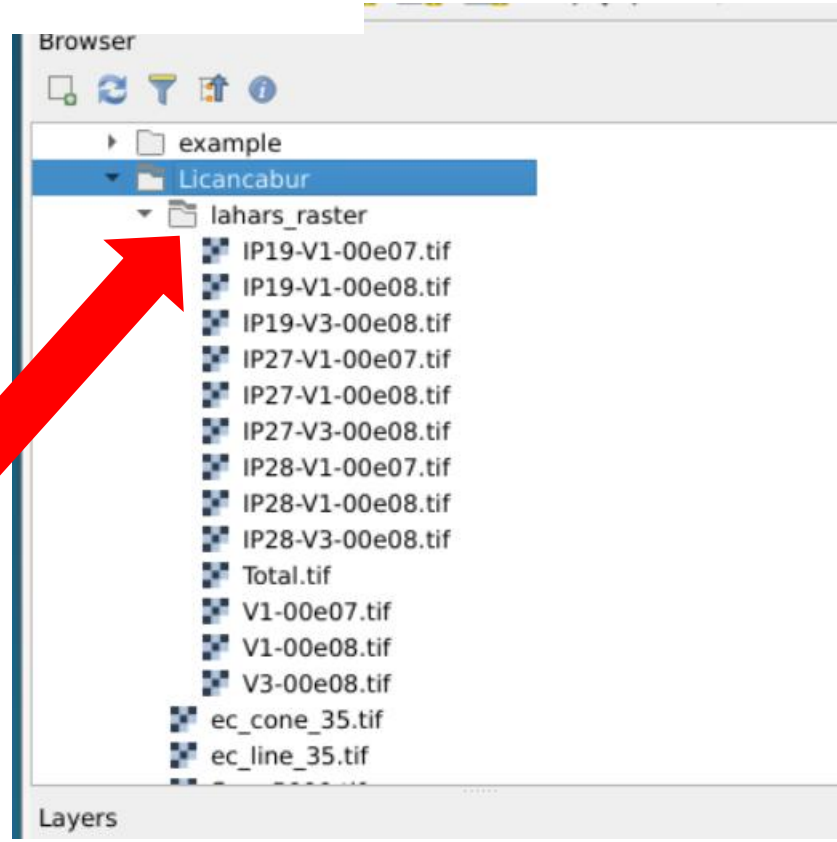


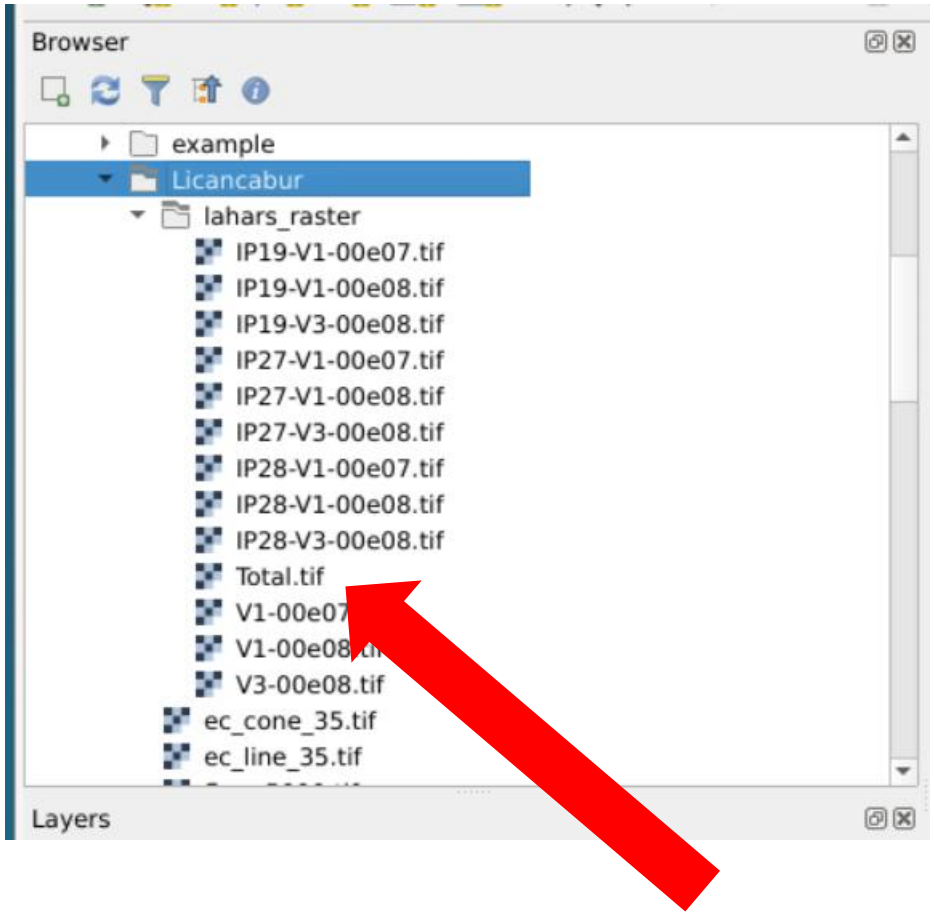
When it’s done, click “Continue”

Let’s look at the new files.



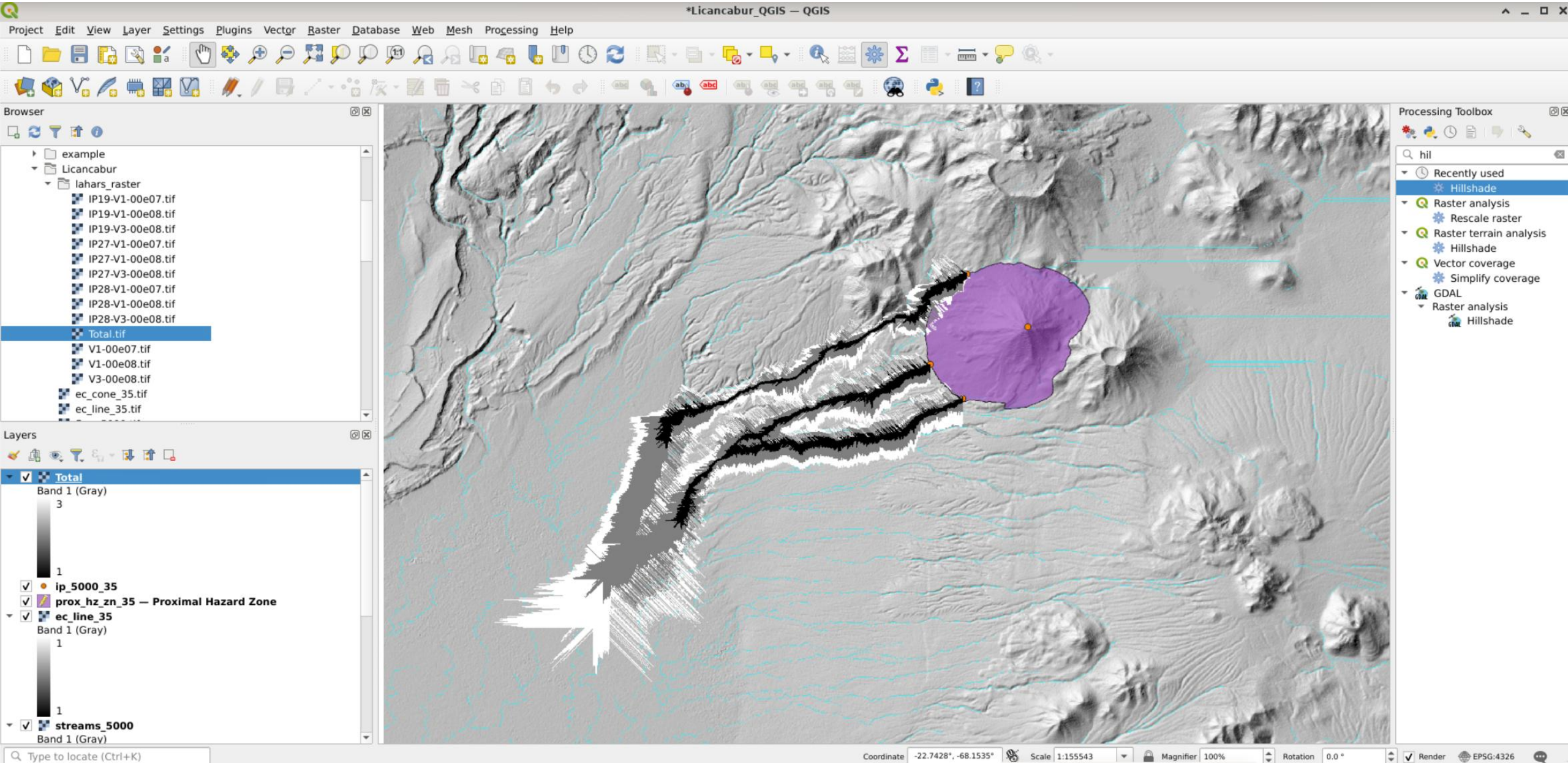
- Reopen Licancabur_QGIS
- Right click on the “Licancabur” folder and click “Refresh” to see your new files
- Open the folder called “lahars_raster”

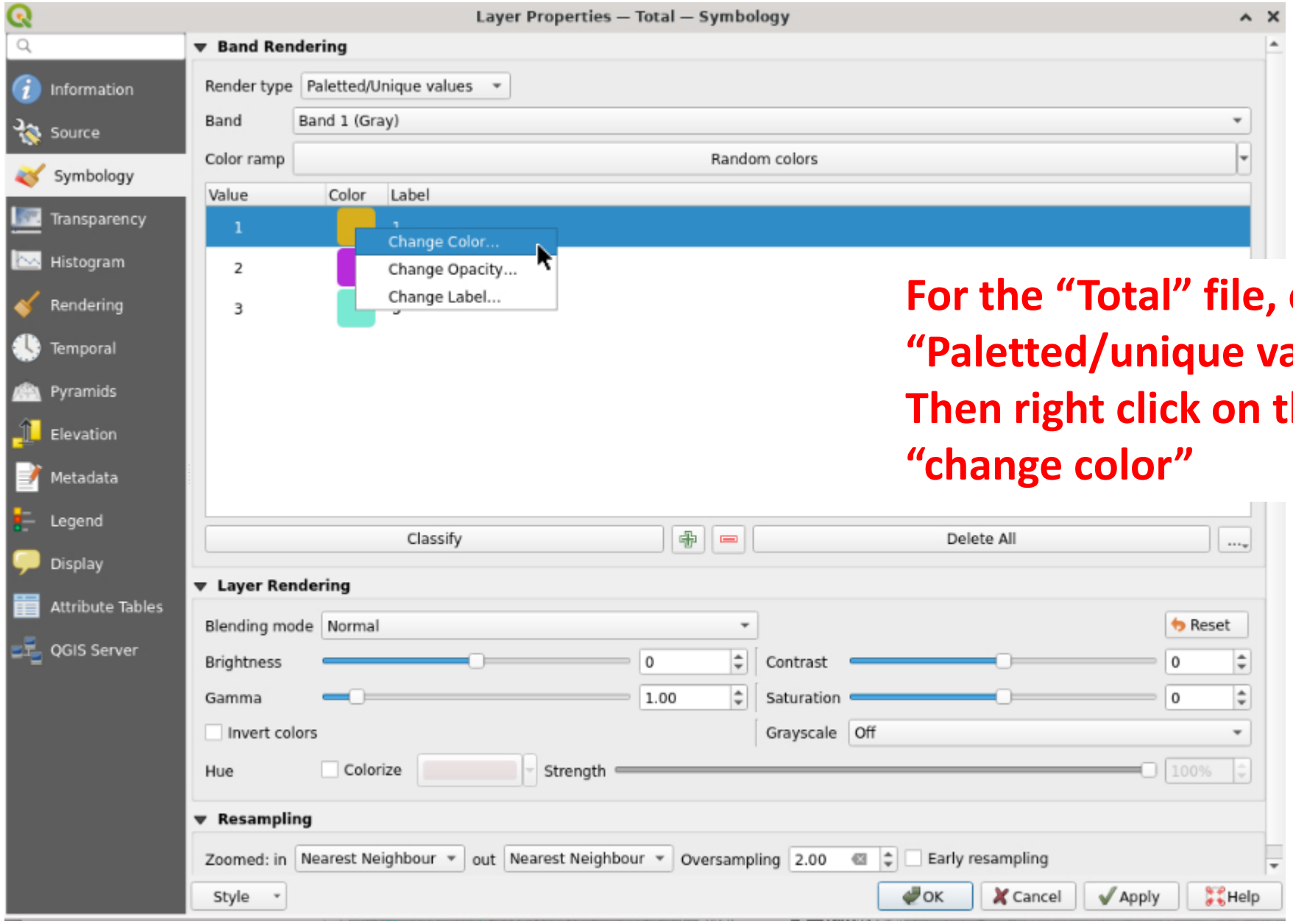




- LAHARZ made a bunch of different files
- There is a separate file for each volume/initiation point combination (these start with “IP”
- LAHARZ also made a file for each volume, merging all the flows from each initiation point (these start with “V”)
- And finally, there is a “Total” file that has all the volumes from all the initiation points in one file
- Add the “total.tif” file to your map

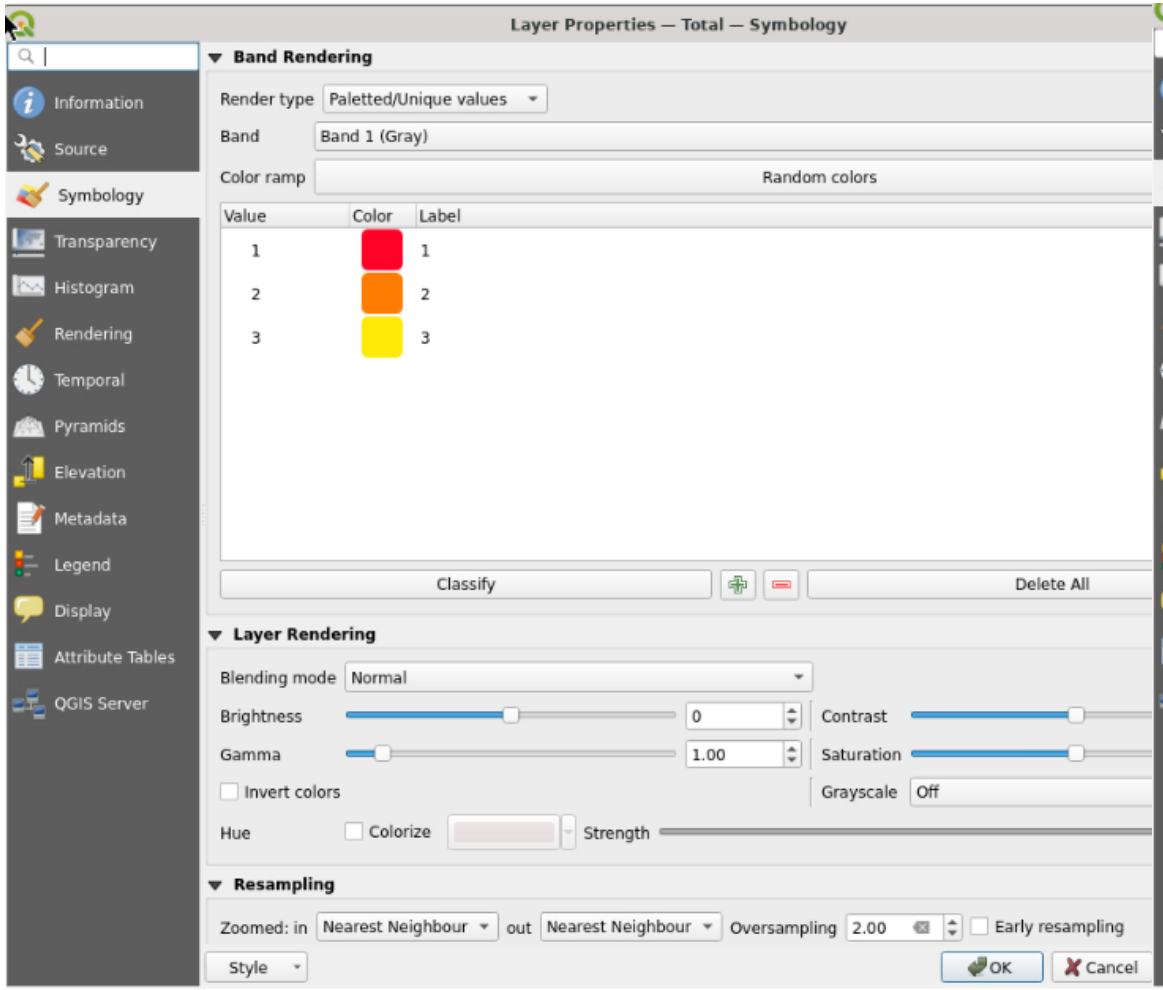
Let’s change the colors and make the map prettier.

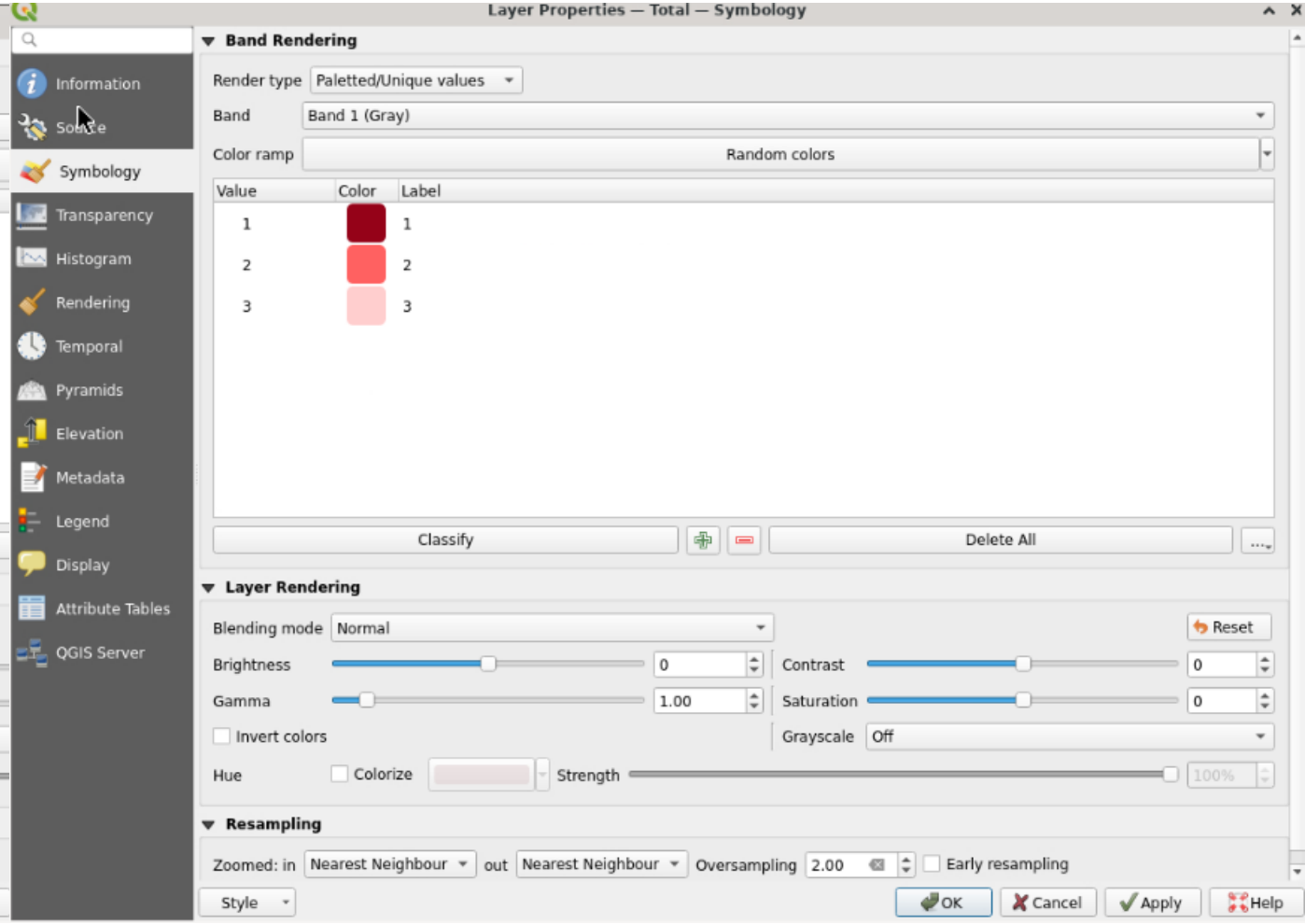


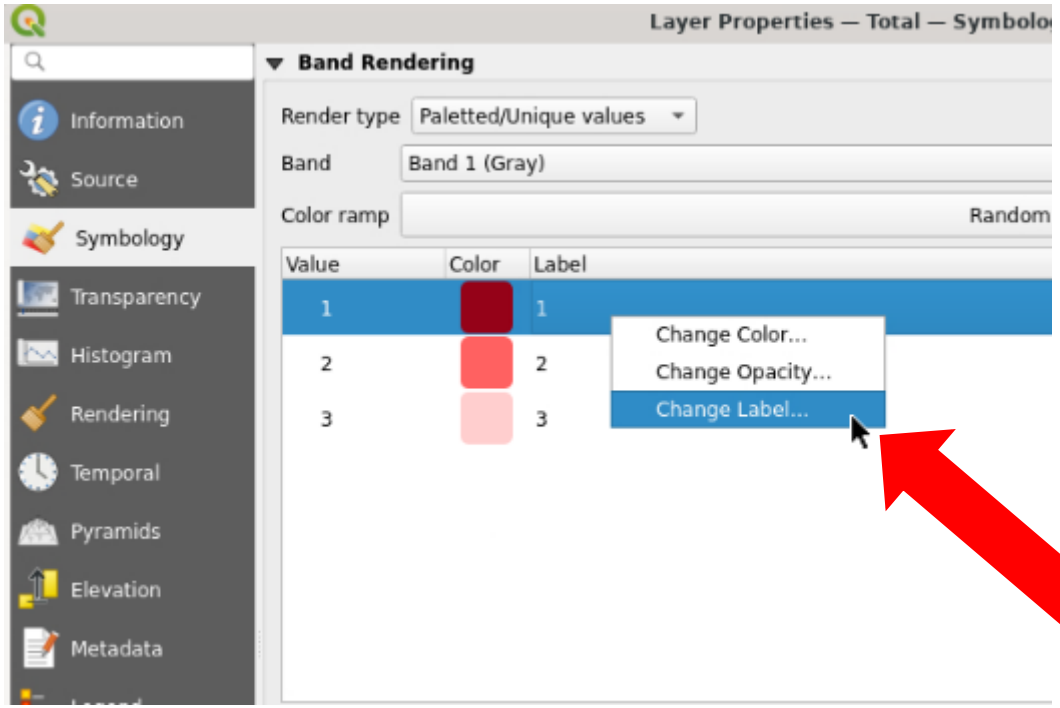


For the “Total” file, change the “render type” to “Paletted/unique values” and click “classify”. Then right click on the listed value and choose “change color”

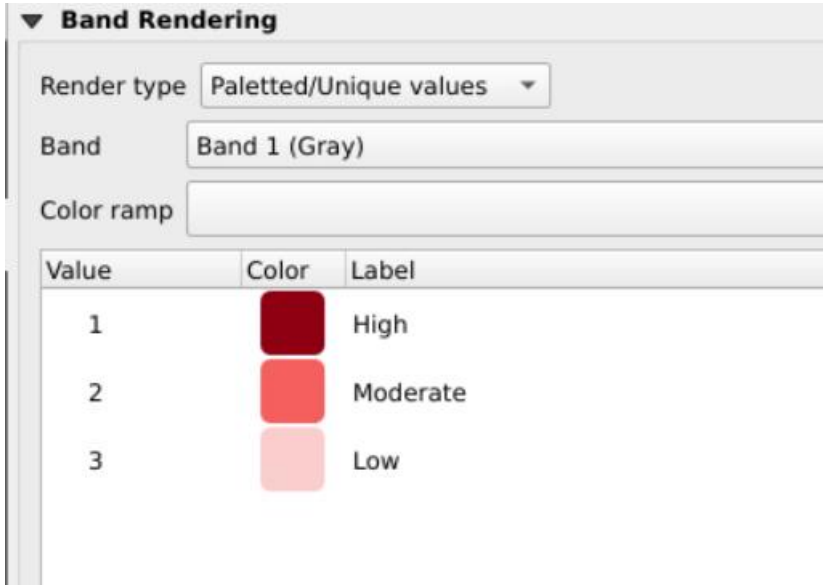
- The lowest value (1) is the lowest volume (1 x 10^7).
- The lowest volume is the highest hazard zone because smaller lahars occur more, a lahar of any size would reach this area, lahars would reach this area first, etc.
- Choose a color scheme that makes sense, like red-orange-yellow or dark red to light red.
- We'll talk more about color schemes later!

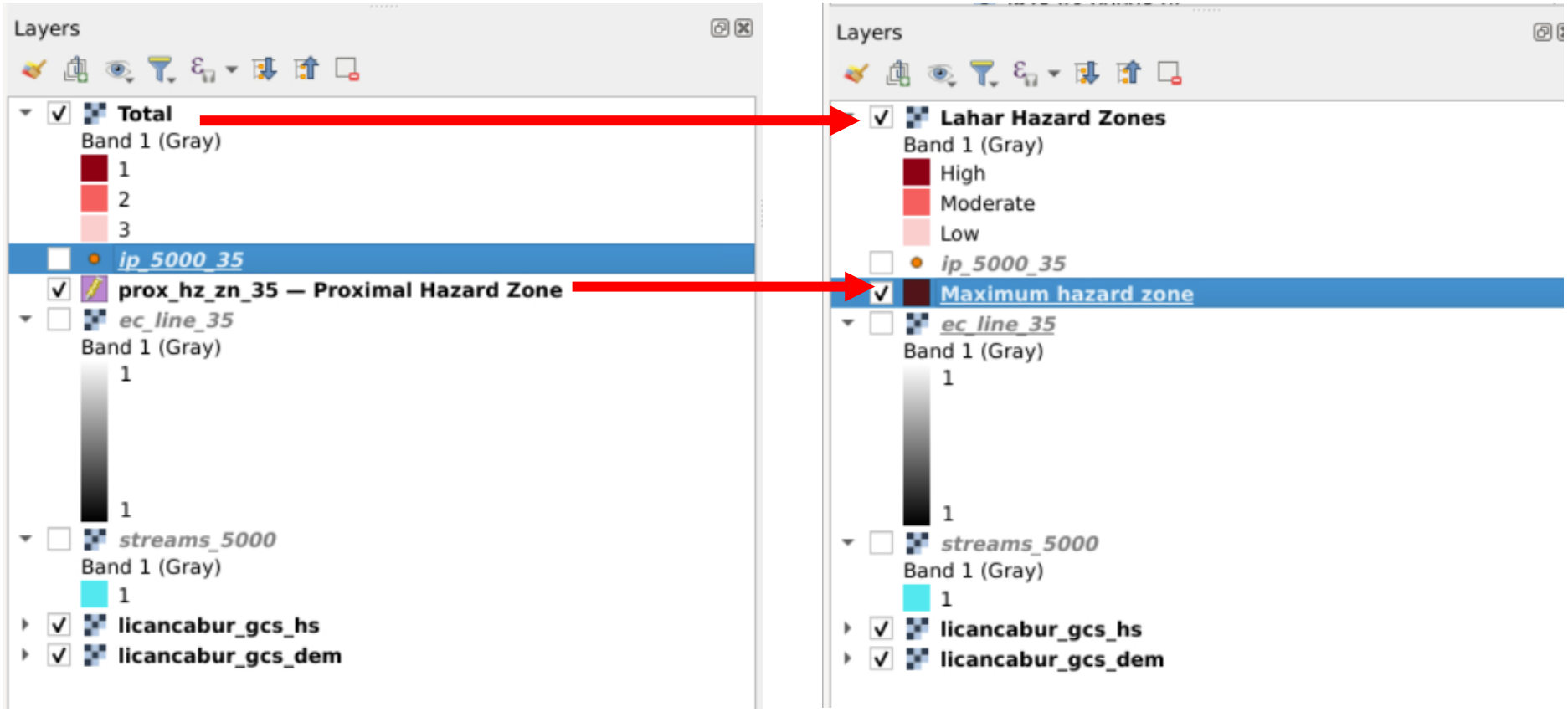






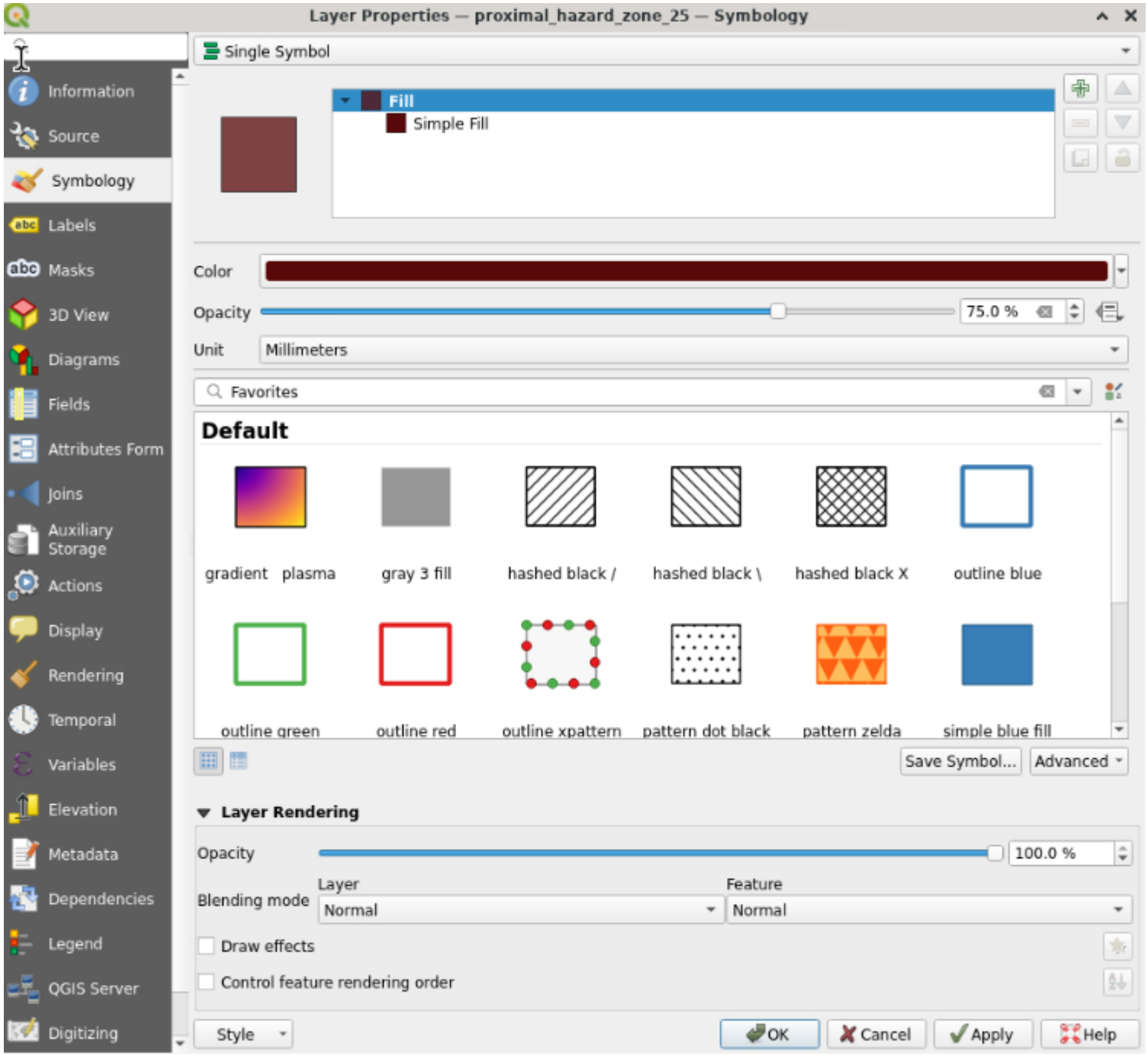
You can also change the label for each layer to: High, Moderate, and Low





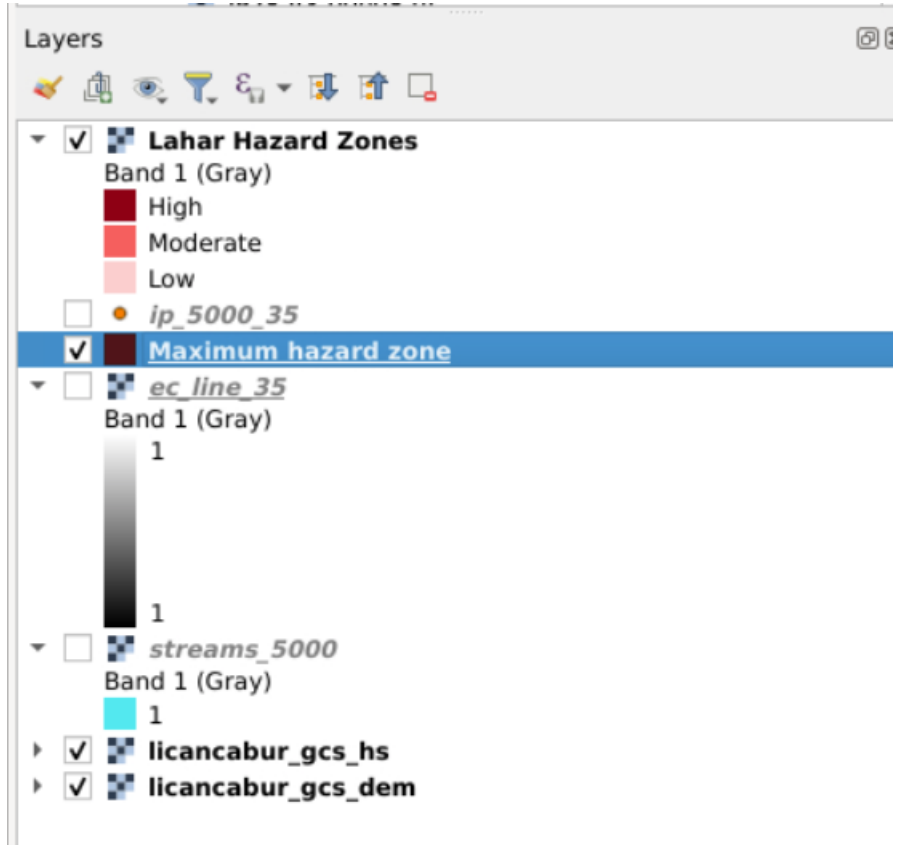
Let's rename some of the layers so they will look good in the legend.

Rename the files to "Lahar Hazard Zones" and "Maximum Hazard Zone" by right clicking and choosing "Rename"

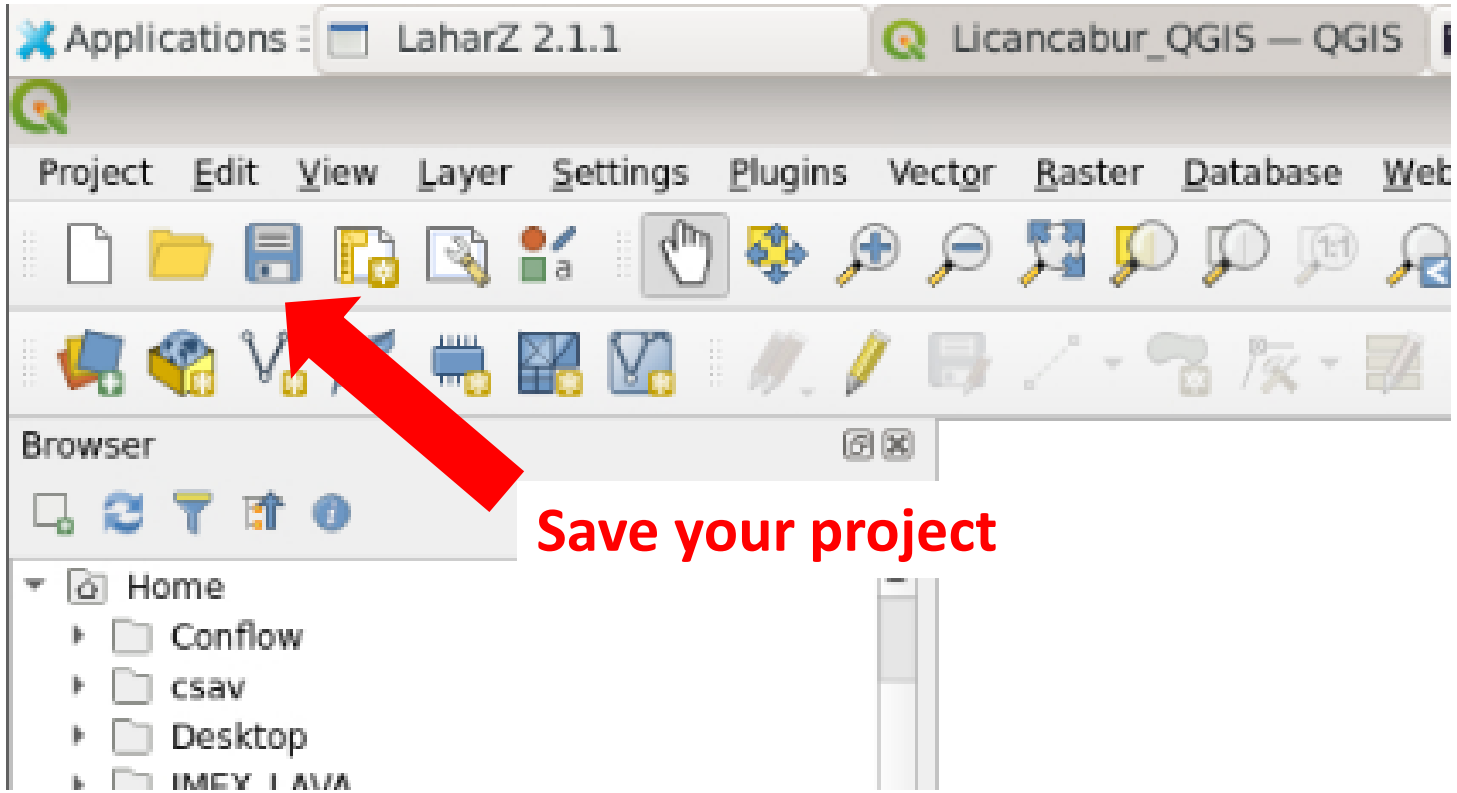


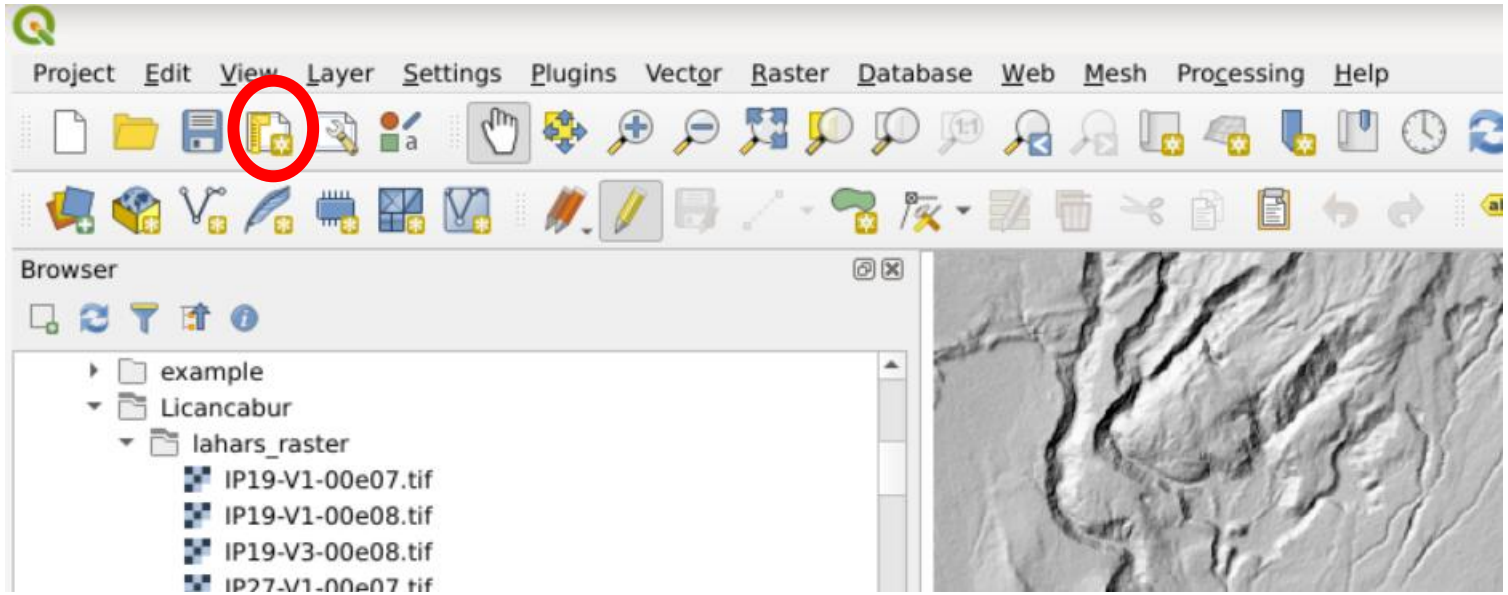
Change the color and opacity of the Maximum Hazard Zone layer

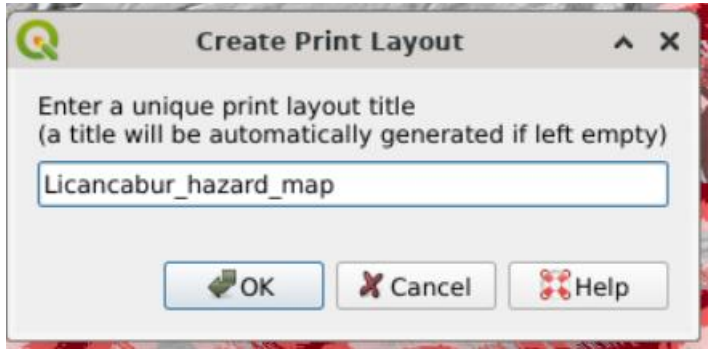
Let’s make the map neater



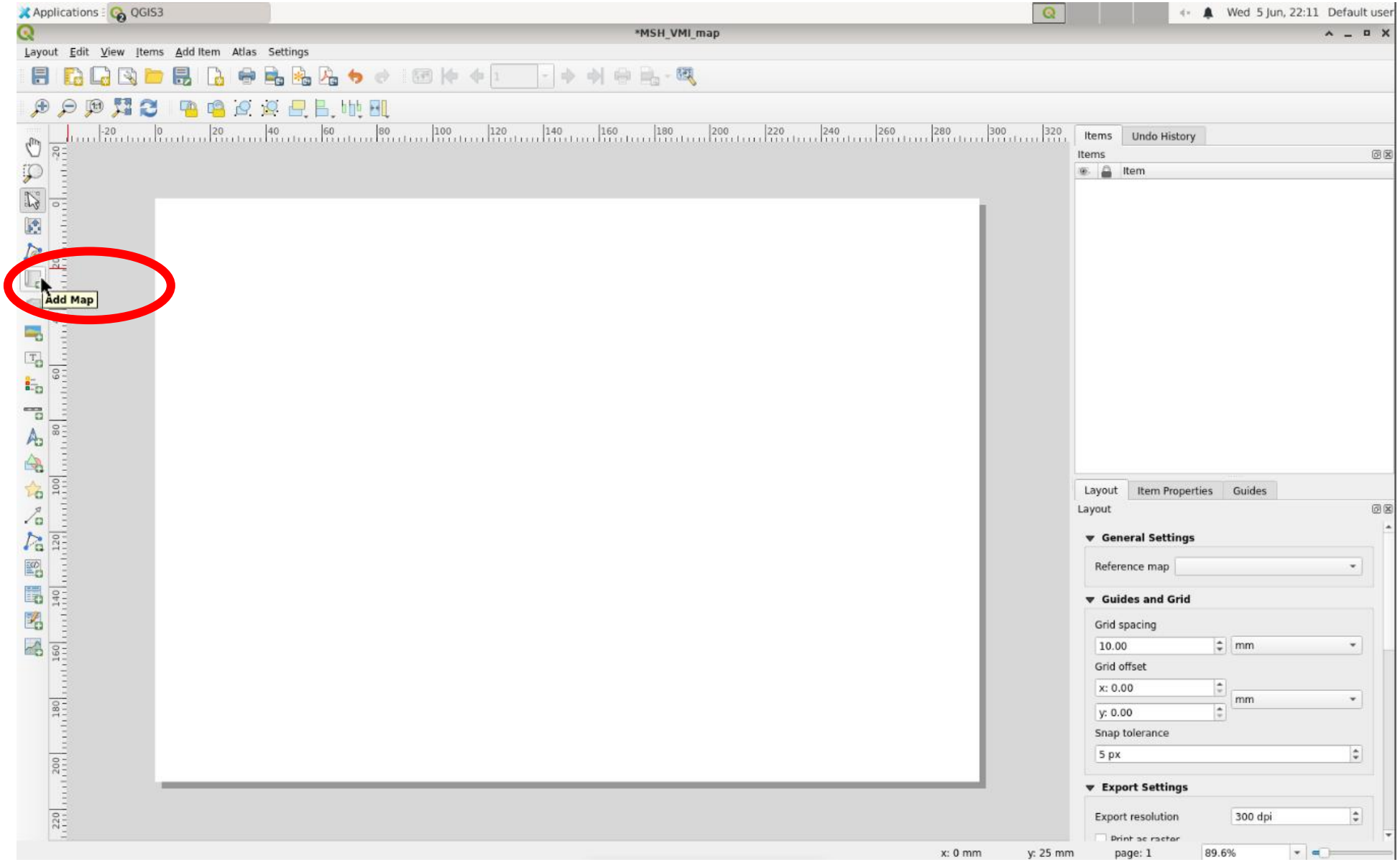
- Turn off layers that aren’t needed


Applications
LaharZ 2.1.1
Licancabur_QGIS — QGIS
Project
Edit
View
Layer
Settings
Plugins
Vector
Raster
Database
Web
Browser
Save your project
Home
Conflow
csav
Desktop

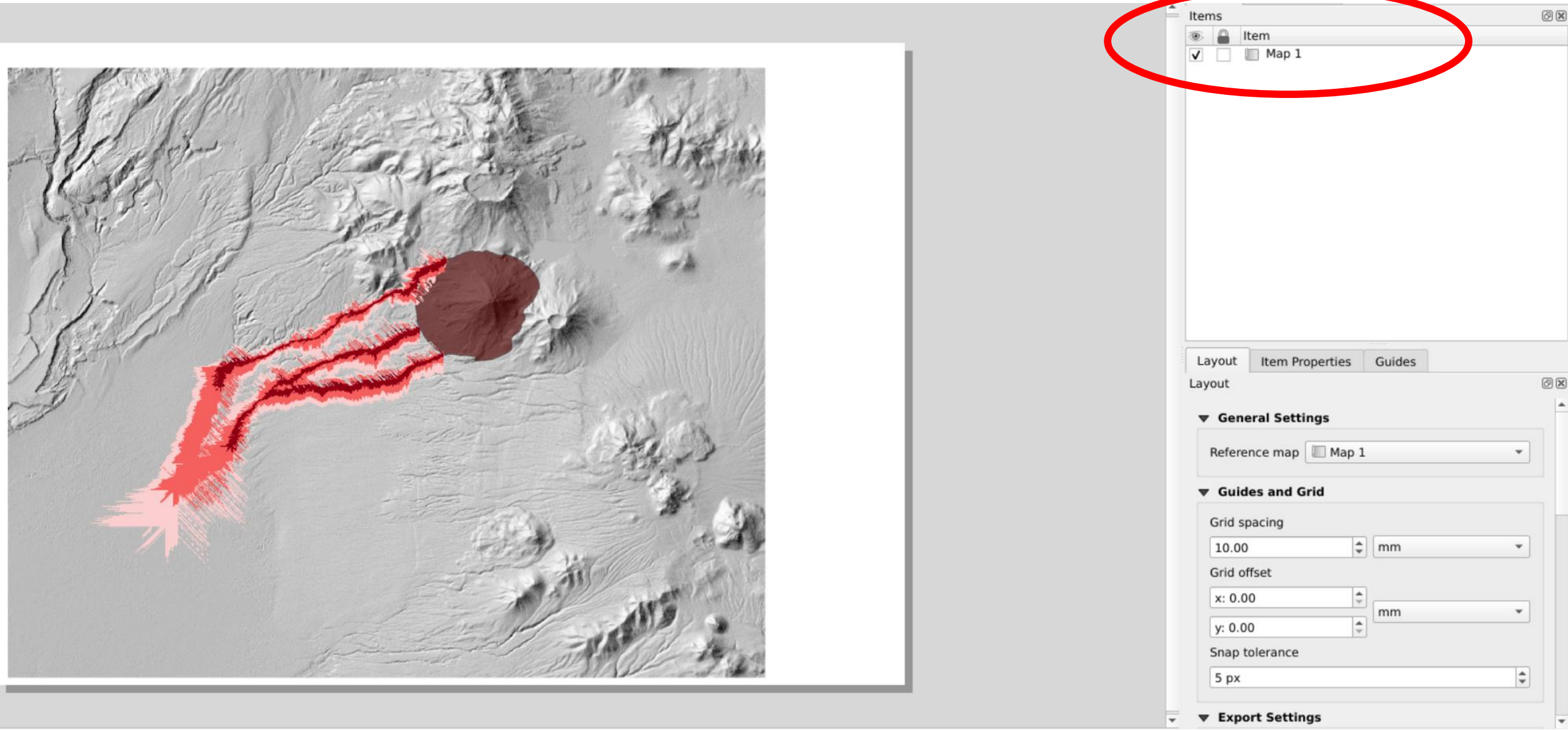




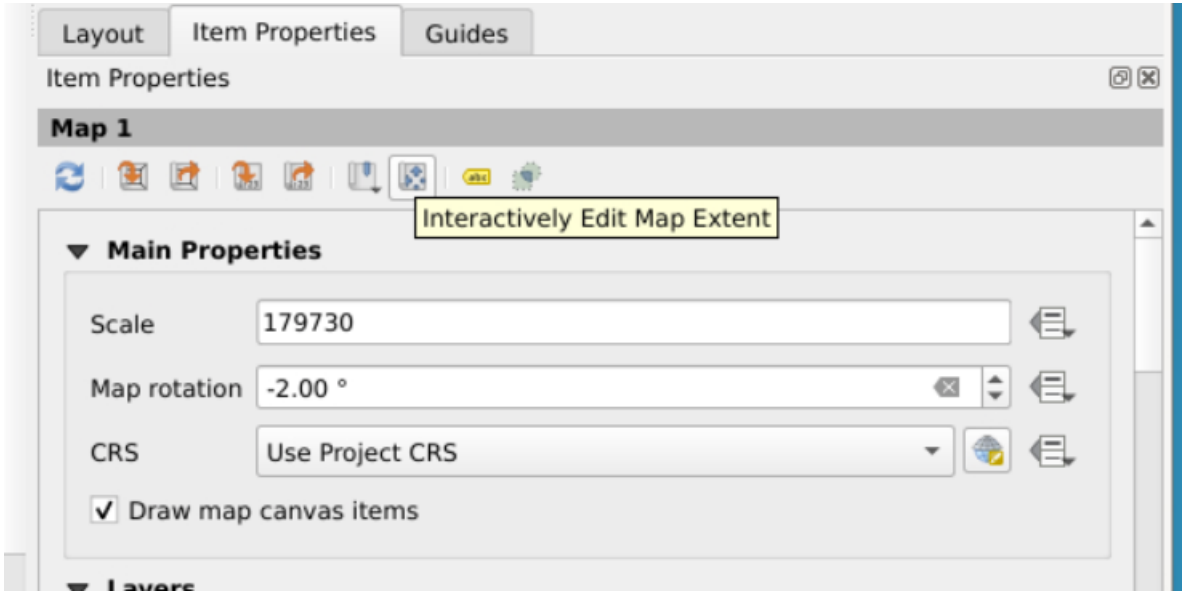
- Let's make a map!
- Click the "New Print Layout" button
- Enter a name for the new map. I chose "Licancabur_hazard_map"
- Click "OK"



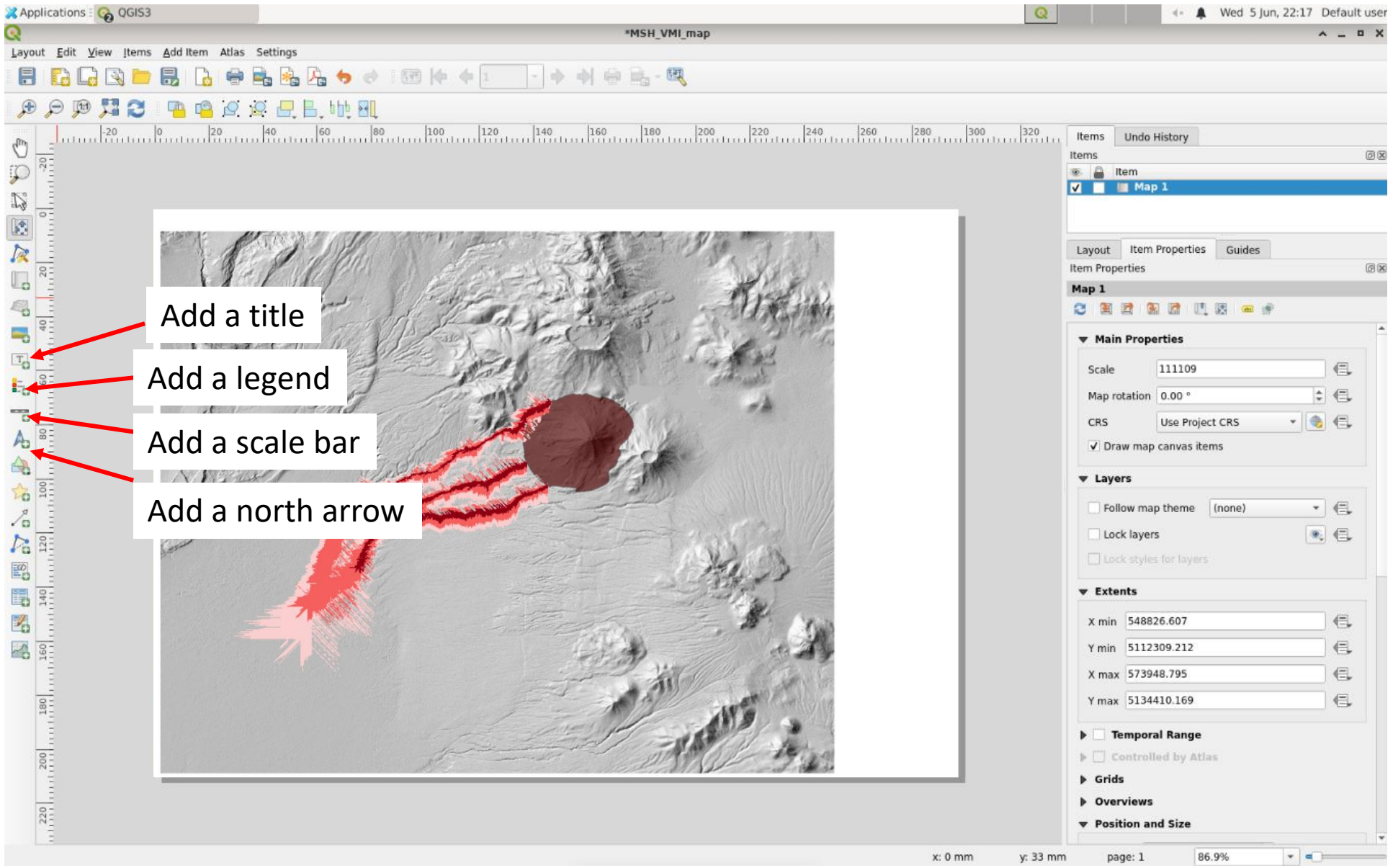
- First you need to add your map to the layout
- Click the “Add map” button
- Then draw a rectangle on the blank page to add the map

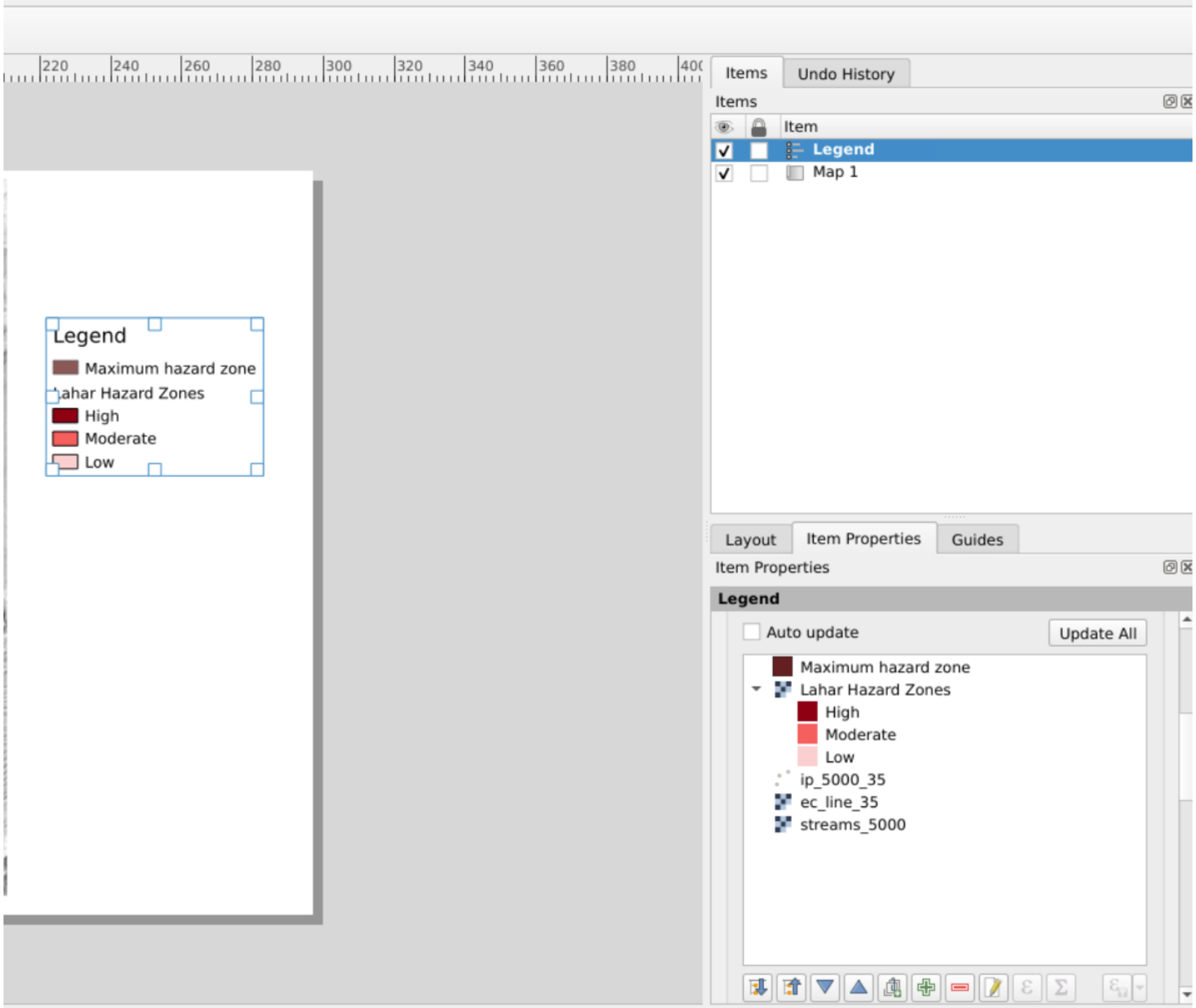


- You can adjust the scale and extent and other properties by selecting the map layer and “Item properties”

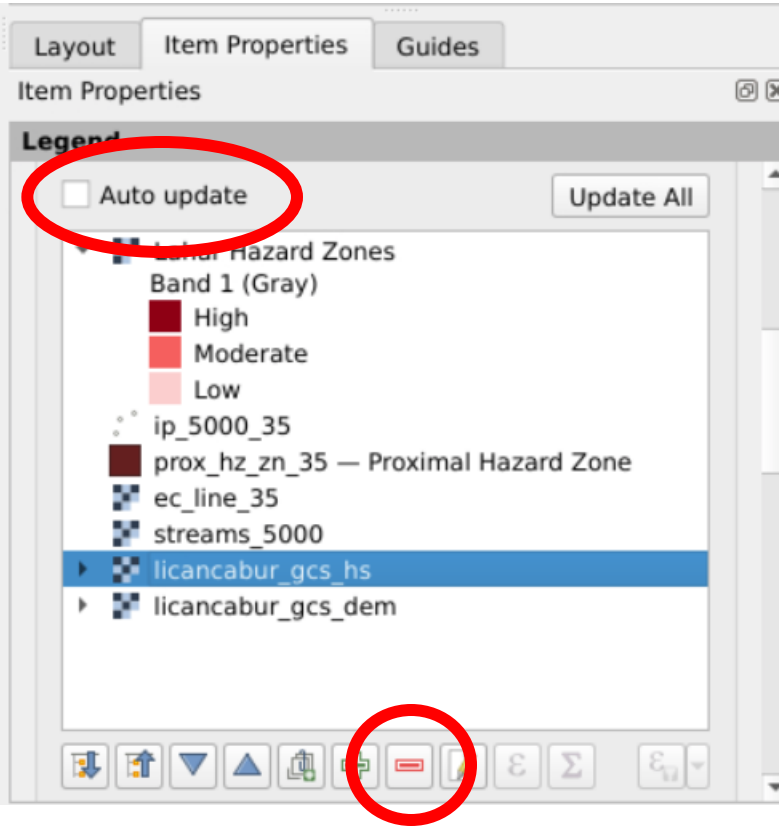


- You can adjust the scale and extent more easily by using the “Interactively Edit Map Extent” button

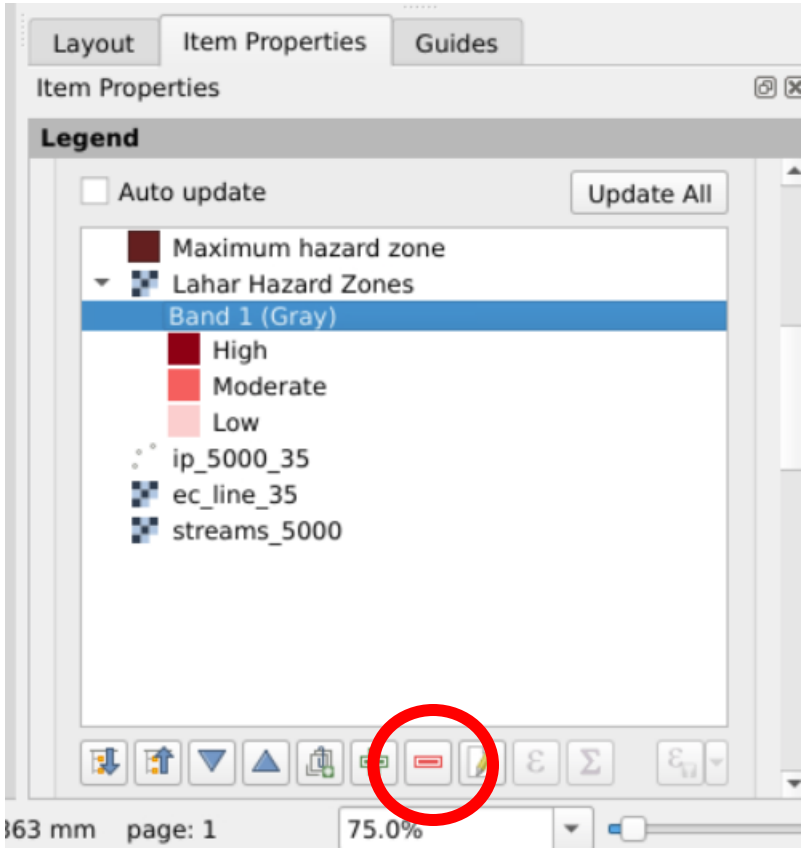

Applications
QGIS3
Wed 5 Jun, 22:17
Default user
*MSH_VMI_map
Layout
Edit
View
Items
Add Item
Atlas
Settings
Add a title
Add a legend
Add a scale bar
Add a north arrow
Items
Undo History
Item
Map 1
Layout
Item Properties
Guides
Item Properties
Map 1
Main Properties
Scale
111109
Map rotation
0.00 °
CRS
Use Project CRS
Draw map canvas items
Layers
Follow map theme
(none)
Lock layers
Lock styles for layers
Extents
X min
548826.607
Y min
5112309.212
X max
573948.795
Y max
5134410.169
Temporal Range
Controlled by Atlas
Grids
Overviews
Position and Size
x: 0 mm
y: 33 mm
page: 1
86.9%



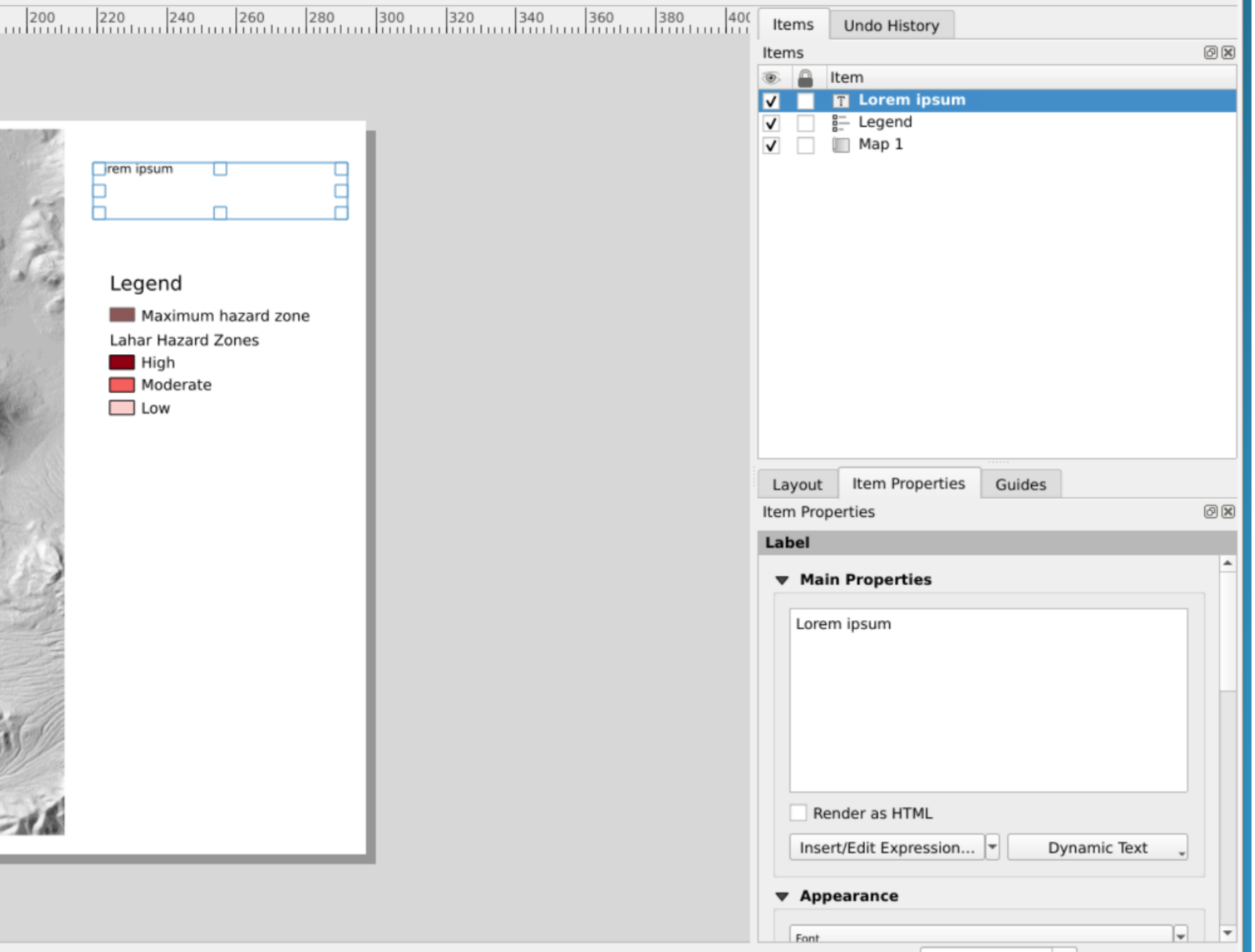
- Add a legend and edit the properties



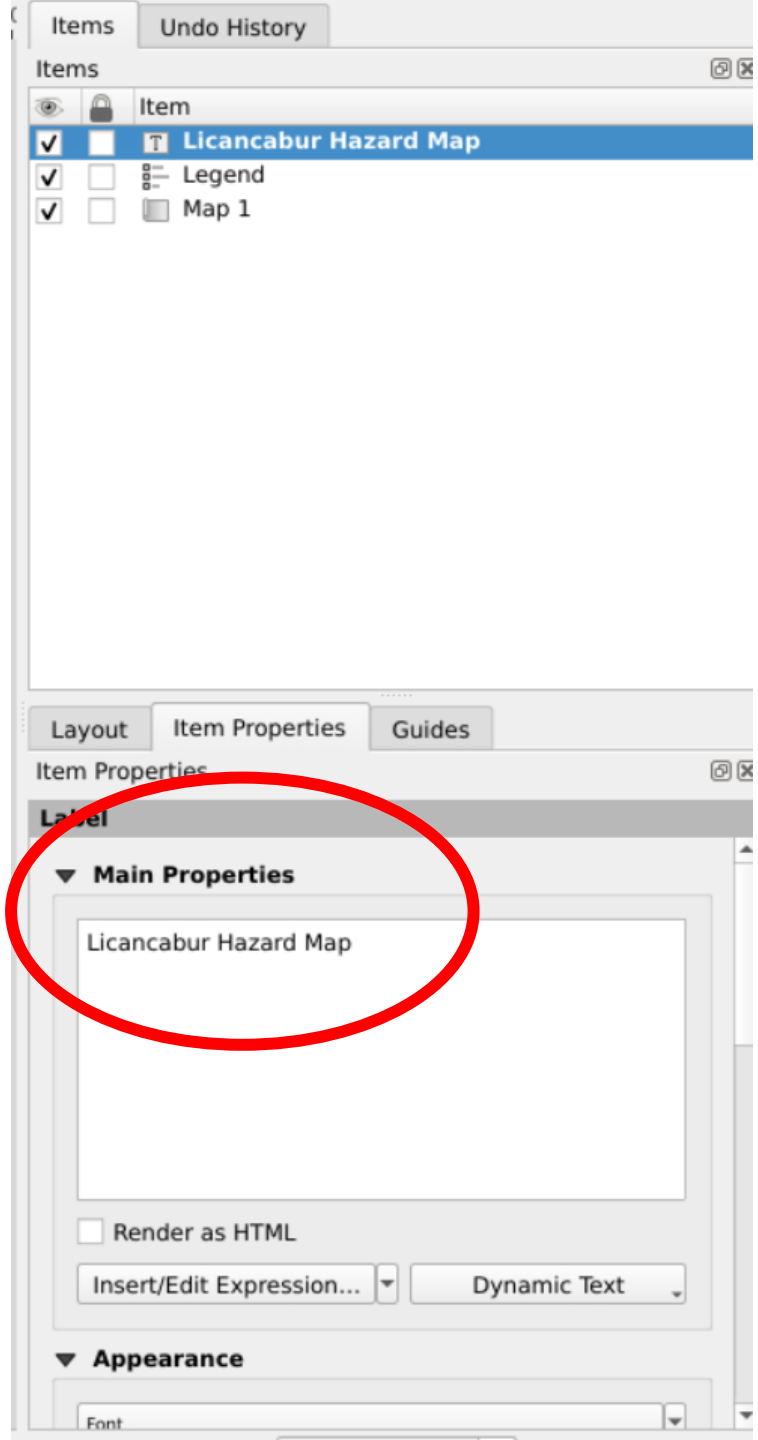
- Turn off auto update so you can make changes to the legend items
- We don't need the DEM on the legend, so select the hillshade and the dem and click the red minus button

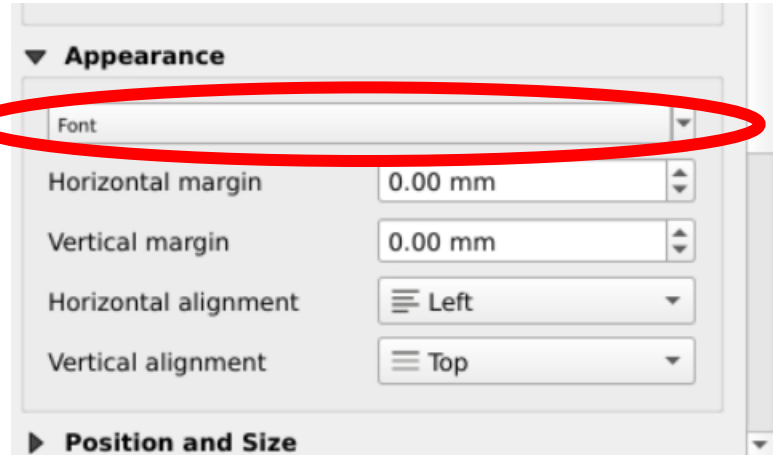


- We also don't want the "Band 1 (Gray)" on the legend, so select it and click the red minus button

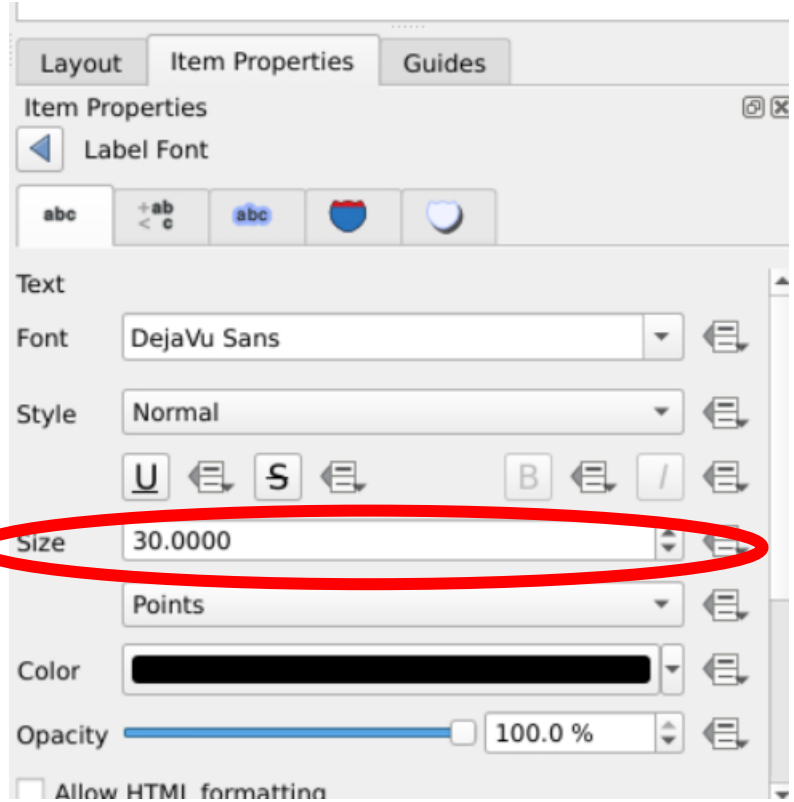


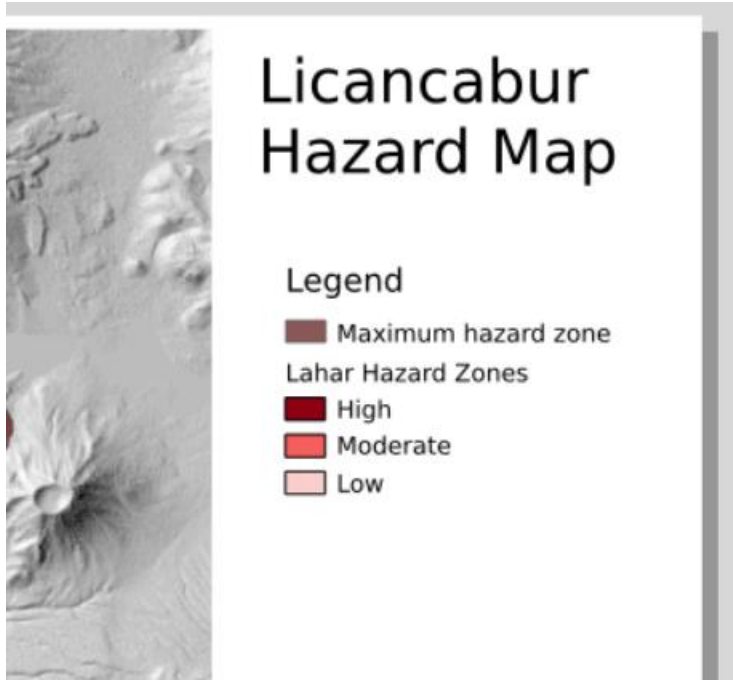
- Add a title and change the properties



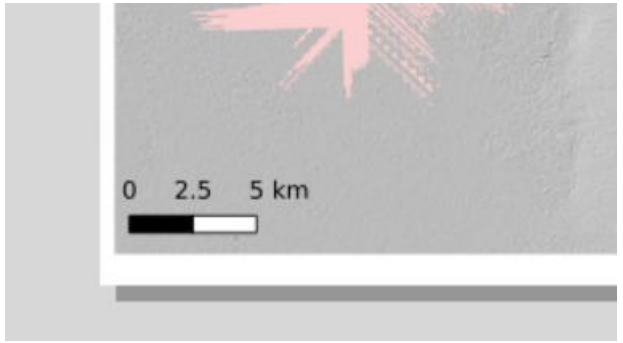


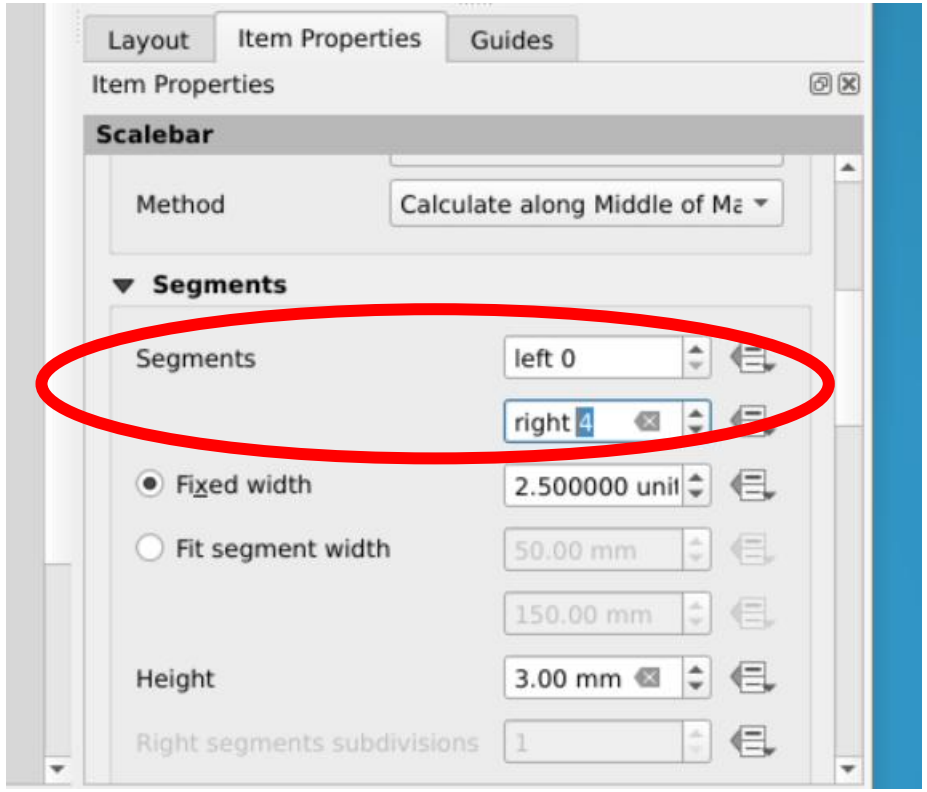
- Change the words for the title to “Licancabur Hazard Map”
- Change the font size to 30

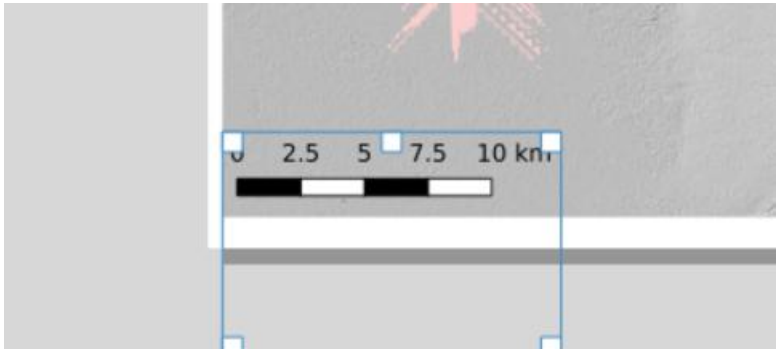


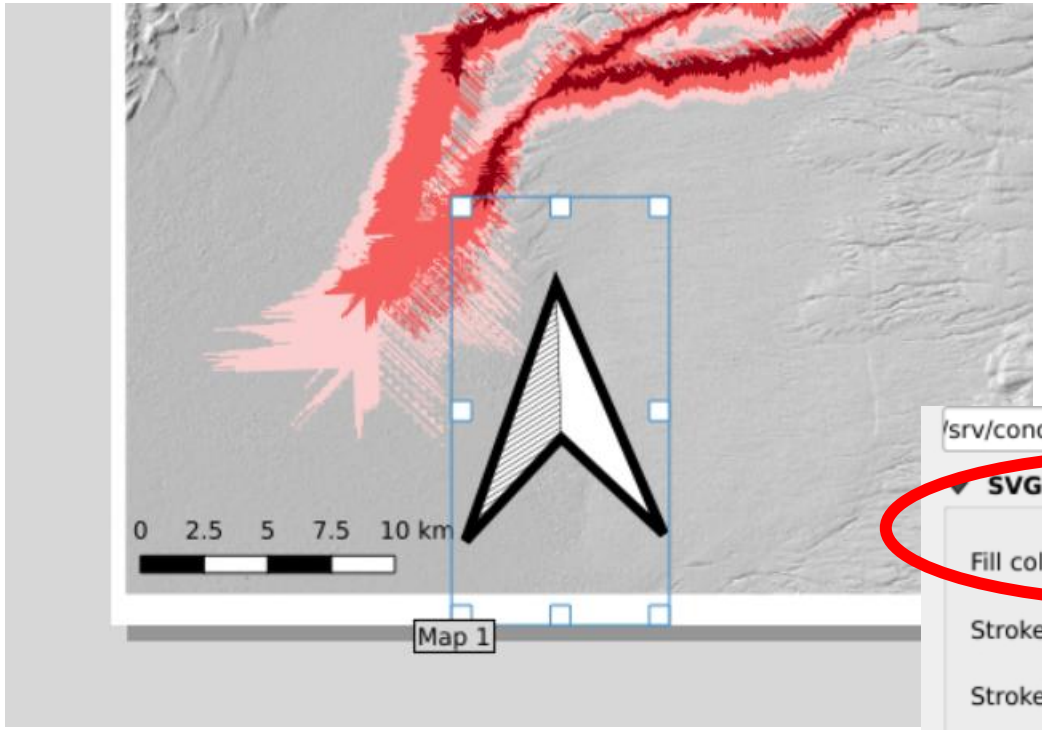


- Add a scale bar and edit the properties
- To make the scale bar longer, change it from 2 segments to 4 segments





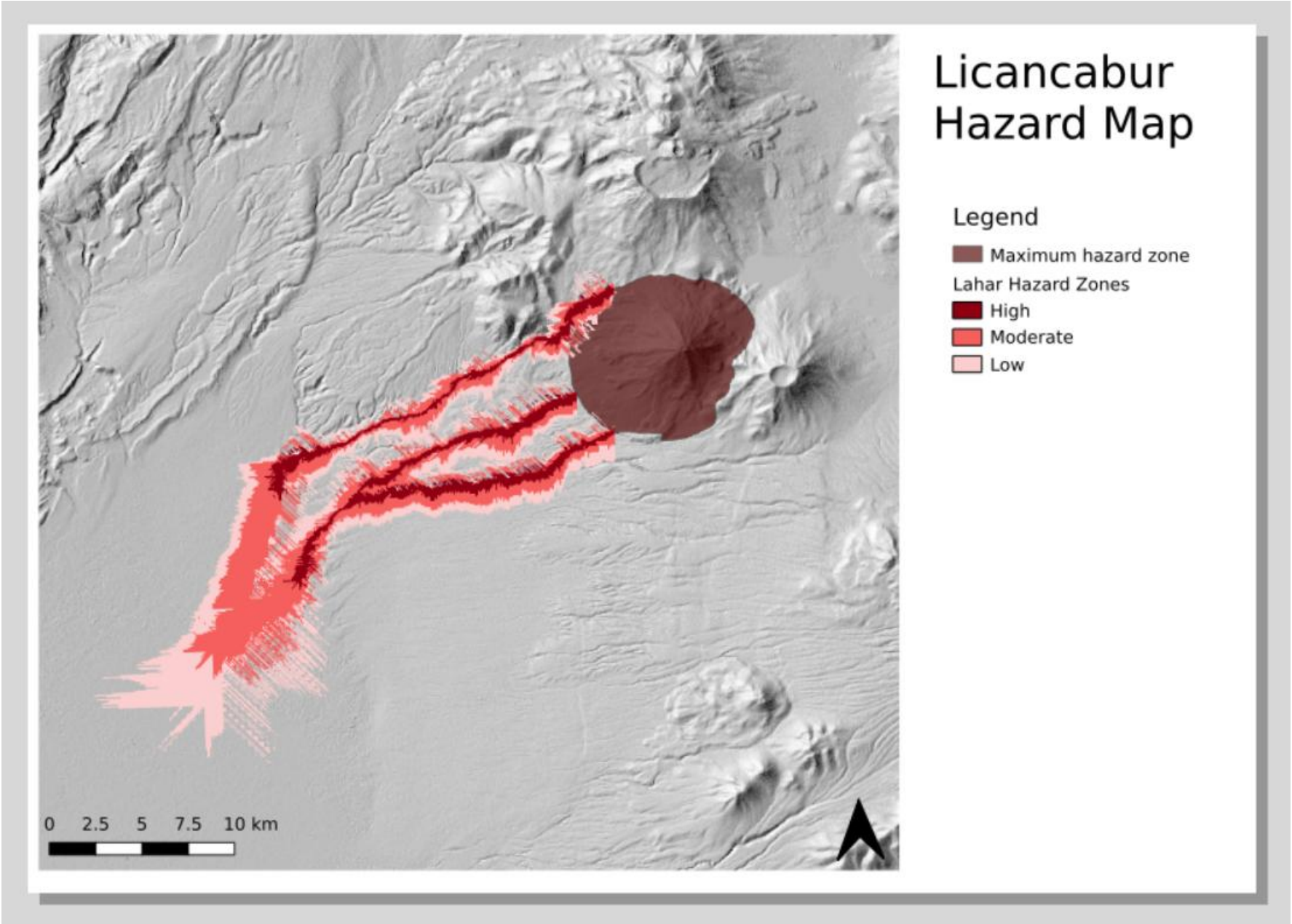






- Add a north arrow
- Adjust the size and placement
- Chage it to all black by changing the fill color to black

You’re done for now!!!



Congratulations!

¡Felicidades!

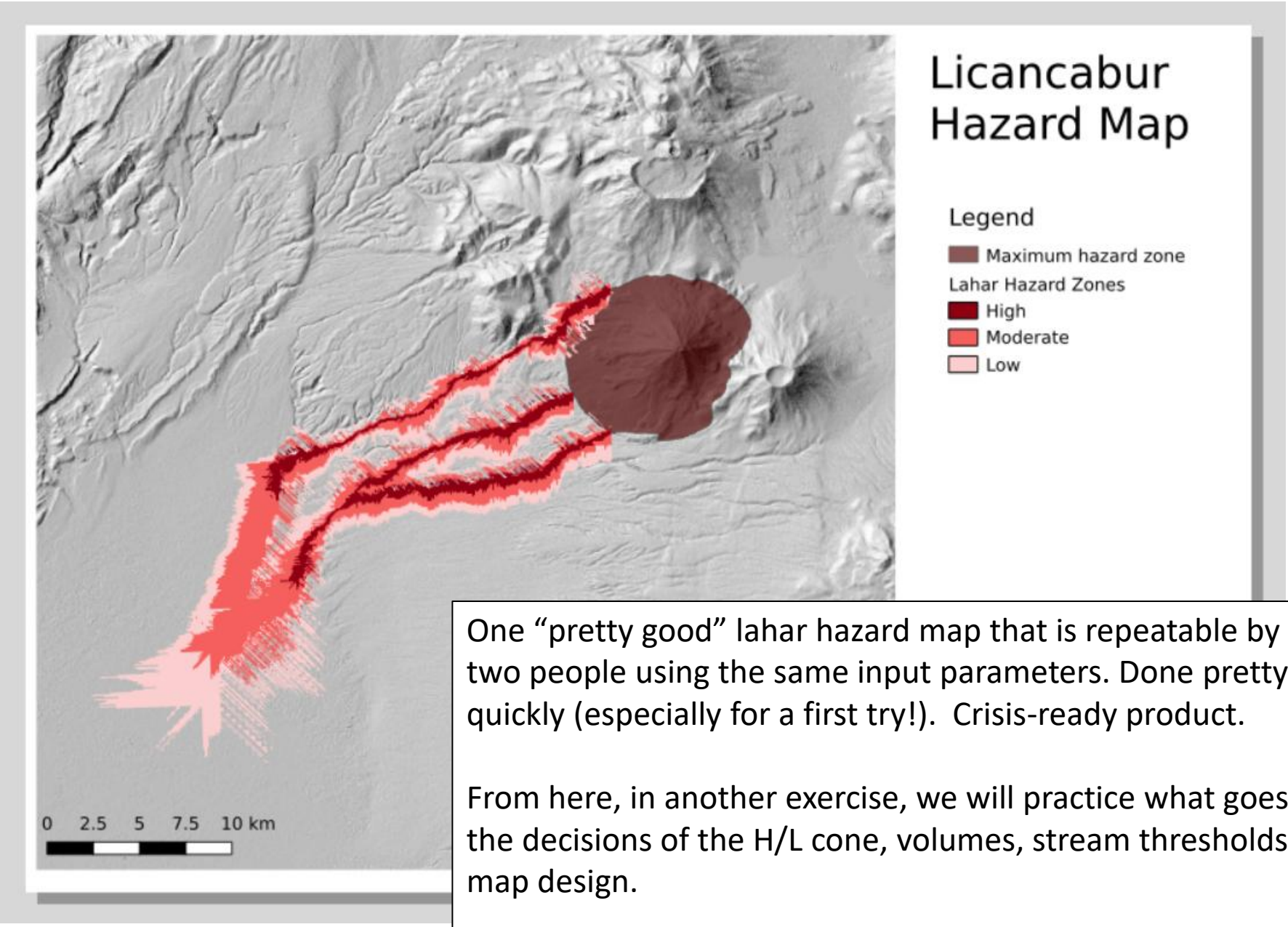
تهانينا

Selamat!

Binabati kita!

Baie geluk!

Félicitations!



One “pretty good” lahar hazard map that is repeatable by any two people using the same input parameters. Done pretty quickly (especially for a first try!). Crisis-ready product.

From here, in another exercise, we will practice what goes into the decisions of the H/L cone, volumes, stream thresholds, and map design.

Later, we will add other hazard zones for PDCs and ballistics.